

SparkBridge MBA go-to-market handoff, 2026-09-07

Everything the GTM program lead needs on day 1. Built from the SparkBridge repo (docs/gtm/), the SE training (branch training-se-v1) and the live site at mqtt.greenguard-usa.com/sparkbridge. Prices verified against the live pricing page 2026-09-06. Contacts re-verified 2026-09-07.

Read in this order

Day	File	Why
0	01-plan/mba-90-day-gtm-sprint.pdf	The playbook. Section 25 is the access checklist; get it done before day 1.
1 to 2	02-training/SparkBridge-SE-Training-v1.pdf, keep SparkBridge-price-cheatsheet.pdf open	Product, Ignition basics, sizing, licensing, discovery, competitive, partners, demo, objections. Quiz at the end (80% pass).
3	12-site-pages/*.pdf and 03-reference-architecture/	Compare, Architecture, Pricing, Benchmarks, Roadmap, Use cases, Architecture review intake, plus the 40-site reference architecture. Read the live pages too; the PDFs are snapshots.
3	04-partner-program/	Ratified terms and the live partners page. Quote these and nothing else.
4	10-competitive-cheat-sheet.md	Living document; add a row per conversation.
5	09-objection-library.md, 08-discovery-call-and-report.md	The twelve seeded objections and the call script.
6	06-targets/	The 31 firms, Dan's approve column in the Google Sheet, scoring columns to fill.
8	07-outreach/	First email, follow-up, four-touch sequence. Nothing goes out for an unapproved row.
weekly	11-reports/	Friday report, KPI sheet, 30/60/90 review.
as needed	14-articles/	Eight article outlines for touch 3; finish one only when a prospect's pain matches its title.
internal	05-design-partner-offer-INTERNAL.md, 13-product-themes-INTERNAL.md, 15-crm-and-product-calendar.md	End-customer offer (never quoted to an integrator as their discount), the four product themes for sorting feedback, HubSpot rules and the release calendar.

Three rules that override everything else

1. Dan approves every target row before the first touch. Claude researches and drafts; the MBA sends; Dan can take sending back at any time.
2. Prices come from mqtt.greenguard-usa.com/sparkbridge/pricing only. Partner terms come from [/sparkbridge/partners](https://mqtt.greenguard-usa.com/sparkbridge/partners) only. No per-tag numbers, no discounts beyond the program, no roadmap promises.
3. Say the caveat out loud (plan section 6). No "certified", no reference customer, early access and pilot labels as the site shows them, "certificate issuance" not "mTLS" until 3.0.0.

Sources of truth when this bundle and something else disagree

Live site > this bundle. SparkBridge repo docs/gtm/ > this bundle's copies. Google Sheet approve column > CSV approve column.

Files marked INTERNAL

Not for prospects. The design partner offer is an unpublished end-customer offer; the product themes document is internal strategy.

SparkBridge 90-Day Go-to-Market Sprint

MBA Program Owner Playbook, adjusted 2026-09-07

Objective: validate SparkBridge's market position with real Ignition integrators, convert qualified firms into Architecture Partners, and create the pipeline for the first production deployments.

Primary owner: MBA / Go-to-Market Program Lead. Duration: 90 days. Start: the day the MBA has the access checklist in section 25.

Changes from the 2026-09-07 draft

The draft was written without sight of what already exists. These are the adjustments; everything not listed stands as written.

1. Target list exists. 31 firms, contacts re-verified 2026-09-07, 2 to 3 named people per firm, email pattern per firm. Days 6 to 14 become "score and extend", not "build". Outreach can begin day 8.
 2. Partner program terms are ratified and live. Section 16 quotes them exactly. The MBA may quote those terms and nothing else.
 3. Sending. Assumption: the MBA sends outreach from their own greenguard-usa.com mailbox and LinkedIn, only to rows Dan has approved in the target sheet. Dan can revert to sending himself; the rest of the plan does not change.
 4. Success metrics. The "target" tier is the KPI set Dan accepted from the final pre-launch review. Minimum and exceptional tiers rescaled around it.
 5. Verticals. Four priority verticals, chosen from where the target list has depth. Others opportunistic. Re-rank at day 30 on replies.
 6. Training exists with a certification. Section 5 uses it instead of an ad hoc test.
 7. Honest-limits rules added (section 6). They are what makes the answer believed.
 8. Prices added (section 17). The live pricing page is the only source of truth.
 9. CRM conventions added (section 13). HubSpot properties and pipeline are fixed.
 10. Architecture review ownership named: Dan plus Claude. Pre-read before, written review within two business days after.
 11. Product calendar added (section 24). 3.0.0 lands mid-October inside the sprint.
 12. Touch 3 assets named. Nothing else is sent.
 13. Leadership review at day 30, 60 and 90 with Dan; Claude compiles the numbers.
 14. Access checklist added (section 25).
 15. The seven discovery questions from the SE training are the required opening set on every call.
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1. Mission

The MBA's job is not primarily to sell software licenses in 90 days.

The job is to run a disciplined market-validation and partner-acquisition sprint answering:

1. Which integrators have the distributed-estate problem SparkBridge solves?
2. Which verticals have the strongest pain?
3. Which messages generate technical interest?
4. What objections repeatedly appear?
5. Where does SparkBridge fit into real architectures?
6. Which integrators can become long-term channel partners?
7. Can we create 1 to 2 serious production deployment opportunities?

Target outcome:

```
30 approved targets
      v
8 to 12 replies
      v
5 to 8 technical conversations
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3 to 5 architecture reviews
2 to 3 Architecture Partners
1 to 2 production pilots
Production proof

2. Role definition

Title: SparkBridge Go-to-Market and Partner Development Lead.

This is not a cold-calling sales role. It combines market research, partner development, customer discovery, technical sales coordination, messaging validation, competitive intelligence, CRM discipline and go-to-market experimentation.

3. Core principle

Do not sell a module. Sell the conversation:

What changes when your Ignition deployment reaches 40 sites?

Core messaging, in the words the site uses:

- The problem: scaling Ignition creates backend complexity.
- The architecture: Edge moves data. Provider carries it. Host turns it into one system.
- The operating principle: publish once. Add consumers when you need them.
- The differentiation: MQTT is the transport. Operating the estate is the problem.
- The adoption story: nothing gets torn out on day one. Keep the plant. Replace the fragile connections between plants.

Never lead with "we have a cheaper MQTT module". That makes SparkBridge a commodity. Lead with architecture.

4. What success looks like at day 90

Metric	Minimum	Target (accepted KPIs)	Exceptional
Approved targets researched and scored	30	30	50
Personalized first touches sent	20	30	45
Replies	5	8 to 12	16+
Technical conversations	4	5 to 8	12+
Architecture reviews delivered	2	3 to 5	7+
Architecture Partners signed	1	2 to 3	4 to 5
Production pilots started	0	1 to 2	2, one in measured production
Objection database, vertical ranking, repeatable process	yes	yes	plus one written measured deployment

5. Week one: product immersion

Days 1 to 2: SparkBridge SE training

Use 02-training/SparkBridge-SE-Training-v1.pdf (16 modules) with the price cheat sheet open throughout. Order and timings are in the training README inside the PDF. Modules 00a and 00b cover Ignition itself; do not skip them.

Certification, as defined by the training: quiz passed at 80% or better (SparkBridge-SE-quiz.pdf), the demo script run once for Dan or an engineer, one discovery call shadowed.

Oral check before any outreach:

"An integrator has 35 remote Ignition sites, one central application, and wants to add a historian. Explain how SparkBridge changes the architecture."

The answer must contain: sites publish once over Sparkplug to a broker; Host builds one namespace; the historian is a consumer at Host (SparkVault) so nothing changes at the 35 sites; store-and-forward covers the WAN gaps on source timestamps; site 36 is one more Edge.

Also be able to explain SparkBridge in 30 seconds, 2 minutes and 10 minutes.

Day 3: website and sales asset immersion

Read in this order, all in 12-site-pages/ as PDFs and live on mqtt.greenguard-usa.com/sparkbridge: Overview (live), Compare, Architecture (Choose Your Starting Point table), Pricing, Partners (04-partner-program/), Roadmap, Benchmarks. Then the 40-site reference architecture PDF (03-reference-architecture/).

Required assignment, one page: what problem SparkBridge solves; who has it; what it replaces; what it does not replace; why an integrator would care; why they might not; when SparkBridge should not be recommended. Knowing when not to sell is the point.

Day 4: competitive training

Training module 08 and 10-competitive-cheat-sheet.md. The category distinction to hold:

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MQTT modules -> connect Ignition to MQTT
SparkBridge -> architecture for operating distributed Ignition estates
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Keep the cheat sheet as a living document. Every conversation adds a row or corrects one.

Day 5: messaging certification

Be able to answer, unprompted:

- "What is SparkBridge?" A distributed data architecture for Ignition that separates plant data production from the applications and systems that consume it.
- "Isn't this just MQTT?" MQTT is the transport, and the right one. SparkBridge is what remains after the transport: one namespace, consumers that scale on the shared stream, replay on source timestamps into history and warehouse, enrollment, fleet, policy.
- "Why not existing MQTT modules?" If the requirement is connecting Ignition to MQTT, they are a good answer and we say so. SparkBridge matters when many sites, independent consumers, fleet operations and governance have to grow without revisiting plants.
- "Do I have to replace my Ignition system?" No. Nothing gets torn out on day one.

6. Honest limits the MBA carries into every conversation

These are the caveats the site publishes. Say them before the prospect finds them.

- No published production reference yet. Every number was measured on named hardware with reproduction steps in the reference architecture. The partner program exists to produce the first case study.
- Never say "certified". The Sparkplug Compatible listing is in progress; we ran the TCK harness.
- Early access and pilot labels: SparkRecord (early access 0.1.0), SparkValidate (pilot), GitOps / Enterprise Governance (pilot). Buy the labeled products for what they do today.
- Enrollment in 2.9.0 issues certificates but the broker does not yet require the client certificate at connect. Say "certificate issuance", never "mTLS enforced". 3.0.0 closes this.
- Mixed-vendor Edge and Host pairing is by specification; we run the exact pairing on the bench before anyone buys.
- Modules are signed with our own code-signing certificate, not an Inductive Automation one; the gateway asks to trust it once.
- Team size: say it honestly if asked. Development is free with no time limit, so a prospect can prove everything on their own estate before spending.

Pattern: answer, then the caveat, then stop.

7. Days 6 to 14: score and extend the target list

The list exists: 06-targets/integrator-targets.csv, 31 firms, refreshed 2026-09-07 with 2 to 3 named people per firm, email pattern and confidence per firm, and Dan's approve column. Google Sheet "SparkBridge Partner Targets" in Drive mirrors it.

Days 6 to 10: score all 31 with the rubric in section 8 (columns score_*, score, tier are in the CSV, blank). Verify any row marked unverified or pattern before it goes out. Rows Dan has not approved are not contacted.

Days 11 to 14: extend to 50 using the same columns. Look for: significant Ignition practice, multi-site customer deployments, distributed SCADA, MQTT or Sparkplug experience, enterprise industrial customers, custom application work, long managed-support relationships. The research notes in 06-targets/integrator-targets-research-notes.md show the evidence standard.

Priority verticals, from where the list has depth and where the design partner offer already points:

1. Distributed manufacturing (DMC, Flexware, Corso, ATS, RoviSys, Tamaki)
2. Water, wastewater, utilities and renewables (SAGE, Perceptive, Brown, NeoMatrix, Ulteig, Vertech)
3. Data centers (Hanley, Thermo, BW Design Group, NFM, BIJC, Vertech)
4. Oil and gas (Streamline, CSE ICON, NFM)

Opportunistic: pharma and validated (SL Controls/NNIT, Agidens, 4IR), food and beverage, mining, logistics (no logistics-specific Ignition integrators surfaced in search).

8. Score every target

Criteria	Score
Large Ignition practice	0 to 5
Multi-site deployments	0 to 5
MQTT / Sparkplug experience	0 to 5
Target vertical fit	0 to 5
Enterprise customer evidence	0 to 5
Likely distributed architecture pain	0 to 5
Public technical sophistication	0 to 5

Maximum 35. Tier A 25 to 35: direct personalized outreach. Tier B 18 to 24: secondary outreach. Tier C under 18: monitor.

Most important question per firm: what evidence suggests this company has a distributed systems problem? Do not contact someone merely because they use Ignition.

9. Days 8 to 30: outreach sprint

Philosophy: no brochures, no feature dumps, no long introductions, no immediate demo ask. Lead with an architectural question. The objective is curiosity and a technical conversation.

Primary message (full text and follow-up in 07-outreach/):

What changes when your Ignition deployment reaches 40 sites?

We built SparkBridge around that question. Sites publish once, so applications, historians, warehouses and other consumers can be added centrally without revisiting every plant.

We are looking for a small group of experienced Ignition integrators to pressure-test the architecture against real distributed estates. If you have a topology worth discussing, we would like to compare approaches.

Rules: one named person per firm at a time, the firm-specific hook from the CSV in sentence one, plain text, no tracking pixels, UTM only on the two links. Dan approves the row before the first touch. Every send logged in HubSpot the same day.

10. Outreach sequence

Touch	When	Content
1 Architecture question	Day 0	Short personalized email or LinkedIn message. Goal: a reply.
2 Specific observation	Day 3 to 5	Reference their distributed project, industry, MQTT experience or multi-site work.
3 One technical asset	Day 8 to 10	Exactly one: Compare page, 40-site reference architecture PDF, Benchmarks page, or one finished article from 14-articles/. Not everything.
4 Partner program	Only after relevance is established	"We are forming a small group of Architecture Partners and thought your team may be a strong fit." Link to /sparkbridge/partners.

A "no thanks" closes the deal as lost and cancels the remaining touches. Silence after touch 4 goes to a 60-day hold.

11. The first discovery call

Integrator speaks 70%, SparkBridge 30%. Discovery, not demonstration.

Required opening set (SE training module 07), every call, answers into HubSpot before the call ends:

1. How many Ignition gateways, and which are redundant pairs? (shape)
2. How do sites reach the center today: Gateway Network, MQTT, VPN, nothing? (shape)
3. What is the busiest central box doing right now, CPU and memory? (the pain number)
4. What happens to history and alarms when a site's WAN drops?
5. How many consumers of site data exist, and who is asking for the next one?
6. Who builds a new site gateway and how long does it take?
7. Is there a broker already, and who owns it?

Then, as time allows: growth (at what site count does architecture become difficult; have customers added historians or warehouses later; have you revisited sites to add a consumer; what becomes painful after 20 sites), competitive (what MQTT stack, why, what you wish it did differently), validation (does SparkBridge solve a problem you actually meet; where does it not fit; what would stop you recommending it). Full script in 08-discovery-call-and-report.md.

Hand off to the architecture review for any estate over ten sites, any HA question, any broker the customer already runs, any 8.3-only feature.

12. Every conversation produces a structured report

Within 24 hours, in HubSpot on the contact and deal (template and field mapping in 08-discovery-call-and-report.md): account, architecture (site count, current shape, MQTT stack, central applications, existing consumers), pain (primary, secondary, business consequence, current workaround), SparkBridge fit (strong / moderate / weak / unknown), objections in the prospect's exact words, next step.

The intelligence database is worth more than the lead count.

13. CRM conventions (fixed)

- HubSpot contact properties: outreach_campaign = sparkbridge-fap-2026-09, outreach_segment = vertical, lifecycle_lead.
- One contact per named person, one company per firm.
- Deals: the "SparkBridge Partners" pipeline (create once in the HubSpot UI; the MCP integration cannot create pipelines) with stages contacted, call booked, review delivered, partner signed, pilot live, production measured. Until it exists, deals go in the default Sales Pipeline tagged "SparkBridge FAP". Deal created at first reply.
- Every deal carries: named owner, next action, next meeting date, architecture status, technical blocker, commercial blocker, target deployment date. No deal sits labeled "interested".

14. Architecture review program

This is the primary conversion mechanism. Offer: draw your estate before you buy anything. Intake is the mailto on mqtt.greenguard-usa.com/sparkbridge/architecture-review (site count, topology, tags per site, update rates, WAN type, history design, applications, pain points, Ignition versions).

SparkBridge delivers: current-state diagram, SparkBridge reference architecture, failure-domain analysis, migration path, rough sizing with the list price for that shape, known limitations and open questions.

Ownership:

- MBA: scheduling, intake, documentation, follow-up, CRM, opportunity progression.
- Dan plus Claude: Claude prepares a one-page pre-read from the intake (architecture number from the Architecture page, list price for the shape, honest limits, partner terms) at least one day before the call; Dan leads the technical conversation; Claude drafts the written review within two business days; Dan approves it before it goes out.

The written review is the deliverable that qualifies a firm as an Architecture Partner.

15. Days 31 to 60: convert conversations into reviews

By day 30 know: which verticals respond, which messages work, which objections repeat, which firms have genuine fit. Double down on evidence. If manufacturing produced five conversations and data centers none, move the effort. This is a market-discovery sprint, not a fixed campaign.

Build the objection library (09-objection-library.md): product, commercial, market, trust. Track objection, frequency, severity, response, product action. Review weekly with Dan.

16. Days 45 to 75: Architecture Partner conversion

Terms (ratified 2026-09-05, live at /sparkbridge/partners). Quote these and nothing else:

- Architecture Partner: invitation only, cap 10. Entry condition: one completed architecture review of an estate of 10 or more sites. Benefits: development licenses (free for everyone, stated so partners know), NFR production-grade key per module for the partner's demo estate renewed yearly, direct engineering access, 90-day deal registration, 20% off list, named escalation with a one-business-day reply on production issues, co-selling on request, one recorded training session per year, quarterly roadmap call.
- Founding Partner: the first five Architecture Partners to take a customer estate into measured production. 40% off list on that first production estate, NFR keys that never expire, shared priority support channel, a vote on public roadmap order, joint case study.
- No discount stacking. Volume pricing is Dan's call, not a discount the MBA offers.
- Design partners are end customers, not integrators: a separate unpublished 50% offer (05-design-partner-offer-INTERNAL.md) used through the integrator. Never quoted to an integrator as their own discount.

Ideal first partner: a technically strong Ignition integrator with a real distributed customer opportunity and enough engineering depth to challenge SparkBridge. One of those beats ten passive partners.

17. Prices the MBA may quote

Only from mqtt.greenguard-usa.com/sparkbridge/pricing; the cheat sheet PDF is a copy verified 2026-09-06. Per gateway, one-time, development free with no time limit, support optional at 20% of list per year, nothing per tag, device or client.

Shape, 40 sites	Required total	Support per year
Customer's own broker: 39 Edge + 1 Host	\$40,300	\$8,060
Provider HA pair + history: 39 Edge + 2 Provider + 2 Host + SparkVault	\$49,780	\$9,956
Complete operating architecture: above + FleetOps + Sentinel + GitOps	\$69,765	\$13,953

Site 41 is one more Edge, \$995. Never invent a per-tag number; explain the model once and show the 40-site example.

18. Production opportunity qualification

Score: real customer, multi-site, active project, architecture pain, executive support, technical fit, timeline under 12 months, each 0 to 5. Prioritize 25 or more.

19. Days 61 to 90: pipeline conversion

Conversation -> Architecture review -> Technical validation -> Partner discussion -> Customer opportunity -> Pilot ,

For each pilot agree the measurement plan up front (sites, tags per site, outage replay, consumer count) so the first write-up can use the same method as the Benchmarks page.

20. Weekly operating rhythm

- Monday: pipeline review (new targets, outreach, responses, meetings, reviews, opportunities) and product intelligence review (objections, competitive findings, feature requests, website confusion).
- Tuesday to Thursday: research, outreach, discovery calls, follow-up, review coordination.
- Friday: one-page weekly report to Dan (11-reports/weekly-report-template.md).

21. Leadership review at day 30, 60 and 90

Present evidence, not activity. Claude compiles the numbers from HubSpot beforehand.

Bad: "I sent 80 emails." Good: "Eight integrators responded. Six immediately understood the 40-site problem. Three use remote tag providers. The repeated objection is new-vendor risk, not pricing."

Dan asks: what did we learn, what changed, what should we stop, what should we double down on, is this a product issue or a messaging issue.

22. KPI dashboard

Leading: target quality, personalized outreach, response rate, discovery calls, architecture reviews. Validation: problem acknowledged, fit, partner interest, technical follow-up requested. Commercial: qualified opportunity, pilot, deal registration, partner agreement. Strategic: most promising vertical, most common architecture, most common objection, most common competitor, feature gaps. Template in 11-reports/.

23. What the MBA should not do

- Become a generic SDR: mass automated email, LinkedIn spam, feature dumps, "just checking in", volume as the metric.
- Become product support: document questions, coordinate escalation.
- Promise roadmap items. Never "we can add that". Instead: "I will document that requirement and have the product team evaluate where it fits." Any "when does X ship" goes to the public roadmap page, which has no dates on purpose.
- Discount. New-vendor risk is answered with architecture access, engineering support, NFR licensing, pilot support and transparent evidence, not price.

24. Product calendar during the sprint

- Today: SparkBridge 2.9.0 released 2026-09-06 (both Ignition lines). SparkRecord 0.1.0 early access. SparkS7 0.2.1.
- Mid-October (target week of 2026-10-12): SparkBridge 3.0.0, a breaking release that partners will prove: manual-ack transport, client-certificate enforcement, two-round enrollment. Partners starting in the sprint install 3.0.0, not 2.9.0.
- After: SparkRecord 0.2.0, SparkInject 0.2.0, SparkNotify 0.3.0.
- Externally: no dates. Internally: do not sell 2.9.0 enrollment claims that 3.0.0 corrects (section 6).

25. Access checklist (before day 1)

- greenguard-usa.com mailbox and calendar; a booking link for review calls.
- HubSpot seat with edit rights; the "SparkBridge Partners" pipeline created.
- Edit access to the Google Sheet "SparkBridge Partner Targets" and the Drive folder Claude uploads / Industrial & Drivers.
- LinkedIn from a personal account with the company listed.
- This bundle, and the SparkBridge repo docs/gtm/ if the MBA will edit the source files.
- A standing 30-minute Monday slot with Dan and the Friday report address.

26. Required deliverables at day 90

1. Market validation report: who has the problem, which verticals care, at what scale pain appears, the strongest buying trigger.
2. Target market ranking: vertical, integrator type, customer architecture, deployment size.
3. Objection report: frequency, severity, recommended response, product implications.
4. Competitive intelligence report: Cirrus Link, Gateway Network, HiveMQ, EMQX, Cedalo, custom architectures, as heard in the field.
5. Partner pipeline: active, qualified, in review, production opportunity, closed / no fit.
6. Product feedback report: critical gap, strategic opportunity, nice-to-have, off-architecture distraction (apply the four themes in 13-product-themes-INTERNAL.md).
7. Recommended GTM strategy, from evidence, not assumptions.

27. Timeline

Days	Phase	Output
1 to 7	Learn: training, certification, competitive, messaging	Can explain and position SparkBridge independently
6 to 14	Map: score 31, extend to 50, tier	Ranked, approved target list
8 to 45	Discover: personalized outreach, conversations	Validated pain points, repeatable messaging
31 to 60	Validate: architecture reviews, technical validation	Real-world architecture fit
45 to 75	Partner: partner discussions, production opportunities	2 to 3 serious partner relationships
61 to 90	Convert: alignment, pilot	1 to 2 credible production opportunities

Final instruction

Your job is not to prove SparkBridge is amazing. Your job is to discover where SparkBridge is genuinely valuable. Be curious. Ask technical questions. Listen more than you pitch. Document objections exactly. Look for patterns. Do not defend the product when someone says it does not fit; that answer is valuable.

After 90 days, success means the company knows who to sell to, what problem to lead with, what objections to solve, which partners matter, and where the first production deployments will come from.

SparkBridge cheat sheet

Every product on one page. List price, US dollars, per gateway, bought once and kept. Development and testing free with no time limit. Support optional at 20% of list per year. Nothing is charged per tag, per device or per connected client. This is the only price table in the course; every module links here.

Thesis in one line

Publish once at the site, add consumers at the Host, run the whole estate from one chair. Sites and consumers scale on separate lines. Nothing at a site changes when a consumer is added.

The core

Product	What it does	Runs where	Included with it	Price
SparkBridge Edge	Publishes site tags once on change over Sparkplug B, buffers to disk through a link drop, replays in order on source timestamps	Each site gateway	SparkLink (metered-link budget)	\$995
SparkBridge Host	Primary host. Collects every site into one namespace, builds tags from each site's birth, runs the Host modules	Host gateway, never at a site	SparkCalc, SparkID	\$1,495
SparkBridge Provider	Sparkplug-aware managed broker and policy layer: routing, per-site identities, deny-by-default topic permissions. Optional: any conformant broker the customer already runs works instead	Broker gateway or pair		\$1,995

Modules on Host

Each runs only on a SparkBridge Host gateway, on its own license.

Module	What it does	Status	Price
SparkCalc	Rates, totals and efficiencies calculated once on live data	Released	Included with Host
SparkID	Who can read and who can send commands, generated from intent; read never includes control	Released	Included with Host
Passage	Serves live plant data over OPC-UA to tools the customer already owns. Read only	Released	\$1,995
SparkInject	Production data landed in the warehouse exactly once, source timestamps kept. Per connector family: Snowflake first; Databricks and BigQuery follow	Released (Snowflake)	\$2,995 per family
SparkVault	Tag historian on the plant's own clock in an open database (Timescale); backfills trends after an outage	Released	\$3,995
SparkRecord	Tamper-evident record historian for structured events: Sparkplug records, MQTT JSON events, SparkFlow events, exactly once, SHA-256 audit chain. All destinations included	Early access 0.1.0	\$7,995
FleetOps	Every gateway in one view: versions, drift, certificates, backups	Released	\$4,995
Sentinel	Scored security and health assessment of the plant data network, with evidence and fix per finding. Observes only	Released	\$4,995
Sentinel compliance pack	Same findings as an auditor-ready binder. Add-on to Sentinel	Released	\$2,495
GitOps / Enterprise Governance	Configuration as code for the estate: signed bundles, review, rollback, audit trail	Pilot	\$9,995
SparkFlow	Every point streamed to the whole company past the tag-count ceiling (NATS backbone). Its own license	Released	\$995

Device drivers

Licensed per gateway. Each works with or without SparkBridge. Both Ignition lines (8.1 and 8.3).

Driver	Equipment	Price
SparkLogix	Allen-Bradley ControlLogix and CompactLogix over EtherNet/IP, live tag browse, no translator box	\$695
SparkSNMP	UPSs, switches, PDUs, servers over SNMP. One license covers both Ignition lines	\$695
IEC 60870-5-104	Utility and substation equipment, reads only	\$695
DNP3	DNP3 outstations with event data and source timestamps, reads only	\$695
TI 505 / CTI 2500	Older Siemens 505 and CTI 2500 racks, direct	\$695
SparkADS	Beckhoff ADS/AMS with live TwinCAT symbol browse	\$695
SparkS7	Siemens S7-1200/1500 symbolic addressing	\$695
Spark61850	IEC 61850 MMS client with report control blocks	\$695
SparkBACnet	BACnet/IP browse, read, write	\$695

Ignition enhancements

Licensed per gateway. Each works with or without SparkBridge.

Product	What it does	Status	Price
SparkInflux	InfluxDB history storage and history provider	Released	\$1,495
SparkValidate	Alarm rationalization (ISA-18.2) and validation document set (GAMP 5, Part 11, 62443). Rockwell first	Pilot	\$2,495 gateway module, \$995 Validate Desk CLI
SparkNotify	Alarm notification to Teams, Slack, WhatsApp, Telegram, Twilio SMS/voice, Discord, PagerDuty, acknowledgment tracked back	Released 0.2.0	\$695
SparkGantt	Perspective Gantt component, smooth at 500+ tasks	Released	\$695

Placement rule

Layer	What installs there
Site (EDGE)	Edge, SparkLink, any protocol driver the site needs
Broker	Provider, or the customer's conformant broker (HiveMQ, Mosquitto 2.1 tested)
Center (HOST)	Host, SparkCalc, SparkID, Passage, SparkInject, SparkVault, SparkRecord, FleetOps, Sentinel, GitOps, SparkFlow
Workstation	SparkID CLI, Sentinel scanner, Validate Desk CLI
Any gateway	Drivers, SparkInflux, SparkNotify, SparkGantt, SparkValidate module

Forty sites, three ways (list)

Shape	Required	Required total	Support / yr
Customer's own broker	39 Edge + 1 Host	\$40,300	\$8,060
Provider HA pair + history	39 Edge + 2 Provider + 2 Host + SparkVault	\$49,780	\$9,956
Complete operating architecture	Above, less Vault as required, plus Vault, FleetOps, Sentinel, GitOps	\$69,765	\$13,953

Site 41 is one more Edge, \$995. No number above moves when the estate gets busier.

Licensing terms, briefly

Unit	One license per production gateway, per product
Term	Perpetual
Development	Free, no time limit. Paid modules run on Ignition's two-hour module trial until a key file names them; Reset Trial restarts the clock
Support	Optional, 20% of list per year: updates and direct engineering support. Software keeps running if it lapses
Activation	One signed key file per gateway in the Ignition data directory, picked up within 30 seconds, no restart
Moving	Follows the role, not the hardware
Partners	Architecture Partners 20% off list with 90-day deal registration. First five Founding Partners 40% on their first production estate. No stacking

SparkBridge cheat sheet

Every product on one page. List price, US dollars, per gateway, bought once and kept. Development and testing free with no time limit. Support optional at 20% of list per year. Nothing is charged per tag, per device or per connected client. This is the only price table in the course; every module links here.

Thesis in one line

Publish once at the site, add consumers at the Host, run the whole estate from one chair. Sites and consumers scale on separate lines. Nothing at a site changes when a consumer is added.

The core

Product	What it does	Runs where	Included with it	Price
SparkBridge Edge	Publishes site tags once on change over Sparkplug B, buffers to disk through a link drop, replays in order on source timestamps	Each site gateway	SparkLink (metered-link budget)	\$995
SparkBridge Host	Primary host. Collects every site into one namespace, builds tags from each site's birth, runs the Host modules	Host gateway, never at a site	SparkCalc, SparkID	\$1,495
SparkBridge Provider	Sparkplug-aware managed broker and policy layer: routing, per-site identities, deny-by-default topic permissions. Optional: any conformant broker the customer already runs works instead	Broker gateway or pair		\$1,995

Modules on Host

Each runs only on a SparkBridge Host gateway, on its own license.

Module	What it does	Status	Price
SparkCalc	Rates, totals and efficiencies calculated once on live data	Released	Included with Host
SparkID	Who can read and who can send commands, generated from intent; read never includes control	Released	Included with Host
Passage	Serves live plant data over OPC-UA to tools the customer already owns. Read only	Released	\$1,995
SparkInject	Production data landed in the warehouse exactly once, source timestamps kept. Per connector family: Snowflake first; Databricks and BigQuery follow	Released (Snowflake)	\$2,995 per family
SparkVault	Tag historian on the plant's own clock in an open database (Timescale); backfills trends after an outage	Released	\$3,995
SparkRecord	Tamper-evident record historian for structured events: Sparkplug records, MQTT JSON events, SparkFlow events, exactly once, SHA-256 audit chain. All destinations included	Early access 0.1.0	\$7,995
FleetOps	Every gateway in one view: versions, drift, certificates, backups	Released	\$4,995
Sentinel	Scored security and health assessment of the plant data network, with evidence and fix per finding. Observes only	Released	\$4,995
Sentinel compliance pack	Same findings as an auditor-ready binder. Add-on to Sentinel	Released	\$2,495
GitOps / Enterprise Governance	Configuration as code for the estate: signed bundles, review, rollback, audit trail	Pilot	\$9,995
SparkFlow	Every point streamed to the whole company past the tag-count ceiling (NATS backbone). Its own license	Released	\$995

Device drivers

Licensed per gateway. Each works with or without SparkBridge. Both Ignition lines (8.1 and 8.3).

Driver	Equipment	Price
SparkLogix	Allen-Bradley ControlLogix and CompactLogix over EtherNet/IP, live tag browse, no translator box	\$695
SparkSNMP	UPSs, switches, PDUs, servers over SNMP. One license covers both Ignition lines	\$695
IEC 60870-5-104	Utility and substation equipment, reads only	\$695
DNP3	DNP3 outstations with event data and source timestamps, reads only	\$695
TI 505 / CTI 2500	Older Siemens 505 and CTI 2500 racks, direct	\$695
SparkADS	Beckhoff ADS/AMS with live TwinCAT symbol browse	\$695
SparkS7	Siemens S7-1200/1500 symbolic addressing	\$695
Spark61850	IEC 61850 MMS client with report control blocks	\$695
SparkBACnet	BACnet/IP browse, read, write	\$695

Ignition enhancements

Licensed per gateway. Each works with or without SparkBridge.

Product	What it does	Status	Price
SparkInflux	InfluxDB history storage and history provider	Released	\$1,495
SparkValidate	Alarm rationalization (ISA-18.2) and validation document set (GAMP 5, Part 11, 62443). Rockwell first	Pilot	\$2,495 gateway module, \$995 Validate Desk CLI
SparkNotify	Alarm notification to Teams, Slack, WhatsApp, Telegram, Twilio SMS/voice, Discord, PagerDuty, acknowledgment tracked back	Released 0.2.0	\$695
SparkGantt	Perspective Gantt component, smooth at 500+ tasks	Released	\$695

Placement rule

Layer	What installs there
Site (EDGE)	Edge, SparkLink, any protocol driver the site needs
Broker	Provider, or the customer's conformant broker (HiveMQ, Mosquitto 2.1 tested)
Center (HOST)	Host, SparkCalc, SparkID, Passage, SparkInject, SparkVault, SparkRecord, FleetOps, Sentinel, GitOps, SparkFlow
Workstation	SparkID CLI, Sentinel scanner, Validate Desk CLI
Any gateway	Drivers, SparkInflux, SparkNotify, SparkGantt, SparkValidate module

Forty sites, three ways (list)

Shape	Required	Required total	Support / yr
Customer's own broker	39 Edge + 1 Host	\$40,300	\$8,060
Provider HA pair + history	39 Edge + 2 Provider + 2 Host + SparkVault	\$49,780	\$9,956
Complete operating architecture	Above, less Vault as required, plus Vault, FleetOps, Sentinel, GitOps	\$69,765	\$13,953

Site 41 is one more Edge, \$995. No number above moves when the estate gets busier.

Licensing terms, briefly

Unit	One license per production gateway, per product
Term	Perpetual
Development	Free, no time limit. Paid modules run on Ignition's two-hour module trial until a key file names them; Reset Trial restarts the clock
Support	Optional, 20% of list per year: updates and direct engineering support. Software keeps running if it lapses
Activation	One signed key file per gateway in the Ignition data directory, picked up within 30 seconds, no restart
Moving	Follows the role, not the hardware
Partners	Architecture Partners 20% off list with 90-day deal registration. First five Founding Partners 40% on their first production estate. No stacking

00 Orientation

What SparkBridge is, who buys it, and the frame every product fits into. Thirty minutes.

Learning objectives

- State the thesis in one sentence without notes.
- Name the three core roles and the four themes.
- Place any product in the line on the product map.

What SparkBridge is

SparkBridge is a line of modules for Inductive Automation's Ignition platform. GreenGuard USA built it for companies that run Ignition at many sites and want one system instead of forty. The customer already owns Ignition. We sell the modules that run on their gateways.

Two core products carry the line. **Edge** moves plant data reliably off each site. **Host** turns many sites into one system. **Provider** is the broker between them, sold when the customer does not already run a conformant MQTT broker. Everything else is an optional module that consumes the data Host holds, or a standalone driver or tool that works with or without SparkBridge.

The thesis

Publish once at the site, add consumers at the Host, and run the whole estate from one chair. Sites and consumers scale on separate lines. Nothing at a site changes when a consumer is added.

Every conversation comes back to this. When a prospect asks why the architecture matters, the answer is the thesis and then the number that proves it (module 05).

Six reasons the home page gives

1. Survive outages: Edge buffers to disk and replays in order.
2. Preserve source timestamps: a reading keeps the time the equipment gave it through buffer, replay, history and warehouse.
3. Scale horizontally: Host consumers share the stream with MQTT 5 shared subscriptions.
4. Eliminate polling: report by exception over Sparkplug B.
5. One namespace: Host builds the tag tree from each site's birth, no mapping file.

6. No per-tag pricing: per gateway, per product, bought once.

The four themes

Every module and every roadmap row lands in exactly one theme. Use them to organize a pitch and to answer "what else do you have" without reciting the catalog.

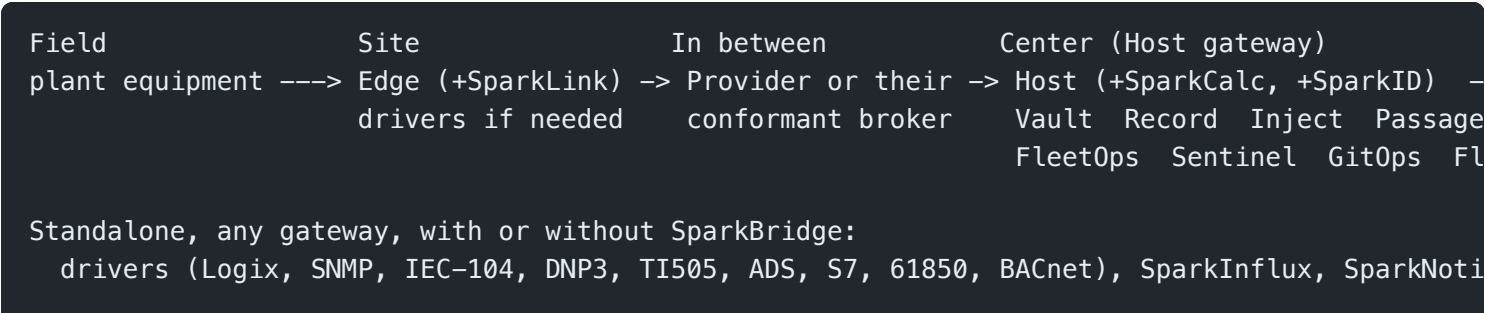
Theme	Promise	Products
Operate	Run a 40 to 200 site estate from one team without visiting sites	FleetOps, zero-touch enrollment, capacity intelligence, GitOps / Enterprise Governance
Trust	Prove where a number came from and that nobody changed it on the way	SparkID, Sentinel, data lineage (trace), data contracts, SparkValidate
Move Data	Land the same published data in every system the company runs, exactly or at-least once, with source timestamps intact	SparkVault, SparkRecord, SparkInject, Passage, SparkFlow, SparkNotify
Scale	Grow past what one Ignition server or one broker can hold, without a new integration project	Host pools and device sharding, Provider HA, federated estates

Who buys

Buyer	What they feel	Where to start
Controls or SCADA lead at a multi-site manufacturer	The hub gateway is the overloaded box; every new dashboard makes the sites work harder	Host + Edge, then SparkVault
Water, wastewater, utilities	Cellular and radio links drop; history has gaps; data bill climbs	Edge with SparkLink, then Vault
Oil and gas	Records must survive an audit; EFM and batch events	SparkRecord, SparkVault
Data centers and facilities	Network and building gear are not in the same picture as the process	SparkSNMP, SparkBACnet drivers
Validated industries	Weeks writing a validation package by hand	SparkValidate
System integrator with an Ignition practice	Wants a repeatable estate design and margin	Partner program (module 09)

Almost nobody buys everything on day one. Most plants buy one product for one problem, then add the next when it shows up.

The product map



Say this

- "Modules on the Ignition gateways you already license."
- "Each site publishes once. Every consumer subscribes at the center."
- "Per gateway, per product, bought once. Nothing per tag."
- "GreenGuard USA built it and measured it. The numbers are on the technology page."

Never say this

- "Replaces Ignition." It runs on Ignition.
- "Suite", "package", "bundle". There is no package. Each item has its own price.
- "Not included", "sold separately", "never bundled". Show the price line instead.
- Any hype word: seamless, robust, leverage, unlock, streamline.
- A reference customer we do not have. See module 11.

Check

1. Which two products are the core, and which is the optional third?
2. A prospect asks for "the security stuff." Which theme is that and which three products belong to it?
3. Where does SparkVault install: site, broker or Host?

► Answers

00a Ignition basics

For an SE who has sold automation but never installed Ignition. Sixty minutes plus a twenty minute lab. Inductive Automation (IA) publishes the platform; SparkBridge runs on it. Verify current edition names and prices against [inductiveautomation.com](https://www.inductiveautomation.com) before quoting anything about IA.

Learning objectives

- Explain gateway, Designer, client, tag, tag provider, UDT, driver, module in one sentence each.
- Describe how a module is installed and how the two-hour module trial works.
- Explain Sparkplug B to a controls engineer in under a minute.
- Avoid the "Edge" and "Host" naming collisions.

What Ignition is

Ignition is a SCADA, HMI and IIoT platform sold by Inductive Automation. One install is a **gateway**: a Java service on a server or industrial PC, configured through a web page (default port 8088, path /web/config). Engineers build screens and logic in the **Designer**, a desktop application that talks to the gateway. Operators use **Perspective** (browser and mobile) or **Vision** (Java desktop) clients. The gateway talks to PLCs through **drivers**, stores tag history in a SQL database, runs alarms and scripts, and connects to other gateways over the **Gateway Network**.

IA licenses per gateway, not per tag, client or device. That is why "unlimited tags, unlimited clients" is the phrase customers know. SparkBridge follows the same shape: per gateway, per product.

Editions

Edition	Purpose	Note for SparkBridge
Ignition	"Build any industrial application and deploy it anywhere." Per server, perpetual, unlimited tags, users and devices	Where Host and the Host modules run. Edge runs here at a site too
Ignition Edge	"Embed Ignition on edge-of-network devices." Per server, perpetual, unlimited tags and devices, at most 2 concurrent client views, 35 days internal data storage, no database connectivity, bulk discounts	IA's product named Edge, not ours. Ask "Ignition Edge edition, or the SparkBridge Edge module?" when the word comes up. Check Edge edition module support with engineering before promising SparkBridge Edge on it
Ignition Cloud Edition	"Add the power of the cloud to your Ignition system." Hosted on AWS or Azure marketplaces, pay as you go, usage-based	A Host can live here in principle; treat as a question for engineering
Maker	Free, non-commercial, home use	Do not sell against it or on it

IA support plans (annual, percent of retail at purchase): BasicCare 16%, TotalCare 20%, PriorityCare 24%. A major-version upgrade without a support plan is 65% of current retail. Redundancy needs both nodes licensed with matching modules. Source: [inductiveautomation.com/pricing/edition](https://www.inductiveautomation.com/pricing/edition), read 2026-09-06. Our 20% optional support sits next to TotalCare on the same sheet.

Vocabulary an SE must get right

Term	Meaning	Why an SE cares
Tag	A named value with quality and timestamp, in a folder tree	Sparkplug metrics map to tags; Host builds tags from births
Tag provider	A named tree of tags on a gateway; a remote tag provider exposes another gateway's tree over the Gateway Network	Remote providers are the fan-in load at the hub (module 00b)
UDT	User Defined Type: a template for a device with members and parameters	Edge admits a UDT whole or not at all; Sparkplug Templates carry it
Device connection / driver	The gateway-side connection to a PLC through a protocol driver module	Our nine drivers are this kind of module
OPC UA	Ignition's internal server exposes driver data; external clients can read it	Passage serves the Host namespace over OPC-UA to external tools
Tag history / historian	Tags marked for history are written to a SQL database through a history provider; 8.3 adds a Core Historian	SparkVault and SparkInflux are history providers
Alarm pipeline	Alarm events routed through notification pipelines	SparkNotify adds channels; SparkValidate rationalizes the alarms
Project	A set of screens, scripts and resources on a gateway	Not licensed; unlimited
Module	A signed .modl file installed at Config > Modules; adds drivers, components or services	Every Spark product is one
Gateway Network (GAN)	Gateway-to-gateway connection for remote providers, history and EAM	The traditional way to build an estate, and what we replace at scale
Store and forward	Gateway buffering of history writes when the database is unreachable	Different from Edge's disk buffer, which sits at the publish path
Redundancy	Master and backup gateway pair	Host active plus warm standby is our equivalent at the center

8.1 and 8.3

8.1 is the long-term support line most estates run. 8.3 changed how the gateway stores configuration (resources as files, a system.config API), added a secrets management system, event streams as project

resources, a Core Historian, gateway deployment modes, bundled JDBC drivers, and removed the serial modules into the core platform. On Siemens: symbolic S7 addressing that worked in the 8.3 early access was removed from the 8.3.0 release over last-minute licensing issues, and returned as the Siemens Enhanced driver, a separately purchased module not active by default. Module 04 has the detail and the SparkS7 pitch. Every Spark product ships on both lines, signed, and the 8.3 build is proven on a live 8.3 gateway before release. The customer's line decides which download they take; the price is the same.

How a module is installed and licensed

1. Upload the signed .modl at Config > Modules. The gateway shows the certificate and asks to trust it.
2. Without a key file a paid Spark module runs on Ignition's two-hour module trial. Reset Trial on the gateway restarts the clock. The data path pauses when the clock runs out. SparkLink, SparkCalc and SparkID never pause.
3. The customer drops one signed key file per gateway into the Ignition data directory. The module picks it up within 30 seconds, no restart, and the status page changes from Trial to Licensed.

Development and testing is therefore free with no time limit: reset the trial as often as needed. A license is bought when a gateway goes into production.

MQTT and Sparkplug B in sixty seconds

MQTT is a publish and subscribe protocol. Publishers send messages to topics on a **broker**; subscribers receive what they subscribed to. Sparkplug B is a specification on top of MQTT that fixes the topic namespace and payload so industrial systems interoperate. The pieces:

- **Edge node** publishes an **NBIRTH** with every metric's name, type and metadata, then **NDATA** with only what changed, by alias. Devices under it use **DBIRTH** and **DDATA**.
- The **primary host** is the one application that owns the namespace. Edge nodes watch its STATE topic, buffer while it is away and rebirth when it returns.
- Each metric carries its own timestamp, so source time survives the trip.
- Store and forward at the edge is the mechanism that makes an outage a delay instead of a loss.

IA's own MQTT modules (Cirrus Link MQTT Transmission, Engine and Distributor) are the reference customers compare us to. SparkBridge Edge is a Sparkplug edge node, Host is a primary host, Provider is a Sparkplug-aware broker. All three interoperate with conformant third parties by specification, and the site says honestly which pairings have run on the bench.

Naming collisions

- **Edge.** IA sells "Ignition Edge" (an edition). We sell "SparkBridge Edge" (a module). Always say the full name once per conversation.
- **Host.** In Sparkplug the "primary host application" is a role. Our product named Host is that role plus the modules. IA's Cirrus "MQTT Engine" plays the role on their side.
- **Provider.** In Ignition a "tag provider" or "history provider" is a data source. Our Provider is a broker. Say "SparkBridge Provider, the broker" the first time.

Lab, twenty minutes

Use the GreenGuard lab gateway (Ignition 8.1.54 at localhost:8088 on the Mac, or the 8.3 gateway on gg-wsl). Read only; change nothing.

1. Open Config > Modules. Find a SparkBridge module, note its version and the certificate it was signed with.
2. Open the module's status page (Status > SparkBridge or the Config > SparkBridge tab). Note whether it says Trial or Licensed and when the trial resets.
3. Open Designer, browse the default tag provider, find a UDT instance and open its definition.
4. On the Host page, open a connected edge node and read its last birth: metric count, timestamp, device list.
5. Write down the one sentence you would say to a prospect about each of the four things you saw.

Say this

- "Modules on the Ignition gateways you already license."
- "Same per-gateway model as Ignition itself."
- "Free to develop and test, for as long as you need."

Never say this

- "Replaces Ignition" or "SCADA in the cloud."
- "Works on any edition." Confirm the edition first.
- Anything about IA pricing from memory. Point to IA's site.

Check

1. What does the gateway do when a paid module's two-hour trial runs out, and what fixes it without a restart?
2. What is the difference between Ignition store and forward and the SparkBridge Edge buffer?
3. Which Sparkplug message carries metric names, and which carries only changes?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 00 Orientation](#) ·

00b Ignition at scale

How Ignition estates are actually built at 1, 5, 20 and 40 sites, and where each shape starts to hurt. This module is the bridge between knowing Ignition and knowing why SparkBridge exists. Forty-five minutes.

Learning objectives

- Draw the four deployment shapes and name the boxes someone has to license, patch and watch.
- Say what breaks in each shape at 5, 20 and 40 sites.
- Map a described estate to a shape and to the SparkBridge required set.

The four shapes

1. Single gateway

PLCs ----> one gateway (drivers, tags, history, screens, alarms)

One plant, one server. Nothing to fix. Most customers start here and it is fine at one site.

2. Hub with remote tag providers over the Gateway Network

```
site gateway 1 --GAN--+
site gateway 2 --GAN--+--> hub gateway (remote tag providers, central screens, history)
site gateway N --GAN--+
```

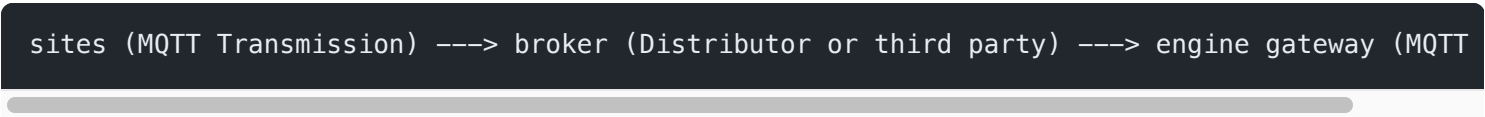
Each site keeps its gateway. The hub subscribes to each site's tag tree as a remote provider. Every remote provider is a live subscription load on both ends. Every consumer at the hub (screens, history, a report) is more subscription load back onto the sites.

3. Front-end and back-end split with redundancy

```
sites --GAN--> backend A (history)    backend B (applications)    backend C (integrations)
                  |                    |                    |
                redundant pair        redundant pair        redundant pair
                  \_____ front-end gateways (clients) ____/
```

When the hub saturates, the estate is split into backends by application family, each holding its own copy of the estate, each redundant. Site count multiplies backend load; consumer count multiplies site load. History is written per backend and reconciled by hand.

4. Module-based MQTT



Sites publish over Sparkplug and buffer through outages. Polling and most fan-in are gone. What is left to custom engineering is everything after the transport: consumer scale-out (a second consumer usually means a second engine), provisioning (broker URL, credentials and certificates typed at each site), configuration drift, fleet health, security policy written as ACL files by hand.

What breaks, by site count

Sites	Shape 2 or 3 (traditional)	Shape 4 (module MQTT)	SparkBridge
5	One hub, remote providers. Fine	Fine	5 Edge, 1 Provider or their broker, 1 Host. Also fine
20	Hub CPU and memory climb with every remote provider; history moves to a second backend	Transport fine; a second consumer needs a second engine; site setup is manual	Same three roles. Host memory grows about 300 bytes per wire metric per consumer
40	The hub is the overloaded box. A second hub splits the estate and every consumer now needs both	Fleet operations are the problem: which sites are offline, buffering, drifted, expiring	Provider HA pair, Host active plus warm standby (takeover measured at 244 ms across 25 nodes). One namespace
50 and beyond	A backend per consumer family; integration projects	Custom tooling for everything after the broker	Add Host consumers on the shared stream; sites never change

The real case on the use-cases page: one central server carrying 40 sites at 50 to 75% CPU with 14 GB of memory, unable to ride out a network outage, getting worse with every dashboard. Per-site scale in the migration plan that produced the reference architecture: Irving 8,700 published leaves, Mesa 22,400, Dickinson 8,500, fleet average about 13,000 per site.

The six problems that arrive with site count

Problem	Traditional	Module MQTT	SparkBridge
Gateway Network fan-in	One hub carries every site connection and saturates first	Removed: sites publish to a broker	Removed: sites publish to Provider; Host consumers share the stream
Backend sprawl	A backend per application family	One engine; a second consumer usually means a second engine	One Host pool; consumers are modules on Host or subscribers of the same stream
History duplication	Site history plus central history, reconciled by hand	Depends where history is enabled	History is one consumer. SparkVault writes once, centrally, on source timestamps
WAN outages	History during the gap depends on per-site store-and-forward configuration	Store and forward at the site	Disk buffer at 2,697 messages/s durable; 400,000 of 400,000 replayed in order with the historical flag so trends backfill
Timestamps	Arrival time unless every hop is configured to keep source time	Source timestamps by Sparkplug	Source timestamps through buffer, replay, history and warehouse
Provisioning and fleet	Manual gateway build, GAN approval, provider setup per site; status pages one at a time	Credentials typed at each site; broker client list	One-time enrollment code over a signed channel; FleetOps versions, drift, certificates, backups in one view

Where a module-MQTT cell reads the same as ours, that is the Sparkplug standard doing its job for both. The difference is what remains for the customer's team to build afterwards. Say that out loud; it is on the compare page and it is honest.

Two per-gateway bills on one sheet

IA charges per gateway, unlimited tags and clients. SparkBridge charges per gateway, per product, nothing per tag. An SE can put both on one sheet for a 40-site estate: 40 Ignition site gateways the customer already owns or will buy from IA, plus 39 SparkBridge Edge, 1 or 2 Host, and Provider if they have no broker. Neither line moves when tag count grows. That single fact removes the objection "what happens to the price when we add tags."

Discovery questions that reveal the shape

1. How many gateways run today, and which are redundant pairs?
2. Where is history written, and how many databases hold a copy of the same tag?
3. How do sites reach the center: Gateway Network, MQTT, VPN, nothing yet?
4. What happens to history and alarms during a WAN outage at a site?
5. Who builds a new site gateway, and how long does it take from box to data at the center?
6. How many consumers of site data exist: screens, historians, reports, warehouse feeds, analytics?
7. What is the CPU and memory on the busiest central box right now?

Answers to 1, 3 and 6 place the estate in a shape. Answers to 4 and 7 give the number that opens the SparkBridge conversation.

Say this

- "Site count multiplies hub load. Consumer count multiplies site load. We cut both lines."
- "Sparkplug removes polling for everyone. What we add is what comes after the transport."

Never say this

- "Ignition doesn't scale." It does; specific shapes hurt at specific sizes.
- "MQTT modules are wrong." They are the right transport, and we interoperate with them.

Check

1. A prospect runs 18 sites on a hub with remote tag providers and a separate history backend. Which shape, and what is the first symptom they will describe?
2. Same prospect. Write the SparkBridge required set and its list total.
3. A prospect already runs MQTT Transmission at every site into HiveMQ. What is left for them to build, and which two of our products address it first?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 00a Ignition basics](#) ·

01 Edge, Provider, Host

The three roles. What each does, what runs where, and the facts an SE can state without checking. Forty-five minutes. Prices on the [cheat sheet](#).

Learning objectives

- Explain each role in two sentences.
- Answer "do we need Provider" correctly.
- State the five measured numbers that belong to the core.

SparkBridge Edge, \$995 per site gateway

Edge is the module installed at each site. It watches a folder of tags the customer chooses, sends each value once when it changes with the timestamp the equipment gave it, and keeps recording to a disk buffer when the link drops. On reconnect the backlog arrives in order and lands on the time it actually happened.

Fact	Detail
Where it runs	The site gateway, nowhere else. Standalone; nothing else installs at the site
Tag structure	Read from the site's own folder including UDTs, announced to Host in the birth. No mapping file. Folders at a chosen depth become devices
Tag budget	250,000 per gateway. A UDT is admitted whole or skipped whole. Subscriptions install in 10,000-path chunks
Edge nodes	Up to eight per gateway, each with its own connection, tag set and buffer
Buffer	50 MB default per edge node; size cellular and satellite sites at 500 MB. Measured 2,697 messages/s durable
Write-back	Off by default; allowed only for listed devices and points
Broker	Provider, or any conformant MQTT broker (HiveMQ, Mosquitto 2.1 tested), TLS and per-site credentials
Enrollment	Zero-touch: one-time code over a signed TLS channel; credentials written owner-only, never typed at the site. 2.9.0 adds CSR/CA and mTLS
Ignition	8.1.19 or newer, and the 8.3 line
Included	SparkLink: measures what a metered site spends, attributes it to the tag, eases off gradually when spend runs ahead, never touches points marked safety-relevant

SparkLink story worth telling: in a real three-day capture one vibration tag with no change threshold set was taking 50.7% of a site's monthly data budget, 47 times the next largest consumer.

SparkBridge Provider, \$1,995 per gateway

Provider is SparkBridge's Sparkplug-aware broker and policy layer: routing, per-site identities, deny-by-default topic permissions, no client count. It runs as an embedded Moquette or as a HiveMQ CE companion service. Two Provider nodes make the HA pair in the reference architecture (4 cores, 8 GB each).

Do we need Provider?

No, if the customer already operates a conformant MQTT broker they are happy to keep. Edge and Host connect through it. Yes, if they have no broker, want SparkID's generated permissions enforced without writing ACL files, or want us to own the HA pair. Say it positively: "Provider, or the broker you already run."

SparkBridge Host, \$1,495 per Host gateway

Host is the primary host. It collects what every Edge reports, learns each site's tag structure from its birth and projects one live tag tree that mirrors the plant. Add a device at a site and it appears at Host. When a site drops, Host marks that site's points bad rather than holding the last number, and asks for a fresh birth on return. A sequence gap triggers the same check.

Fact	Detail
Where it runs	Host gateway, never at a site. Single Host, or active plus warm standby, or a pool
Speaks	Sparkplug B and 3.0 over MQTT 3.1.1 and MQTT 5, as the primary host
Consumers	MQTT 5 shared subscriptions: four Host consumers deliver about 20x one. BIRTH, DEATH and STATE are subscribed non-shared so alias tables stay correct
Memory	About 300 bytes per wire metric per consumer; 500,000 wire metrics in under 4 GiB
CPU	Projector shards equal core count. Six cores sustained 339,527 updates/s end to end on the Linux reference rig
Takeover	Standby promotes in 244 ms across 25 nodes; edges never rebirth
Capacity route	Host reports its own ceiling from benchmark evidence and the shard count it runs (/capacity), not a fixed number
Fleet route	/fleet shows buffering and headroom per edge; Prometheus metrics
Recommended box	8 cores, 16 GB; 32 GB if SparkVault shares it
Included	SparkCalc and SparkID
Ignition	8.1.19 or newer, and the 8.3 line

Host alone, no modules added, replaces the work of stitching forty sites together by hand. The modules are how the customer opens that data, puts numbers on it, secures it, runs it and keeps it. Each needs a Host gateway underneath it.

Cross-cutting facts, since 2.7.0

- Tiered route authorization: every status and control route on every Spark product uses the same read, write and admin token policy. Loopback trust is off unless turned on. A forwarded header cannot impersonate the local host.
- One signed key file per gateway, checked every thirty seconds, applied with no restart. An invalid or tampered key fails closed.
- Data lineage: six trace hops shipped in 2.8.0; wire hops arrive in 3.0.0. Data contracts shipped in 2.8.0.

The five numbers

Number	Claim	Source page
2,697 messages/s	Durable disk buffering at Edge	technology, compare
400,000 of 400,000	Replayed in order after an outage with the historical flag	compare, benchmarks
244 ms	Host standby takeover across 25 nodes	compare, architecture
500,000 under 4 GiB	Wire metrics held in Host memory	compare, technology
About 20x	Four shared-subscription consumers versus one	compare, benchmarks

Cite the page. Every figure was measured on named hardware and the reproduction steps are in the reference architecture document.

Say this

- "Edge at each site. Host once. Provider, or the broker you already run, in between."
- "A dropped link is routine. Edge was built around that fact."
- "Host marks a site bad instead of stale. Nothing quietly drifts."

Never say this

- "Unlimited tags at Edge." The budget is 250,000 per gateway.
- "Provider is required." It is not, if they have a conformant broker.
- "Mixed-vendor broker pairing is certified." It is by specification; say which pairings ran on the bench (HiveMQ, Mosquitto 2.1).

Check

1. A site has 300,000 candidate tags. What does Edge do?
2. The customer runs HiveMQ Enterprise. Which of the three core products do they buy?
3. What happens at Host when a site's WAN drops, and what happens when it returns?

► [Answers](#)

[Previous: 00b Ignition at scale](#) ·

02 Data consumers

The Move Data theme: the modules that take what Host holds and land it somewhere useful. Each runs on the Host gateway on its own license. Forty-five minutes. Prices on the [cheat sheet](#).

Learning objectives

- Match each consumer to the customer sentence that triggers it.
- Explain SparkVault versus SparkRecord in one breath.
- State each module's status label exactly as the site does.

SparkVault, \$3,995, released

Trigger: "A trend runs flat through an outage and then jumps. Nobody can defend the report built on it."

SparkVault is the family's tag historian: time series, trend readback and replay, in an open database (TimescaleDB) served back as an Ignition history provider. History is one consumer of the backbone; adding it changes nothing at the sites. Every reading lands on the time it actually happened, so the backlog fills in where it belongs after an outage. Any past day can be replayed to test a dashboard or walk through an incident.

Proof: a gateway disconnected for four hours and eleven minutes while production continued; zero values lost, every reading on its original time. Host-side only; nothing installs at the edge, which is why there is no "site package."

SparkRecord, \$7,995, early access 0.1.0

Trigger: "Our batch and event records have to survive a rerun, a migration or an auditor."

SparkRecord is the structured record lane: Sparkplug B records (Cirrus Transmission Record objects, DataSet and Template metrics), JSON events on arbitrary MQTT topics, and SparkFlow events over NATS (the NATS lane ships in 0.2.0). Schema is discovered from the payload and migrated additively only. Delivery is exactly once: source and receipt timestamps kept separately, producer identity on every row, deduplication by content hash, ledger written in the same transaction. Every record id is a SHA-256 over canonical JSON chained to the one before; the module never issues UPDATE or DELETE. Destinations: PostgreSQL, SQL Server, TimescaleDB, Snowflake, S3-compatible object storage, all included.

Vault or Record

SparkVault stores continuous telemetry: timestamp, metric, value, quality. SparkRecord lands structured events: a batch, a transaction, an alarm record, with a tamper-evident chain. Most estates need the first; regulated ones usually need both.

Say "early access 0.1.0" every time. Do not promise the NATS lane or a date beyond what the roadmap page says.

SparkInject, \$2,995 per connector family, released for Snowflake

Trigger: "Analysts reconcile duplicate rows in the warehouse by hand and never trust the totals." Or: "We pay a per-row subscription to see our own production data."

SparkInject streams plant data into the customer's own Snowflake account, exactly once, with source timestamps kept and an offset token per row. The customer owns the account, the storage and the bill; we are not in the middle. A restart or a dropped connection cannot cause a gap or a duplicate. Proof: 8,375 readings queued through a four-hour outage and landed with nothing missing and nothing counted twice.

Connector families: Snowflake first. Databricks and BigQuery follow. Kafka, ADX and S3 sinks were live-proven in the 2.7.0 bundle; quote them only after checking the current products page for what is listed as available. One family per license.

Passage, \$1,995, released

Trigger: "We have a historian or SCADA package that will never be replaced and it still needs the same live data."

Passage serves the Host namespace over OPC-UA so the tools the customer already owns connect the way they already do. No custom connector per system. Read only: it cannot write or command, which is a selling point to the security team.

SparkCalc, included with Host, released

Trigger: "The same efficiency number is hand-built in a PLC, a dashboard and a spreadsheet, and gives three different answers."

Rates, totals and efficiencies calculated once, in one place, on live data, then used everywhere. Runs at a site or centrally.

SparkFlow, \$995, released, its own license

Trigger: "Screens and analytics for the whole company need every point, live, and the tag system is the ceiling."

SparkFlow streams every point Host decodes to a NATS backbone, past what the tag system carries, lossless and replayable. Built for the ultra-large estate. Priced on its own, on any shape.

SparkNotify, \$695, released 0.2.0, any gateway

Listed here because it moves alarms rather than data: one alarm to Teams, Slack, WhatsApp, Telegram, Twilio SMS or voice, Discord and PagerDuty from one configuration, acknowledgment tracked back into Ignition. Secrets encrypted at rest on 8.3. Works with or without SparkBridge. Email channel is removed in 3.0.0; do not sell it.

The consumer add story

Each of these is a subscription at Host or a module on it. No change at the forty sites. When a prospect asks about a future analytics or AI project: subscribe to the shared stream, read SparkVault, or take the Snowflake copy. A second corporate historian: Passage already serves OPC-UA, add a client. That sentence is the thesis applied.

Say this

- "History is a consumer, not another integration project."
- "Exactly once, on the plant's own clock, in a database you own."
- "SparkRecord is early access. Here is what 0.1.0 does today."

Never say this

- "SparkVault replaces the site historian." It is central; a site may keep local history for local screens.
- "Inject supports every warehouse." Per family; Snowflake today.
- "Passage can write." It cannot.

Check

1. A pharma customer wants batch records with an audit trail and also long-term trends. What do they buy?

2. A customer uses OSIsoft PI and wants it fed from the same data. Which module, and what is the security answer?
3. What does a customer buy to land data in both Snowflake and Databricks?

► **Answers**

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 01 Edge, Provider, Host](#) ·

03 Operate and trust

The Operate and Trust themes: running the estate from one chair, and proving where a number came from. Forty-five minutes. Prices on the [cheat sheet](#).

Learning objectives

- Explain FleetOps, Sentinel, SparkID, GitOps and SparkValidate by the question each answers.
- State which are included, which are pilot, and what is roadmap.

FleetOps, \$4,995, released, Host

Question it answers: "Forty gateways in the field and nobody knows which version, which drifted, or which certificate expires next month."

One view of every gateway: versions, drift, certificates, backups. A fix is made once and applied everywhere instead of forty visits. Worth it from roughly ten sites up. True fleet management (config templates, drift remediation) is a 3.1 program; say "drift is shown per gateway" and stop there. 2.9.0 registers enrolled sites into FleetOps automatically.

Sentinel, \$4,995, released, Host; compliance pack \$2,495

Question it answers: "Nobody can say what the plant network is exposing, and an auditor is asking for evidence."

A scored security and health assessment of the plant data network, ranked by risk, with the evidence behind each finding and the fix for it. It only observes and changes nothing. The compliance pack turns the same findings into an auditor-ready binder. The scanner also runs from a workstation.

SparkID, included with Host, released

Question it answers: "One compromised site could write another site's data, or send a command to equipment it has no business touching."

SparkID generates deny-by-default topic permissions from intent: who can read, who can send commands, scoped to one site on purpose. A read permission never includes control. Safe credential rotation. Provider enforces the rules; Sentinel scores the result. Emits HiveMQ credentials; a Mosquitto dynsec emitter is on the internal list, not promised.

Proof point from the reference architecture: a leaked site credential can publish only its own group and node, never a command. Proven live.

GitOps / Enterprise Governance, \$9,995, pilot, Host

Question it answers: "Who changed what, when, and can we roll it back?"

Configuration as code for the whole estate: signed bundles distributed fleet-wide, review and rollback, an audit trail. First form shipped in 2.8.0. Git adapter and two-key approvals come after 3.0 and are partner-driven. Always say "pilot." Sell it to the customer who already asks for change control, not as a default line.

Host features in these themes (no separate price)

Feature	What it does	Status
Zero-touch enrollment	New site bootstraps identity with a one-time code over a signed TLS channel	2.8.0 slice 1; 2.9.0 CSR/CA, mTLS, HiveMQ credential emit
Capacity intelligence	Host reports its own ceiling from measured evidence and shard count; sparkline; degraded flag	2.8.0, extended 2.9.0
Data lineage (trace)	Six hops from equipment to consumer	2.8.0; wire hops in 3.0.0
Data contracts	Declared shape per stream; strict mode projects bad quality on violation	2.8.0
Tiered route authorization	Read, write, admin tokens on every route, every product; admin fails closed	2.7.0, hardened 2.9.0

SparkValidate, \$2,495 gateway module or \$995 Validate Desk CLI, pilot, any gateway

Question it answers: "We spend weeks hand-typing a validation package that restates what the control system already knows."

SparkValidate reads existing exports (Rockwell .ACD and L5X first, Ignition alarm configuration and journal) and builds one traceable model. It generates 13 documents and protocols with a requirements traceability matrix: URS to FRS to DS to risk assessment to OQ, every ID pointing back to a rung, a tag path or a view. Risk class drives test depth. Alarm rationalization per ISA-18.2 comes from what the alarm journal shows the plant actually doing. Frameworks: GAMP 5, 21 CFR Part 11, IEC 62443. The CLI works offline; the gateway module executes tests against the running system and seals the evidence.

Pilot release, Rockwell systems first. It does not require SparkBridge.

How the three security products fit together

SparkID generates the rules --> Provider enforces them --> Sentinel scores the result and

That sentence sells all three and answers the security architect's question of who does what.

Say this

- "A read permission never includes control."
- "Sentinel observes and changes nothing. The fix is yours to apply."
- "GitOps is a pilot. Here is what ships today and what comes after 3.0."
- "Certificate issuance at enrollment shipped in 2.9.0; reconciliation across the fleet is roadmap."

Never say this

- "FleetOps pushes config to every site." It shows drift; templates and remediation are 3.1.
- "Sentinel is a firewall" or "hardens the network." It assesses.
- "SparkValidate covers Siemens." Rockwell first; say what the site says.

Check

1. A customer asks for change control with approvals. Which product, and what caveat?
2. Which of these are included with Host: SparkID, Sentinel, FleetOps?
3. What does Sentinel change on the network when it runs?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 02 Data consumers](#) ·

04 Drivers and tools

Nine device drivers and four Ignition enhancements. Each is licensed per gateway and works with or without SparkBridge. Thirty minutes. Prices on the [cheat sheet](#).

Learning objectives

- Name the driver for any equipment a prospect mentions and the box it removes.
- Explain the per-gateway rule for standalone items without negative phrasing.

The rule

Drivers and tools are bought on their own, per gateway, at their own price. A site connecting one kind of equipment needs that driver and usually not the others. They are absent from every "included with" list on the site, and the price line says the rest. Do not add "sold separately" or "never bundled"; show the price.

Device drivers, \$695 each per gateway

Driver	Equipment	What it removes	Proof
SparkLogix	Allen-Bradley ControlLogix, CompactLogix over EtherNet/IP	The translator box; direct connect with live tag browse	433 tests, lab smoke on real hardware, field QA in progress. v0.2.0
SparkSNMP	UPSs, switches, PDUs, servers	A separate network monitoring tool for the plant; the network shows up next to the process	One license covers both Ignition lines
IEC 60870-5-104	Utility and substation equipment	The protocol converter and its failure points. Reads only, no control path	Released
DNP3	DNP3 outstations, event data, source timestamps	Another vendor's converter. Reads only	Released
TI 505 / CTI 2500	Older Siemens 505 and CTI 2500 racks	The translator box, its license and its patching	Running on the GreenGuard lab gateway against a CTI 2500
SparkADS	Beckhoff TwinCAT over ADS/AMS, live symbol browse	An OPC UA bridge on the PLC side	Verified against a real TwinCAT runtime on both lines; certificate pinning on by default. First commercial ADS driver for Ignition
SparkS7	Siemens S7-1200/1500 symbolic addressing	Raw DB offsets; plants keep their tag names	Classic S7comm and S7comm-plus (symbolic, optimized DBs, TLS, certificate pinning) verified on S7-1500 in PLCSIM Advanced V7 with a real TIA Portal V20 project through both the 8.1 and 8.3 gateways: browse, reads, subscriptions, writes. Real S7-1500 hardware still pending; say so if asked
Spark61850	Substation IEDs over IEC 61850 MMS, report control blocks	Flattening through a DNP3 gateway; logical-node model intact	Verified byte for byte against libiec61850, buffered reports on a live gateway
SparkBACnet	Building automation over BACnet/IP: browse, read, write,	A separate BMS integration; building equipment in the	Verified against the reference bacnet-stack

	segmented Complex-ACK	same tag tree as the process	
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On SparkS7 and Ignition 8.3: what actually happened

Symbolic S7 addressing worked in the Ignition 8.3 early access builds. It was removed from the 8.3.0 release. IA staff said on the forum (September 2025) that "some licensing issues are preventing us from doing that" and that the issues "came up pretty last minute, and were largely out of Inductive Automation's hands." Symbolic addressing then reappeared as the Siemens Enhanced driver: a separate module, purchased and licensed on its own, not activated by default at install, built on a third-party library. Users complained about the surprise cost for a feature they expected in the platform, about a fixed-delay polling bug that roughly doubled update latency (IA said fixed in 8.3.2 or 8.3.3), about larger tag groups blocking faster ones, a 250-device connection limit, a GLIBC 2.34 requirement from 8.3.3 that rules out Ubuntu 20.04 and Debian 10/11, and S7-1200 G2 support arriving only in 8.3.3. Symbolic access needs S7-1200/1500 and does not cover S7-300/400.

The pitch that stands: "The free 8.3 Siemens driver has no symbolic addressing. SparkS7 gives symbolic S7-1200/1500 addressing on both 8.1 and 8.3 from one \$695 per-gateway license, verified on an S7-1500 in PLCSIM Advanced with a real TIA project, on both gateway lines. Plants keep their tag names instead of raw DB offsets." If asked about real hardware: PLCSIM Advanced runs the same CPU firmware; a site pilot on the customer's own S7-1500 is the offer, not a claim. Do not characterize IA's driver quality; state the timeline and the price line, and let the customer compare. If a prospect already bought Siemens Enhanced, SparkS7 still matters for 8.1 gateways and for one driver across both lines.

Ignition enhancements

Product	Buyer sentence	Status	Price
SparkInflux	"We standardized on InfluxDB and Ignition history has to go there." History storage and history provider, 8.1 and 8.3	Released	\$1,495
SparkValidate	"Weeks of validation paperwork." Module 03 covers it	Pilot	\$2,495 / \$995 CLI
SparkNotify	"The on-call person is in Teams, not in a Vision client." Seven channels, acknowledgment tracked back	Released 0.2.0	\$695
SparkGantt	"Production schedule on a Perspective screen that does not bog down." Virtualized, smooth at 500+ tasks, drag limits enforced on the gateway	Released	\$695

Where a driver enters a SparkBridge deal

A driver sale is often the first purchase. The site gets a direct connection today; SparkBridge Edge comes later when the estate grows. Drivers pass the "benefits multiple customers" test and rarely the architecture ones, so they stay off the SparkBridge roadmap table. Keep them on a separate quote line from the estate.

Say this

- "Licensed per gateway. Works with or without SparkBridge."
- "The translator box comes out, along with its license and its patching."
- "Verified against the reference stack; the report is in the release bundle."

Never say this

- "Included with Edge" or "included with Host" for any driver or tool.
- "Never bundled", "sold separately." Show the price.
- "Writes to the substation." IEC-104 and DNP3 are reads only.

Check

1. A data center has UPSs on SNMP and chillers on BACnet, on one gateway. Quote it.
2. A utility asks whether the DNP3 driver can send controls. Answer.
3. A customer with SparkBridge Host asks whether SparkNotify is included. Answer in the positive form.

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 03 Operate and trust](#) ·

05 Sizing an estate

How to turn a site count into a required set, a server spec and a list price in ten minutes. Forty-five minutes.
Prices on the [cheat sheet](#); full method in docs/REFERENCE-ARCHITECTURE-40.md.

Learning objectives

- Produce the required set and total for any site count.
- Recommend the Host and Provider hardware from the sizing rules.
- Explain HA in one paragraph.

The arithmetic

Required infrastructure first, optional consumers second, so the required number stands on its own. One site hosts the center, so an estate of N sites usually needs N minus 1 Edge licenses plus the Host.

Shape	Lines	Required	Support, optional, per year
40 sites on the customer's broker	39 x Edge \$38,805; 1 x Host \$1,495; broker \$0 from us	\$40,300	\$8,060
40 sites with Provider HA and history	39 x Edge \$38,805; 2 x Provider \$3,990; 2 x Host \$2,990; 1 x SparkVault \$3,995	\$49,780	\$9,956
Complete operating architecture	Required subtotal \$45,785 (39 Edge, 2 Provider, 2 Host) plus SparkVault \$3,995, FleetOps \$4,995, Sentinel \$4,995, GitOps \$9,995 pilot	\$69,765	\$13,953

Site 41 is one more Edge, \$995, an enrollment code, and nothing else. Optional consumers on any shape, none touching the sites: FleetOps \$4,995, Sentinel \$4,995, SparkInject \$2,995 per family, SparkRecord \$7,995, Passage \$1,995. Every figure is list before volume pricing; tell us the site count and we run the numbers.

Generic formula

```
required  = (N - 1) x 995  + hosts x 1,495  + providers x 1,995
hosts      = 1 (single) or 2 (active + warm standby)
providers  = 0 (their broker) or 2 (HA pair); 1 only for a lab
optional  = sum of Host modules chosen
```

support = 0.20 x (required + optional) per year, optional
partner = list x 0.80 (Architecture) or x 0.60 first production estate (Founding); never bo

Sizing rules

Layer	Rule
Edge per site	One license per site gateway. Up to 8 edge nodes. 250,000-tag budget
Wire metrics fleet-wide	Sites x published leaves after metadata folding. 40 x about 7,000 = about 280,000 in the reference estate
Host memory	About 300 bytes per wire metric per consumer. 500,000 wire metrics under 4 GiB. 280k x 4 consumers is about 340 MB of session state plus the managed provider
Host CPU	Projector shards equal core count. Six cores sustained 339,527 updates/s; four logical cores bend at 55,464
Host box	8 cores, 16 GB; 32 GB if SparkVault shares it
Provider	Two HiveMQ CE nodes, 4 cores, 8 GB each. No client-count license
Buffer	50 MB default per edge node is hours to days at typical change rates; 500 MB for cellular and satellite

What happens when things fail

Failure	Behaviour	Measured
WAN drop at a site	Edge buffers to disk; Host marks the site's points bad	2,697 msg/s durable
WAN returns	Reconnect within a 10 s cap, rebirth, ordered replay with the historical flag; SparkVault backfills	400,000 of 400,000; 40,035 upd/s replay on the six-core rig
Host primary crash	Standby promotes after the debounce; edges never rebirth	244 ms across 25 nodes
One Provider node down	Edges and Host walk to the other server	About 0.6 s server walk; MultipleBroker TCK pass
Broker outage entirely	Every site buffers; nothing lost up to buffer size	Store-and-forward suite
A site's credential leaks	It can publish only its own group and node, never a command	Provider ACL, proven live

Worked example: 12 sites, water district

Cellular links, no broker today, wants history that survives outages and a view of every gateway.

Line	Qty	Each	Total
Edge	11	\$995	\$10,945
Provider (single; HA pair when they ask)	1	\$1,995	\$1,995
Host	1	\$1,495	\$1,495
Required			\$14,435
SparkVault	1	\$3,995	\$3,995
FleetOps	1	\$4,995	\$4,995
Total, list			\$23,425
Support, optional, per year			\$4,685

Buffer at 500 MB per site because the links are cellular. SparkLink comes with Edge and answers the data-bill question. Host box: 8 cores, 16 GB, 32 GB since Vault shares it.

Say this

- "Required number first, on its own line. Consumers second."
- "None of these numbers move when the estate gets busier."
- "Not sure of the shape? We size it from your topology in the free architecture review."

Never say this

- A per-tag or per-client figure of any kind.
- "Package" or "suite" totals. Build every quote from lines.
- A discount you have not been given. Volume pricing is Dan's call.

Check

1. 25 sites, they run HiveMQ, want Host HA and SparkVault. Required set and total.
2. Same estate adds Sentinel and the compliance pack. New total and support.
3. What is the Host memory estimate for 25 sites at 13,000 leaves each with 3 consumers?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 04 Drivers and tools](#) ·

06 Licensing

The terms, the mechanics, and what to say when support lapses. Thirty minutes. Prices on the [cheat sheet](#).

Learning objectives

- State the seven terms without notes.
- Walk a customer from trial to licensed without a restart.
- Answer the support-lapse and hardware-move questions.

The terms

Unit	One license per production gateway, per product
Term	Perpetual. Bought once and kept. Not a subscription
Development	Free, no time limit, nothing held back. Development and test gateways are not counted
Support	Optional, 20% of list per year. Covers updates and direct engineering support. Twelve months from purchase, renewal re-issues the key with a new date
Metering	None. Tags, devices and connected clients are never counted
Activation	One signed key file per gateway, applied with no restart
Moving	Follows the role, not the hardware. Move it when you rebuild a server
Volume	Every price is list. Site count goes to Dan for volume pricing
Partners	Architecture Partners 20% off list with deal registration; the first five Founding Partners 40% on their first production estate. Module 09

How the mechanics work

1. **Install.** The customer uploads the signed .modl at Config > Modules. Every Spark product ships signed with the SparkBridge Code Signing certificate on both Ignition lines.
2. **Trial.** Without a key, a paid module runs on Ignition's two-hour module trial. Reset Trial on the gateway restarts it as often as needed. When the clock runs out the data path pauses. SparkLink, SparkCalc and SparkID never pause.
3. **Purchase.** Stripe one-time payment on the site (Buy buttons), or invoice. No subscriptions, ever.
4. **Key.** We issue one signed key file per gateway listing every entitlement bought for it, emailed from admin@greenguard-usa.com. The customer drops it in the Ignition data directory as sparkbridge-

license.key. The module rechecks every 30 seconds, the status page changes from Trial to "Licensed to ...", no restart. CLI tools take `--license <file>`.

5. **Integrity.** A tampered or invalid key fails closed; the product pauses exactly as an expired trial does. Keys are normally issued for any gateway name; a key can be bound to one gateway name on request.
6. **Renewal.** Support renewal re-issues the key with a new support-until date. Nothing changes if it is not renewed.

Questions customers ask

What happens if support lapses?

The software keeps running exactly as before. They stop receiving new versions and support until they renew.

Can we add modules later?

Yes. Each module goes on a Host gateway they already own, on its own license, when the need shows up.

Is development really free?

Yes. Reset Trial as often as needed; a key is bought when a gateway goes into production.

We rebuild the Host on new hardware.

The license follows the role. Move the key file.

Redundant pair: one license or two?

Two Host licenses for active plus warm standby; the pricing page examples show it that way. Two Provider licenses for the HA pair.

We have 8.1 at sites and 8.3 at the center.

Same price, separate downloads per line. One key per gateway either way.

Do you count Ignition Edge edition gateways?

A production gateway is a production gateway. Confirm edition support with engineering before quoting Edge on an Ignition Edge edition install.

Internal: who does what

- Keys are issued by Dan from the signing key on his Mac after payment lands; SEs never issue keys.
- A copy of every issued key is kept in Drive under GreenGuard/SparkBridge/licenses/<customer>.
- Trial-expiry proofs and key-swap proofs were run live on the lab gateway for 2.7.0; the record is in the release bundle if a customer's security team asks.

Say this

- "Per gateway, per product, bought once, kept for good."
- "One key file, thirty seconds, no restart."
- "Nothing stops running if support lapses."

Never say this

- "Subscription", "annual license", "per seat."
- "We can meter it if you prefer." We cannot and will not.
- "Perpetual updates." Updates come with support.

Check

1. A customer edits the key file to add an entitlement. What happens?
2. How long between dropping a key file and the module reporting Licensed?
3. A prospect wants a monthly price. Answer.

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

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07 Discovery

Qualifying questions by vertical, and the routing table from a customer sentence to the product and price. Forty-five minutes. Prices on the [cheat sheet](#).

Learning objectives

- Run a twenty-minute discovery call that places the estate in a shape and surfaces the first purchase.
- Route any customer sentence to one product line.
- Know when to hand off to the architecture review.

The seven questions, every call

1. How many Ignition gateways, and which are redundant pairs? (shape)
2. How do sites reach the center today: Gateway Network, MQTT, VPN, nothing? (shape)
3. What is the busiest central box doing right now, CPU and memory? (the pain number)
4. What happens to history and alarms when a site's WAN drops? (Edge, Vault)
5. How many consumers of site data exist, and who is asking for the next one? (Host, Passage, Inject)
6. Who builds a new site gateway and how long does it take? (enrollment, FleetOps)
7. Is there a broker already, and who owns it? (Provider or theirs)

Then one vertical question from the table below. Write the answers into HubSpot on the contact before the call ends.

By vertical

Vertical	Ask	Likely first purchase
Multi-site manufacturing	"Which plant is the hub, and what breaks when it goes down?"	Host + Edge, then SparkVault
Water, wastewater, utilities	"What links reach the lift stations, and what does a month of data cost?"	Edge with SparkLink; Vault when the trend gaps come up; IEC-104 or DNP3 driver at substations
Oil and gas	"Who audits your EFM and batch records, and how do you prove a row never changed?"	SparkRecord early access, SparkVault
Pharma, food, validated	"How many weeks does the validation package take, and who types it?"	SparkValidate; SparkRecord for the audit chain
Data centers, facilities	"Is the UPS and the chiller in the same screen as the process?"	SparkSNMP, SparkBACnet on the site gateway
Substations, transmission	"Do the IEDs speak 61850 or are they flattened through a DNP3 gateway?"	Spark61850, IEC-104, DNP3
Beckhoff or Siemens plants	"Are you on raw DB offsets or symbolic names, and is an OPC UA bridge in the path?"	SparkS7, SparkADS
Integrator	"How many estates over ten sites are you scoping this year?"	Partner program, module 09

Routing table

If they say	What changes	What they buy
A remote site on cellular or satellite; link drops; data bill climbs	Report by exception, disk buffer, per-tag spend list	Edge \$995 with SparkLink. Standalone
Forty sites on one central server that is already stretched thin	Sites hold their own data in an outage; the center stops working harder per dashboard	Host \$1,495 plus Edge \$995 per site
A trend runs flat through an outage and then jumps	Every reading keeps its real time; backlog lands where it belongs	SparkVault \$3,995
Analysts reconcile duplicate rows in the warehouse	Exactly once into their own warehouse	SparkInject \$2,995 per family
A historian or SCADA package that will never be replaced	Existing tools connect the way they already do	Passage \$1,995, read only
Three different answers for the same efficiency number	Calculated once, used everywhere	SparkCalc, included with Host
Nobody can say what the network exposes; auditor asking	Scored report with evidence and fix per finding	Sentinel \$4,995; compliance pack \$2,495
One compromised site could write another site's data	Read never includes control; access scoped per site	SparkID, included with Host; Provider \$1,995 if no broker
Forty gateways and nobody knows versions or certificate dates	One view; fix once, apply everywhere	FleetOps \$4,995, worth it from about ten sites
Batch or event records must survive an audit	Append-only, exactly once, SHA-256 chain	SparkRecord \$7,995, early access
Who changed the configuration and can we roll it back	Signed bundles, review, rollback, audit trail	GitOps / Enterprise Governance \$9,995, pilot
An older rack behind a translator box	Box comes out with its license and patching	TI 505 / CTI 2500 driver \$695
A new dashboard or analytics project must not slow the plant	New consumer adds no load at the source	Host \$1,495, or Passage \$1,995 for standards-based tools
Every point live to the whole company, past the tag ceiling	Lossless replayable stream	SparkFlow \$995

When to hand off

Any estate over ten sites, any HA question, any broker the customer already runs, any 8.3-only feature: offer the free architecture review. Engineering sizes it from the topology and returns a written shape with the required number. The review is also the entry condition for the partner program, so an integrator gets it before their first production quote.

Disqualifiers

- No Ignition and no plan to buy it. We sell modules on Ignition.
- One site, no growth, no outage problem, no warehouse. A driver may still fit.
- Wants per-tag or per-client pricing to compare line by line. Explain the model once; do not invent a per-tag number.

Say this

- "Find the row that sounds like your plant. The last column is what fixes it and what it costs."
- "Most plants buy one product for one problem, then add the next when it shows up."

Never say this

- "You need the whole line." Nobody buys everything on day one.
- "We'll figure out pricing later." Every row has a list price today.

Check

1. A plant manager says "our reports are wrong after every storm." Route it.
2. An IT security lead says "I need to know what this MQTT thing is exposing before I approve it." Route it.
3. A customer runs Cirrus Transmission into HiveMQ at 30 sites and wants a warehouse feed. Which question do you ask first, and what is the likely purchase?

► Answers

08 Competitive

What the prospect is comparing us to, what each does well, and where the difference is. Honest by rule: the compare page names what each approach does well and so do we. Forty-five minutes.

Learning objectives

- Position against the three architectures and the four named vendors without disparaging any.
- Answer "we already run Cirrus" in under a minute.

The three architectures

	Traditional distributed Ignition	Module-based MQTT for Ignition	SparkBridge
Shape	Site gateways, Gateway Network, remote tag providers, backends per application family	Sites publish Sparkplug to a broker; one engine gateway subscribes	Edge publishes once; Provider or their broker; Host pool with shared consumers
Does well	Everything at five sites; familiar to every Ignition integrator	Removes polling and most fan-in; site buffering through outages	Everything after the transport: consumer scale-out, provisioning, drift, fleet, policy
Where it hurts	Hub saturation; backend sprawl; history reconciled by hand; provisioning by visit	Second consumer usually means a second engine; credentials typed per site; ACL files by hand; fleet by status page	Early access and pilot labels on Record, Validate, GitOps; no published case study yet; mixed-vendor pairing by specification, bench-proven only for named brokers

Where a module-MQTT cell reads the same as ours, that is Sparkplug doing its job for both. Say it. It buys credibility for the rows that differ.

Named vendors

Cirrus Link (IA's MQTT modules: Transmission, Engine, Distributor, Recorder)

The reference. Transmission is the edge publisher, Engine the primary host, Distributor the broker. Mature, sold through IA, known to every integrator. Our answer is not "they are wrong." It is: same transport, and here is what remains for your team afterwards. Concretely: Engine consumer scale-out (one wildcard subscriber per engine versus our shared subscriptions, four consumers about 20x one), provisioning (typed credentials versus

a one-time enrollment code), fleet (status pages versus FleetOps), policy (ACL files versus SparkID generated rules), history on source timestamps through replay (SparkVault), and exactly-once warehouse delivery (SparkInject; Cirrus Recorder maps to SparkVault in our line). We interoperate: a Host can be primary host for Transmission edges by specification. Say which pairings have run on the bench and which have not.

Cedalo (Mosquitto Pro / Management Center)

Broker vendor. Mosquitto 2.1 added broker-created MQTT 5 topic aliases, SQLite persistence and dynamic security. Cedalo Pro gates HA, UI, identity, audit and integrations, priced per concurrent client (tiers from about EUR 39 to enterprise). Our Provider has no client count, and SparkID generates the policy. SparkBridge has run against Mosquitto 2.1.2 with the persistence plugin on the bench, four tests passing, and fixed a Paho v5 alias edge case doing it. If the customer loves Mosquitto, keep it; we sell Edge and Host on top.

HiveMQ

Enterprise broker; the Provider HA pair in our reference architecture is two HiveMQ CE nodes. Not a competitor to Edge or Host. If the customer runs HiveMQ Enterprise, Provider is optional and SparkID emits HiveMQ credentials. Say "keep it."

Kymera and other driver vendors

Kymera sells InfluxDB and SNMP modules for Ignition among others. SparkInflux and SparkSNMP compete line for line, per gateway, and ship on both Ignition lines. Compete on the proof in the release bundle and on the price line; do not characterize their product.

Ignition-native (do nothing)

Gateway Network, remote providers, redundancy and store-and-forward are all real and all included with Ignition. At five sites, recommend them. The compare page's 5 to 50 table is the honest ramp: the traditional shape is "fine" at 5 and "the overloaded box" at 40.

The one-minute answer to "we already run Cirrus"

"Good. Sparkplug is the right transport and your sites already publish. What we add sits after the broker: one Host namespace with consumers that scale on the shared stream, history and warehouse delivery on source timestamps with exactly-once landing, an enrollment code for site 41, and a fleet view with drift and certificates. Your Transmission edges can stay; we start with Host on your broker. Where we have not run your exact broker on the bench, we say so and run it before you buy."

Where we are honest about limits

- Sparkplug Compatible listing with the Eclipse working group is administrative and in progress; conformance has been run on the TCK harness, "certified" is their stamp.
- Modules are signed with our own code-signing certificate, not an IA-issued one; the gateway asks to trust it once.
- Mixed-vendor edge and host pairing has not run on the bench; the site says so.
- No published production case study yet. The design partner offer exists for that reason. See module 11.
- SparkRecord, SparkValidate and GitOps carry early access or pilot labels.

Say this

- "MQTT is the transport. Operating the estate is the problem."
- "Keep your broker."
- "Every number on our compare page was measured; the reproduction steps are in the reference architecture."

Never say this

- Anything negative about a named vendor's quality or roadmap.
- "Certified" for Sparkplug until the listing exists.
- "Replaces Cirrus." Coexists, and replaces the work after the transport.

Check

1. Prospect: "Cirrus Engine already gives us one namespace." Reply.
2. Prospect: "Mosquitto is free, why pay for Provider?" Reply.
3. Prospect: "Are you Sparkplug certified?" Reply.

► Answers

09 Partner program

Architecture Partners, Founding Partners, the unpublished design partner offer, and the rules for who sends what. Thirty minutes. Public page: mqtt.greenguard-usa.com/sparkbridge/partners.

Learning objectives

- State both partner tiers and the no-stacking rule.
- Know what an SE may draft and what only Dan sends.

Architecture Partner

Who: an integrator with an Ignition practice that has deployed, or is scoping, an estate of ten or more sites and has completed one SparkBridge architecture review with us. Invitation only. Cap of ten.

Benefit	Terms
Development licenses	Unlimited, every module, no time limit (true for everyone; stated so partners know)
NFR licenses	One production-grade key per module for the partner's own demo estate, renewed yearly at no charge
Architecture support	Direct access to the module authors by email and scheduled call for any estate the partner designs
Deal registration	Registered opportunities protected for 90 days; we do not quote the end customer directly on a registered deal
Discount	20% off list on every license the partner resells or deploys
Escalation	Named contact, reply within one business day on production issues
Co-selling	We join technical calls with the partner's customer on request
Training	One remote architecture and deployment session per year, recorded
Roadmap	Quarterly roadmap call; requests logged and answered

Founding Partner

The first five Architecture Partners to take a customer estate into measured production.

Benefit	Terms
NFR keys	Never expire and survive any later change to the program
Priority support	Shared channel with our engineers; production issues first in the queue
Discount	40% off list on the first production estate
Roadmap vote	Founding partners vote on the order of the public roadmap items
Joint case study	We write it, they approve it, both names or anonymized at their request

What we ask of partners

- One completed architecture review before the first production quote.
- Production issues reported with the gateway diagnostics bundle.
- Permission to list the partner by name on the partners page (optional).
- For Founding: a measured writeup from the production estate, per the table on the partners page.

Program limits

- Ten Architecture Partner seats; five Founding.
- Invitation only.
- No discount stacking: 20% or 40%, never both, never with volume pricing on the same line unless Dan says so.

Design partners (end customers), unpublished

A separate offer for end customers, not integrators, kept in docs/gtm/design-partner-offer.md and not on the site. Proposed: 50% off list on the first production estate, 25% off expansion, direct engineering access, the free review plus a written deployment plan and a two-site pilot with us on the call, priority roadmap input, remote training at go-live. The customer provides at least ten sites in production within six months, permission to publish anonymized measurements, up to four reference calls a year, and approval of one case study. Target three to five deployments, two in production, one published case study. Verticals in order: multi-site manufacturing, water and utilities, oil and gas, data centers, distributed logistics. Every figure is "proposed" until Dan ratifies it for a named customer. Never quote it without Dan.

Outreach rules

- Targets: docs/gtm/integrator-targets.csv, 31 firms, mirrored to the "SparkBridge Partner Targets" Google Sheet. Dan approves each row before contact.
- Claude or an SE drafts only; Dan sends. First touch tagged [FAP], follow-up [FAP-2], template in docs/gtm/outreach-first-email.md, signed Dan Fenter, GreenGuard USA.
- Opener that tested best: "What changes when your Ignition deployment reaches 40 sites?"
- HubSpot: contact props outreach_campaign = sparkbridge-fap-2026-09, outreach_segment = vertical; deals in the default Sales Pipeline tagged "SparkBridge FAP".
- Targets for the first wave: 30 contacts, 8 to 12 replies, 5 to 8 technical calls, 3 to 5 reviews, 2 to 3 partners, 1 to 2 pilots.

Say this

- "Architecture Partners take 20% off list with deal registration; the first five Founding Partners take 40% on their first production estate."
- "The review comes first. It is free and it is how we size your customer's estate."

Never say this

- "40% for all integrators." That was corrected on the site on 2026-09-05.
- Any design-partner figure to a customer before Dan ratifies it.
- "I'll send you the agreement." Dan sends.

Check

1. An integrator wants the Founding discount on their third estate. Answer.
2. A partner registers a deal; the end customer calls us directly for a quote. What do we do?
3. What must happen before a partner's first production quote?

► Answers

10 Demo script

A twenty-five minute live demo on the GreenGuard lab gateways that shows the thesis, not the catalog. Rehearse it until the four beats land without notes. Sixty minutes to learn, twenty-five to run.

Learning objectives

- Run the four beats in order: enrollment, link drop, replay, consumer add.
- Know what to click and what number to point at in each beat.
- Recover when a beat fails.

Lab environment

Piece	Where	Notes
Ignition 8.1.54 gateway	localhost:8088 on the GreenGuard Mac	Runs Host, an Edge node, and every Spark module. Shared with engineering; check the calendar and the trial-proof hold notes before touching Reset Trial or reinstalling anything
Ignition 8.3 gateway	gg-wsl (Windows workstation, WSL Ubuntu)	Same modules on the 8.3 line; used for the "both lines" beat and for demos that need a project edit
Broker	Provider on the lab gateway; HiveMQ CE and Mosquitto 2.1 available on gg-wsl	Use Provider for the policy beat, HiveMQ for the "keep your broker" beat
Timescale	gg-wsl, ports 15432 and 15433 over an ssh tunnel	SparkVault and SparkRecord destinations

Ask engineering for the current demo credentials and the read token before the call. Never demo on a customer gateway.

Before the call, ten minutes

1. Open the gateway status page. Every Spark module green and Licensed or in fresh trial.
2. Open Host status: the demo edge node connected, birth timestamp recent, metric count visible.
3. Open a Perspective trend on two demo tags with history through SparkVault.
4. Open the /fleet route in a second tab, the /capacity route in a third.
5. Have the pricing page open in a fourth tab. That is the whole quote when they ask.

Beat 1: a site enrolls, three minutes

Say: "Site 41 is one more Edge license and one code."

1. On Host, Config > SparkBridge Host > Enrollment: issue a one-time code.
2. On the second gateway (8.3 on gg-wsl), Config > SparkBridge Edge > Settings: paste the code, connect.
3. Point at Host: the new node appears with its birth, tags build themselves in the tree. Point at /fleet: the node is registered.

Number to point at: seconds from paste to tags visible. **Caveat to say:** "CSR and mTLS issuance shipped in 2.9.0; fleet-wide reconciliation is roadmap."

Beat 2: the link drops, five minutes

Say: "Every link goes down sometime. Watch what Host does and what Edge does."

1. Cut the edge node's broker connection (engineering provides the switch: a firewall rule on gg-wsl or the Edge Settings disconnect).
2. Point at Host: the site's points go bad, not stale. Point at the trend: it stops, it does not flatline.
3. Point at Edge status: buffer depth climbing, messages per second, disk.

Number to point at: the buffer count growing; "2,697 messages per second durable on the reference rig."

Beat 3: replay, five minutes

Say: "Now the link comes back. Nothing lost, nothing twice, every reading on the time the equipment gave it."

1. Restore the connection. Edge rebirths, replay starts.
2. Point at the trend: the gap backfills with the real curve.
3. Open the SparkVault or SparkInject row view: source_ts earlier than ingest_ts, is_historical true.

Number to point at: replayed count equals buffered count. "Four hundred thousand of four hundred thousand on the bench."

Beat 4: add a consumer without touching the site, five minutes

Say: "Publish once. Add consumers at the center. The site did not change."

1. Show Passage: connect an OPC-UA client (UaExpert on the Mac) to the Host namespace, browse the site's tags. Try a write; it is refused.
2. Show SparkCalc: one calculated efficiency tag derived from two site tags.
3. Show /capacity: Host reporting its own ceiling and headroom with the new consumer attached.

Number to point at: Edge status unchanged; nothing happened at the site.

Close, five minutes

Switch to the pricing tab. Build their required set from lines in front of them (module 05). Offer the architecture review. Stop.

Optional beats by audience

Audience	Beat
Security lead	SparkID: generate rules from intent, show a site credential refused on another site's topic and on any command topic. Sentinel: run the scan, open one finding with its evidence and fix
Data team	SparkInject: the landed Snowflake row with two timestamps four hours apart; SparkRecord: verify CLI walking the chain, "0 breaks"
Operations	FleetOps: versions, drift, certificate dates across the lab gateways
Validated industry	SparkValidate CLI on a sample .ACD: the traceability matrix from rung to OQ
Driver buyer	The CTI 2500 on the lab gateway through the TI 505 driver; SparkS7 against PLCSIM on gg-wsl

When a beat fails

- **Node does not rebirth.** Say what Host is doing (waiting on STATE), give it the 10-second reconnect cap, then show the /fleet row. Do not restart anything on a shared gateway mid-call.
- **Trial paused a module.** Say exactly what it is: "development trial, two hours, a key file lifts it in thirty seconds." Reset Trial only if the hold notes allow it.
- **gg-wsl unreachable.** Run beats 2 to 4 on the single 8.1 gateway with its local edge node; skip beat 1 and describe enrollment with the products page screenshot.
- **Anything else.** The measured numbers on the technology and benchmarks pages are the fallback demo. Never fake a screen.

Never

- Change a customer's gateway during a demo.
- Promise a roadmap date.
- Run a beat you have not rehearsed that week.

Check

1. What is the one thing that must be visibly unchanged at the end of beat 4?
2. Which two timestamps prove replay kept source time?
3. A prospect asks you to point Edge at their test gateway right now. Answer.

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 09 Partner program](#) ·

11 Objections

Objection cards. Each has the honest answer and the caveat we publish. Read the caveat out loud; it is what makes the answer believable. Thirty minutes.

Learning objectives

- Answer the twelve common objections in under a minute each.
- Never answer with a claim the site does not make.

"What happens to the price when we add tags?"

Nothing. Per gateway, per product. Tags, devices and connected clients are never counted. The 40-site examples on the pricing page do not move when the estate gets busier. *Caveat:* Edge has a 250,000-tag budget per gateway; past that you add an edge node or a gateway, not a fee.

"We already run Cirrus."

Keep the transport. What we add sits after the broker: shared-subscription consumers, history and warehouse on source timestamps, enrollment codes, fleet view, generated policy. Start with Host on your broker. *Caveat:* mixed-vendor edge and host pairing is by specification; we run your exact pairing on the bench before you buy.

"Why pay for Provider when we have HiveMQ / Mosquitto?"

You do not have to. Edge and Host connect through any conformant broker; Provider is for the customer with no broker or who wants generated per-site policy enforced without ACL files, with no client count. *Caveat:* HiveMQ and Mosquitto 2.1 are the brokers that have run on our bench.

"What happens if support lapses?"

The software keeps running exactly as before. You stop receiving new versions and support until you renew. *Caveat:* none; that is the whole answer.

"Early access and pilot: is this production software?"

Edge, Host, Provider, Vault, Inject, Passage, FleetOps, Sentinel, Flow and the drivers are released and live-proven on both Ignition lines. SparkRecord 0.1.0 is early access, SparkValidate and GitOps are pilots, and we label them that way on every page. *Caveat:* buy the labeled ones for the problem they solve today, not for the roadmap.

"Do you have a reference customer?"

Not a published one yet. Every number we print was measured on named hardware with reproduction steps in the reference architecture. The design partner offer exists to produce the first case study, and Founding Partners get a joint one. *Caveat:* never name a customer or imply one. If they need a reference before buying, say so to Dan and route them to the design partner conversation.

"8.1 or 8.3?"

Both. Every product ships signed on both lines and is proven on a live 8.3 gateway before release. Same price. *Caveat:* the 8.3 line is newer; ask engineering about any 8.3-only feature before promising it.

"Are you Sparkplug certified?"

We have run the Eclipse TCK harness and the Sparkplug Compatible listing is in progress. We do not use the word certified until it is granted. *Caveat:* that is the caveat.

"The gateway warned me about the module certificate."

Modules are signed with the SparkBridge code-signing certificate, not one issued by Inductive Automation, so the gateway asks to trust it once. Every release bundle includes the signature and the live proof record. *Caveat:* an IA-issued certificate is on the list; no date.

"Isn't this just MQTT with a markup?"

MQTT is the transport and it is the right one. What you pay for is what remains after the transport: one namespace built from births, consumers that scale on the shared stream, replay on source timestamps into history and warehouse, enrollment, fleet, policy. The compare page lists it row by row and says where the standard alone gets you the same cell. *Caveat:* at five sites, use what Ignition gives you; we say that on the compare page.

"We want a subscription / monthly price."

We sell one-time licenses with optional 20% support. There is no monthly option. *Caveat:* volume pricing by site count is available; that is Dan's call, not a discount an SE offers.

"Who supports this? You are a small company."

Direct engineering support is what the 20% buys: the people who wrote the modules. Every module has a manual, a live proof record and a review record in its release bundle. Development is free so you can prove it on your own estate before you spend anything. *Caveat:* say the team size honestly if asked; do not inflate it.

"Can Passage / IEC-104 / DNP3 write to equipment?"

No. Passage is read only; the utility drivers are reads only. Edge write-back is off by default and allowed only for listed devices and points. Security teams usually count this as a plus. *Caveat:* BACnet and Logix drivers do write; say which is which.

"Your site says Ignition 8.3 dropped symbolic S7, but IA lists a Siemens Enhanced driver."

Both are true. Symbolic addressing was in the 8.3 early access, was removed from 8.3.0 over licensing issues IA said were out of its hands, and came back as the Siemens Enhanced driver, a separately purchased module that is not active by default. SparkS7 gives symbolic S7-1200/1500 addressing on 8.1 and 8.3 from one \$695 per-gateway license, verified on an S7-1500 in PLCSIM Advanced with a real TIA project on both gateway lines (real hardware pilot offered, not claimed). *Caveat:* state the timeline and the price line; do not characterize IA's driver. Module 04 has the detail.

The pattern

Answer, then the caveat, then stop. The caveat is not a weakness; it is the reason the answer is believed.

Check

1. Write the reference-customer answer in your own words, with the caveat.
2. A prospect insists on a per-tag comparison against a competitor. What do you do?
3. Which products carry early access or pilot labels today?

► Answers

SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 10 Demo script](#) ·

Quiz

Forty questions: ten on Ignition, thirty on SparkBridge. Pass mark 80% (32 correct). Closed book except the [cheat sheet](#). Answer key at the bottom.

Ignition (00a, 00b)

1. What is an Ignition gateway, and on what default port is its web configuration served?
2. What is the difference between a tag provider and a remote tag provider?
3. How does Inductive Automation license Ignition: per what unit, and what is not counted?
4. Name two things Ignition 8.3 changed versus 8.1.
5. What does the gateway do when a paid Spark module's two-hour trial expires, and what lifts it?
6. Which Sparkplug message carries metric names and metadata, and which carries only changes?
7. What is the primary host in Sparkplug, and which SparkBridge product plays that role?
8. "Ignition Edge" and "SparkBridge Edge": which is an edition and which is a module?
9. In the hub-with-remote-providers shape, what multiplies hub load and what multiplies site load?
10. What does module-based MQTT leave to custom engineering after the transport? Name three items.

SparkBridge (00 to 11)

11. State the thesis in one sentence.
12. Name the four themes and one product in each.
13. Which two products are the core, and when is Provider optional?
14. What is included with Edge? With Host?
15. List price of Edge, Host and Provider.
16. Edge's tag budget per gateway, and how many edge nodes per gateway.
17. What does Host do to a site's points when that site drops?
18. Five measured numbers that belong to the core, with what each proves.
19. SparkVault versus SparkRecord, in one breath.
20. SparkInject is priced per what, and which family is available first?
21. Can Passage write to equipment?
22. Which of SparkID, Sentinel, FleetOps is included with Host?
23. What does Sentinel change on the network when it runs?
24. Which products carry early access or pilot labels?
25. Price of any protocol driver, and the sentence that describes its licensing.
26. Which two drivers are reads only?
27. Required set and list total for 40 sites on the customer's own broker.
28. Required set and total for 40 sites with Provider HA, Host active plus standby, and SparkVault.
29. What does site 41 cost?
30. Host memory rule of thumb per wire metric per consumer.

31. Support: percentage, term, and what happens if it lapses.
32. How is a license activated, and how long does it take?
33. Does a license follow the hardware or the role?
34. Architecture Partner discount and its condition; Founding Partner discount and its condition.
35. Who sends partner outreach emails?
36. The one-minute answer to "we already run Cirrus," in your words.
37. What do we say when asked for a reference customer?
38. The four beats of the demo, in order.
39. Three phrases the site never uses, and why.
40. A water district with 12 sites on cellular, no broker, wants outage-proof history and a fleet view. Write the quote lines and total.

► [Answer key](#)

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[Previous: 11 Objections](#)

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SparkBridge sales engineer training, internal. Prices verified against mqtt.greenguard-usa.com/sparkbridge/pricing on 2026-09-06.

[Previous: 11 Objections](#)

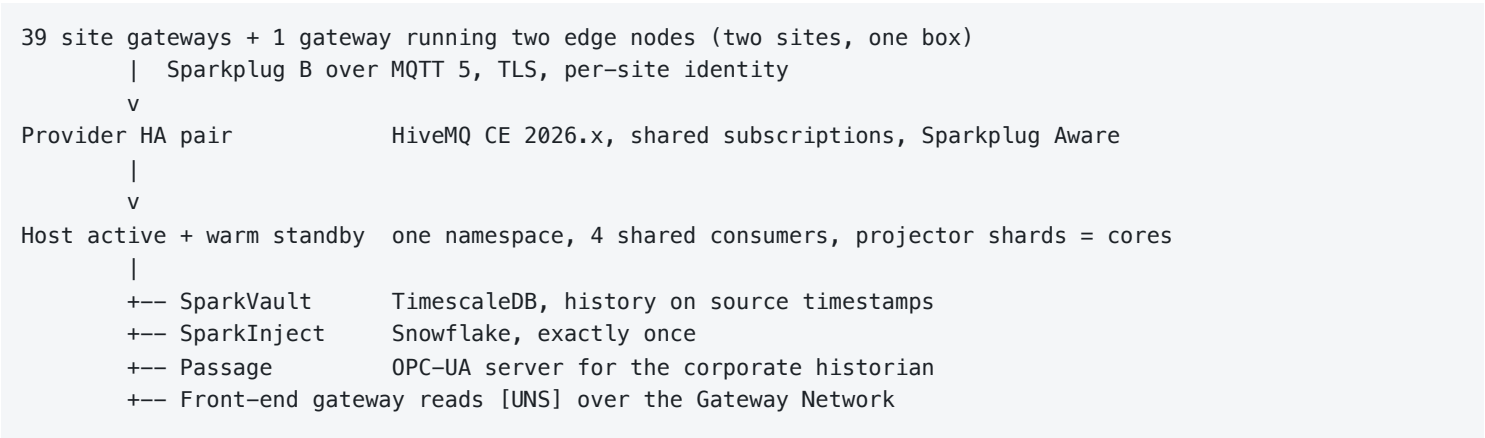
The 40-Site Ignition Reference Architecture

SparkBridge for Ignition. GreenGuard USA. September 2026. Every figure in this document names the machine it was measured on; the full record is at <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>.

1. The estate

Forty production sites, each running an Ignition gateway with 5,000 to 25,000 tags. Sites reach the center over mixed WAN: fiber at the plants, cellular at the remote pads, one satellite link. A front-end gateway serves several hundred Perspective users. Corporate wants history in one place, a warehouse feed, and OPC-UA access for a historian it already owns.

2. The shape



Three roles. Edge moves the data. Provider carries it. Host uses it.

3. Sizing

Layer	Sizing rule	Source
Edge per site	One module license. Up to 8 edge nodes per gateway. 250,000-tag budget per gateway	products page, Edge table
Wire metrics fleet-wide	40 sites x ~7,000 published leaves after metadata folding = ~280,000	migration plan calibration (Irving 8,700, Mesa 22,400, Dickinson 8,500 leaves)
Host memory	~300 bytes per wire metric per consumer; 500,000 wire metrics in under 4 GiB; 280k x 4 consumers ~ 340 MB session state plus the managed provider	PERFORMANCE.md section 7
Host CPU	Projector shards = core count. Six cores sustained at least 339,527 updates/s end-to-end on the Linux reference rig; four logical cores bend at 55,464	PERFORMANCE.md sections 0 and 8
Host recommendation	8 cores, 16 GB, 32 GB if SparkVault shares the box	sizing from the rows above
Provider	Two HiveMQ CE nodes, 4 cores, 8 GB each. No client-count license	docs/BROKER.md, docs/HA.md

Layer	Sizing rule	Source
Store-and-forward	50 MB default per edge node; at 2,697 messages/s durable and typical site change rates that is hours to days of outage. Size cellular and satellite sites at 500 MB	PERFORMANCE.md section 5

4. What happens when things fail

Failure	Behaviour	Measured
WAN drop at a site	Edge buffers to disk; Host marks the site's points bad, not current	2,697 msg/s durable
WAN returns	Reconnect within 10 s cap, rebirth, ordered replay on source timestamps with the historical flag; SparkVault backfills the curve	400,000 of 400,000 replayed; 40,035 upd/s on the six-core rig
Host primary crash	Standby promotes after the debounce; edges never rebirth	244 ms across 25 nodes, 5.7 ms mechanical
One Provider node down	Edges and Host walk to the other server in the set	Edge MultipleBroker TCK PASS, ~0.6 s server walk
Broker outage entirely	Every site buffers; nothing is lost up to buffer size	store-and-forward suite
A site's credential leaks	It can publish only its own group and node, never a command	Provider ACL, proven live

5. Adding consumers later

Each of these is a subscription at Host or a module on it. No change at the forty sites.

- Analytics or AI: subscribe to the shared stream, or read SparkVault, or take the Snowflake copy.
- A second corporate historian: Passage already serves OPC-UA; add a client.
- Estate-wide streaming past the tag ceiling: SparkFlow on Host.

6. Operations

- Site 41: install Edge, enter the enrollment code, \$995. Certificate issuance and automatic FleetOps registration are on the public roadmap.
- Fleet health: Host /fleet (buffering, headroom per edge), FleetOps (versions, drift, certificates).
- Security posture: SparkID generates the rules, Provider enforces them, Sentinel scores the result.
- Capacity: Host /capacity reports its ceiling from the shard count it runs.

7. List price

Item	Qty	Each	Total
SparkBridge Edge	39	\$995	\$38,805
SparkBridge Provider	2	\$1,995	\$3,990
SparkBridge Host	2	\$1,495	\$2,990

Item	Qty	Each	Total
SparkVault	1	\$3,995	\$3,995
Required total			\$49,780 once
Optional: SparkInject (Snowflake)	1	\$2,995	
Optional: Passage	1	\$1,995	
Optional: FleetOps	1	\$4,995	
Optional: Sentinel	1	\$4,995	
Support, optional		20% of list per year	\$9,956 on the required set

Nothing is metered. Development gateways are free and not counted.

8. Reproduce the numbers

```
./gradlew :common:loadTest --console=plain
./gradlew :common:loadTest --tests io.sparkbridge.perf.CapacityLoadTest
./gradlew :common:standbyDrill
```

Send us your topology for a review of your own estate: <https://mqtt.greenguard-usa.com/sparkbridge/architecture-review>

SparkBridge for Ignition, GreenGuard USA. <https://mqtt.greenguard-usa.com/sparkbridge/> . Perpetual per-gateway licensing, free for development. Figures: PERFORMANCE.md sections 0, 5, 7, 8; docs/HA.md; docs/TCK.md.

SparkBridge Architecture Partner program (ratified terms)

Status: RATIFIED by Dan 2026-09-05 and published at mqtt.greenguard-usa.com/sparkbridge/partners (expanded page live 2026-09-06). This file is the internal record of the terms; the live page is what a partner sees. Invitation only. Ten Architecture Partner seats, five Founding.

Goal

Ten integrators who consider SparkBridge part of their standard architecture toolkit for distributed Ignition estates.

Architecture Partner

Who: an integrator with an Ignition practice that has deployed, or is scoping, an estate of 10 or more sites and has completed one SparkBridge architecture review with us.

Benefit	Terms
Development licenses	Unlimited, every module, no time limit (already true for everyone; stated so partners know it)
NFR licenses	One production-grade key per module for the partner's own demo estate, renewed yearly at no charge
Architecture support	Direct access to the engineers who built the modules, by email and scheduled call, for any estate the partner is designing
Deal registration	Registered opportunities are protected for 90 days; we do not quote the end customer directly on a registered deal
Partner discount	20% off list on every license the partner resells or deploys
Engineering escalation	Named contact, reply within one business day on production issues
Co-selling	We join technical calls with the partner's customer on request
Training	One remote architecture and deployment session per year, recorded for the partner's team
Roadmap access	Quarterly roadmap call; partner requests logged and answered

Founding Partner

The first five Architecture Partners to take a customer estate into production on SparkBridge.

Benefit	Terms
Permanent internal development entitlement	The NFR keys above never expire and survive any later change to the partner program
Priority support channel	Shared channel with our engineers; production issues first in the queue
Discounted first production deployment	40% off list on the first production estate

Benefit	Terms
Roadmap participation	Founding partners vote on the order of the roadmap items on the public roadmap page
Joint case study	We write it, they approve it, both names on it, or anonymized at their request

What we ask of partners

- One completed architecture review before the first production quote.
- Production issues reported with the gateway diagnostics bundle.
- Permission to list the partner by name on the partners page (optional, per partner).

Decisions (Dan, 2026-09-05)

1. Partner discount 20% off list. No stacking with volume pricing or any other discount.
2. Founding discount 40% off list on the first production estate, every module in that estate.
3. NFR scope: one production-grade key per module for the partner's own demo estate.
4. Deal registration window 90 days.
5. Invitation only; the page is public so an invited firm can read the terms.
6. Design partners (end customers, 50%) stay a separate unpublished offer: design-partner-offer.md.

FOUNDING ARCHITECTURE PARTNERS

FOR INTEGRATORS WHO DESIGN DISTRIBUTED IGNITION ESTATES

Ten integrators who consider SparkBridge part of their standard toolkit, and five of them with a measured production estate behind the claim. That is the whole goal of the program, and the terms are written to make standardizing on it safe.

TEN SEATS FIVE FOUNDING INVITATION ONLY

ARCHITECTURE PARTNER

ARCHITECTURE PARTNER

Any Ignition integrator that has completed one architecture review with us.

BENEFIT

TERMS

Development licenses

Unlimited, every module, no time limit

NFR licenses

Production-grade keys for the partner demo estate, renewed yearly at no charge

Architecture support

Direct access to the module authors for any estate

● [ASK AN ENGINEER](#)

BENEFIT	TERMS
Deal registration	Registered opportunities protected for 90 days
Partner discount	20% off list
Escalation	Named engineer, one business day on production issues
Co-selling	We join the partner on technical calls on request
Training	One remote session per year, recorded
Roadmap	Quarterly call; requests logged and answered

FOUNDING

FOUNDING PARTNERS

The first five partners to take a customer estate into production.

BENEFIT	TERMS
Permanent development entitlement	NFR keys that never expire
Priority support channel	Shared channel, production issues first
First production deployment	40% off list
Roadmap participation	Founding partners vote the order of the public roadmap



Joint case study

Written by us, approved by them, named or anonymized

HOW IT WORKS

FOUR STEPS, IN ORDER

Every partner enters the same way. The first production estate is what separates the two tiers.

STEP	WHAT HAPPENS	WHAT YOU GET
1. Architecture review	One review of an estate you are designing now. We size it from your topology and write the reference drawing with numbers from the bench.	A drawing, a bill of materials at list, and the honest limits for that shape.
2. Architecture Partner	Review completed, terms signed. NFR keys issued for your demo estate.	Everything in the Architecture Partner table above.
3. First production estate	You take a customer estate live. We run the install with you, log every figure, and write it up.	A one-page measured writeup, named or anonymized, which you may use in your own proposals.
4. Founding Partner	The first five partners to complete step 3.	Everything in the Founding Partners table above, for as long as the firm exists.



WHAT WE ASK

WHAT WE ASK OF PARTNERS

Three things. None of them cost money.

OBLIGATION

WHY

One completed architecture review before the first production quote

The review is where the estate gets sized correctly. Every measured number on this site came from a shape somebody drew first.

Production issues reported with the gateway diagnostics bundle

One business day on escalation is only possible when the report carries the logs, the config and the version on the first message.

Permission to list the firm by name on this page, optional, per partner

Some partners want the listing and some do not. Either answer is fine and can change later.

WHAT THE FIRST FIVE PROVE

MEASURED, NOT CLAIMED

A founding partner's estate becomes one of the production references this site currently lacks. We publish what was measured, at the size it was measured, with the limits printed under the drawing.



Shape	Which reference drawing it matches, site count, Host and Provider arrangement, modules installed
Throughput and buffer	Sustained tag changes per second at Host, store-and-forward coverage per site in hours, measured on the estate
Failure behavior	What happened on the first real outage: replay order, timestamp preservation, standby takeover time
Upgrade in place	Each SparkBridge release applied to the live estate, with the time it took and anything that had to be redone
Honest limits	What the estate does not do yet, in the same table as what it does

The writeup is drafted by us and approved by the partner and the end customer before anything is published. Anonymized is always available.

PROGRAM LIMITS

PROGRAM LIMITS

LIMIT	TERMS
Seats	Ten Architecture Partners. When ten are seated, new firms go on a list and are told where they stand.
Founding seats	Five, in order of first production estate. Sixth and later production estates get the Architecture Partner terms.



Entry	Invitation, or a request through the form below. Firms with an Ignition practice and an estate of ten or more sites in design are the fit.
Discounts	Partner discount is on list price and does not stack with volume pricing. Founding 40% applies to the first production estate, every module on it.
Design partners	End customers who want to be a reference estate directly are a separate offer. Ask.

QUESTIONS

QUESTIONS PARTNERS ASK FIRST

DOES THE CUSTOMER BUY
FROM US OR FROM YOU?

From you, on a registered deal. We do not quote the end customer directly while the registration stands.

WHAT IF THE CUSTOMER
ALREADY RUNS A BROKER?

Edge connects to any conformant MQTT broker. Provider is a managed broker and policy layer you can add or leave out. The review states which.

WHAT DOES SUPPORT COST
THE CUSTOMER?

Optional, 20% of list per year. Nothing stops running if it lapses.

DO NFR KEYS COVER EVERY
MODULE?

Yes, every module, for the partner demo estate. Renewed yearly for Architecture Partners, never expiring for Founding Partners.

CAN THE CASE STUDY BE
ANONYMIZED?

Yes, at the partner's or the customer's request, at any time before or after publication.



START WITH A REVIEW

The program begins with one architecture review of an estate you are designing. Ten seats, five founding, no fee to apply.

REQUEST AN ARCHITECTURE REVIEW

ASK ABOUT THE PROGRAM

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

[Benchmarks](#) · [Roadmap](#) · [Changelog](#) · [Pricing](#) · [Partners](#) · [Contact](#)

Design partner offer and case-study template

Status: draft for Dan. Design partners are end customers, not integrators. Target: 3 to 5 meaningful deployments, 2 in production, 1 published case study. This is the single biggest gap on the scorecard and no feature closes it.

The offer

We provide	Terms
Discount	50% off list on the first production estate (proposed), locked for expansion at 25% off (proposed)
Engineering access	Direct line to the module authors for the life of the deployment
Architecture support	The free architecture review, then a written deployment plan, then a two-site pilot run with us on the call
Priority roadmap input	Design partner requests go to the front of the roadmap
Training	Remote sessions for the operations team at go-live and at the first expansion

Design partner provides	Terms
Production deployment	At least 10 sites within 6 months of pilot start (proposed)
Measurements	Permission to collect and publish anonymized capacity, outage and replay figures from the estate
Reference calls	Up to four reference calls per year with prospects, scheduled at their convenience
Case study	Approval of one anonymized case study; named attribution optional

Ideal verticals, in order

1. Multi-site manufacturing (the 40-gateway shape we already designed for)
2. Water, wastewater, utilities (cellular and radio links, store-and-forward matters)
3. Oil and gas (SparkRecord audit chain, EFM records)
4. Data centers (SNMP plus BACnet drivers alongside the process)
5. Distributed logistics

Anonymized case-study template

Title: A North American {vertical} operator running {N} Ignition gateways

1. Previous architecture. Hub gateway, backends, remote tag providers, where history was written. One diagram.
2. Problem. The box that was overloaded, the outage that lost data, the consumer that could not be added. Numbers: CPU, memory, tag count, sites, outage duration.
3. Deployment. Edge count, Provider shape, Host shape, consumers added, in what order, over what period. What was not migrated and why.
4. Measurements. Before and after: hub load, replay after a real outage, consumer add time, time to provision a new site. Every figure tied to the machine it came from.

5. Results. What the operations team says now. What they would do differently.

Rule: no figure appears in a case study that the design partner has not seen and approved in writing. Nothing is published before a production deployment exists.

Target list

integrator-targets.csv: 31 firms, contacts re-verified 2026-09-07 against each firm's current team page, LinkedIn public pages and 2026-dated press. Mirrored to the Google Sheet "SparkBridge Partner Targets" (Drive > Claude uploads > Industrial & Drivers), which carries Dan's approve column. The sheet is the approval record; the CSV is the working copy.

Columns added 2026-09-07: person2..3, title2..3, email2..3, linkedin2..3, generic_email, email_pattern, email_confidence (published / pattern / none), status (current / replaced / new / unverified), verified_date, refresh_notes, and the seven score_* rubric columns plus score and tier, blank for the MBA.

Email rule: published addresses came from the firm's own site. pattern addresses were built from the firm's observed pattern (rocketreach / leadiq / zoominfo majority format) and have not been sent to; verify with a LinkedIn message or the generic inbox first if a bounce would matter. none means use the generic inbox or LinkedIn.

Refresh results: 7 contacts replaced (Granttek, SAGE, Veritech, Flexware, Perceptive, Hanley, Tamaki), 2 firms newly named (ATS Global, E Tech Group), 1 unverified (SL Controls merged into NNIT, January 2025). DMC's John Williams was checked and is still listed on dmcinfo.com.

Rules: nothing is contacted until Dan marks the row approved. One person per firm at a time; move to person2 only after touch 4 or a bounce.

06-targets/integrator-targets.csv

firm	rank	person	title	person_url	email	linkedin
DMC Inc.	1	John Williams	Regional Manager, PNW Automation	https://www.dmcinfo.com/about/our-team/john-williams/		https://www.linkedin.com/in/jwilliams
Streamline Control (part of SGS)	2	Tom Chenier	Director, IIoT and Middleware	https://streamlinecontrol.com/about-us/who-we-are/	tom.chenier@streamlinecontrol.com	https://www.linkedin.com/in/tom-cher
Corso Systems	3	Cody Johnson	Head of Sales	https://corsosystems.com/contact/	cody.johnson@corsosystems.com	https://www.linkedin.com/in/cody-johnson-499006240/
CSE ICON	4	Damon Vinciguerra	Practice Manager, Real-Time Data	https://www.cse-icon.com/leadership-team/	damon.vinciguerra@cse-icon.com	
Grantek Systems Integration	5	Ian Tooke	Vice President, Enterprise Solutions	https://theorg.com/org/grantek/org-chart/ian-tooke	ian.tooke@grantek.com	
NFM Consulting	6	Michael O'Dell	President	https://www.linkedin.com/in/michaelodell-nfm/	michael.odell@nfmconsulting.com	https://www.linkedin.com/in/michaelc-nfm/
SAGE Group / SAGE Automation	7	Justin Kahl	Head of Engineering Services, SAGE Automation	https://gotosage.com/leadership-team	justin.kahl@gotosage.com	
Vertech	8	Paul Warning	Director of Automation	https://theorg.com/org/verttech	pwarning@verttech.com	
Flexware Innovation	9	Wade Harsy	VP of Engineering	https://www.flexwareinnovation.com/our-team/	wade.harsy@flexwareinnovation.com	
4IR Solutions	10	James Burnand	CEO	https://www.4ir.cloud/about-us	james.burnand@4ir.cloud	https://www.linkedin.com/in/burnand/
BJC Ltd	11	Jonathan (Joff)	Technical Director	https://www.bjc.co.uk/team.html		
NeoMatrix	12	Andrew DeChirico	Engineering Manager	https://neomatrixinc.com/company/team/	adechirico@neomatrixinc.com	
Perceptive Controls (UpT Reliability)	13	Ryan Fisher	President and co-founder	https://www.linkedin.com/in/ryan-fisher-84154313/	rfisher@perceptivecontrols.com	https://www.linkedin.com/in/ryan-fish-84154313/
Hanley Energy	14	Eoin Rigney	Controls Engineering Manager (Ignition	https://theorg.com/org/hanley-energy/org-chart/eoin-rigney	eoin.rigney@hanleyenergy.com	https://ie.linkedin.com/in/eoin-rigney-79344086

			Gold Certified)			
Tamaki Control	15	Adnan Sharief	Regional Manager, North and East US	https://ie.linkedin.com/in/adnan-sharief-36a398190	asharief@tamakicontrol.com	https://ie.linkedin.com/in/adnan-sharief-36a398190
NLS Engineering (now Ulteig)	16	McKenzie Santin	Associate Director, AISP (energy SCADA, NextDAS)	https://www.zoominfo.com/p/Mckenzie-Santin/6117004117	mckenzie.santin@ulteig.com	https://www.linkedin.com/in/mckenzie-santin-38218783/
Brown Engineers	17	Dee Brown PE	Co-Founder / Principal	https://www.linkedin.com/in/dee-brown-pe-5744333/	dbrown@brownengineers.net	https://www.linkedin.com/in/dee-brown-pe-5744333/
RoviSys	18	Anh-Tuan Tran	Group Manager, CPG / IDS Division	https://theorg.com/org/rovisys/org-chart/anh-tuan-tran	anh-tuan.tran@rovisys.com	https://www.linkedin.com/in/anh-tuan-tran-b4195015/
Thermo Systems	19	David Musto	President and CEO	https://thermosystems.com/enhanced-executive-leadership-team-at-thermo-systems/	david.musto@thermosystems.com	
ATS Global	20	Rob Valent	Managing Director / Global Partner Manager	https://www.linkedin.com/in/rvalent/	rob.valent@ats-global.com	https://www.linkedin.com/in/rvalent/
SL Controls	21	Brian McLaughlin	Systems Architect (now NNIT)	https://slcontrols.com/author/brian-mclaughlin/		
Agidens	22	Dries Smeuninx	Senior Project Engineer, Ignition Gold Certified	https://www.linkedin.com/in/dries-smeuninx-30a8a133/	dries.smeuninx@agidens.com	https://www.linkedin.com/in/dries-smeuninx-30a8a133/
Barry-Wehmiller Design Group	23	Adam Welles	Partner; National Lead, Information Solutions (data centers)	https://inductiveautomation.com/blog/user/764	adam.welles@bwdesigngroup.com	https://www.linkedin.com/in/adam-welles-a6515ab/
E Tech Group	24	Kurt Wakeman	Business Director, Food and Beverage / CPG	https://etechgroup.com/about/	kwakeman@etechgroup.com	
Industrial Networking Solutions (INS)	25	Morris (Mo) Moore	Director of Software Services	https://www.linkedin.com/in/morris-mo-moore-481115a/	mmoore@industrialnetworking.com	https://www.linkedin.com/in/morris-moore-481115a/
InflexionPoint (formerly Automated Control Concepts)	26	Chris Monchinski	CTO	https://www.inflexionpoint.ai/pages_company/about.php	c.monchinski@inflexionpoint.ai	
Kanoa	27	Jason Coope	Founder and Principal	https://kanoa.ai/about		https://www.linkedin.com/in/jasoncoc
NZ Controls	28	Brett Bartlett	Director	https://nz.linkedin.com/in/brett-bartlett-6616381b	brett.bartlett@nzcontrols.co.nz	https://nz.linkedin.com/in/brett-bartlett-6616381b

Multi-Dimensional Integration (MDI)	30	Mike Ficchi	Senior Controls Engineer	https://www.linkedin.com/in/mike-ficchi-b73675155	mficchi@mdiadvantage.com	https://www.linkedin.com/in/mike-ficchi-b73675155
Merit Controls	31	Lars Johnson	Chief Technology Officer	https://meritcontrols.com/team/	ljohnson@meritcontrols.com	
Fairfield Control Systems	32	Peter McMorow	Technical Director	https://www.fairfields.co.uk/fcs/about/team/	pmcmorrow@fairfields.co.uk	

SparkBridge partner outreach: integrator target list

Compiled 2026-09-03 from web search only (Inductive Automation news/blog/ICC pages, Cirrus Link partner pages, HiveMQ, integrator sites). Every URL below appeared in a search result or was fetched directly. HQ marked "not confirmed" where sources did not state it. Sorted by strength of evidence (direct published MQTT/Sparkplug/UNS work + Ignition tier + distributed multi-site verticals).

#	Firm	HQ	Evidence (URL + reason)	Primary vertical(s)	Contact route
1	DMC, Inc.	Chicago, IL, USA (8 US offices)	https://www.dmcinfo.com/our-work/mqtt-sparkplug-implementation-with-ignition/ : multi-site Sparkplug UNS with Parris-method encoding into central Ignition historian; also ICC 2025 session with Sila on UNS + HiveMQ (https://inductiveautomation.com/resources/icc/2025/scaling-smart-architecting-for-industrial-growth)	Distributed manufacturing, water skids, battery materials	https://www.dmcinfo.com/about/partners/integrator/ ; John Williams (Regional Mgr, Automation) presented at ICC 2025
2	Streamline Control (part of SGS)	Calgary, AB, Canada	https://streamlinecontrol.com/wp-content/uploads/2023/05/MQTT-Ignition-Success-Story.pdf : MQTT+Ignition replaced legacy SCADA; site claims 500+ Sparkplug B edge nodes, 90%+ bandwidth reduction, MQTT since 2015 (https://streamlinecontrol.com/technology-we-work-with/ignition-ia/)	Oil & gas, pipelines, utilities, renewables	https://streamlinecontrol.com/about-us/cc
3	Corso Systems	USA (HQ not confirmed)	https://corsosystems.com/posts/unified-namespace-implementation-using-ignition/ : step-by-step UNS with Sparkplug B, HiveMQ, Cirrus Link; multiple MQTT/UNS posts (https://corsosystems.com/posts/mqtt-and-unified-namespace-integrations)	Manufacturing, F&B	https://corsosystems.com/mqtt ; sales co site: Cody Johnson
4	CSE ICON	The Woodlands, TX, USA (also Lafayette, LA)	https://www.cse-icon.com/category/inductive-automation/ : IA Enterprise Integrator (Jan 2025) plus Cirrus Link IIoT Integrator certification (Feb 2025)	Oil & gas, enterprise SCADA	contact@cse-icon.com (from site)
5	Grantek Systems Integration	Canada/USA (HQ not confirmed)	https://grantek.com/grantek-recognized-as-a-cirrus-link-certified-iiot-integrator-by-inductive-automation/ : Cirrus Link certified IIoT integrator; IA Enterprise Integrator (https://inductiveautomation.com/news/inductive-automation-designates-grantek-enterprise-integrator)	F&B, energy, pharma, CPG	https://grantek.com/ (root seen)
6	NFM Consulting	San Antonio, TX, USA	https://nfmconsulting.com/knowledge/ignition-mqtt-sparkplug-b-integration/ : published Sparkplug B integration and Edge IIoT guides; managed Ignition SCADA for distributed sites	Oil & gas field automation, data centers, water, smart grid	https://nfmconsulting.com/services/management/ ; info@nfmconsulting.com
7	SAGE Group / SAGE Automation	Adelaide (Tonsley), SA, Australia	https://icc.inductiveautomation.com/discover-gallery-detail/2022/sagegroup : Alexandrina Council, 300+ remote sites on Perspective + MQTT Sparkplug B over LTE with Red Lion edge; Premier Integrator (https://www.sageautomation.com/news/pages/sage-achieves-premier-integrator-status-with-inductive-automation)	Water/wastewater, councils, utilities	https://www.sageautomation.com/contact
8	Vertech	Phoenix, AZ, USA	https://www.verttech.com/blog/using-mqtt-with-ignition-scada : MQTT with Ignition via Cirrus Link (author Roger Larson); IA cites Vertech dashboards across six solar sites; #3 top-selling integrator 2025 (https://inductiveautomation.com/blog/lessons-from-the-top-selling-ignition-integrators-of-2025)	Solar, data centers, water, manufacturing	https://www.verttech.com/ (root seen)
9	Flexware Innovation	Fishers, IN, USA	https://inductiveautomation.com/resources/customerproject/fully-automated-distillery-uses-ignition-to-integrate-scada-mes-erp-cloud-analytics-into-single-platform : UNS on Cirrus Link Transmission/Distributor/Engine feeding Canary/Flow; #1 top-selling integrator 2025	Distributed manufacturing, F&B, multi-site rollouts (SparkSCADA/SparkMES)	https://flexwareinnovation.com/spark-scada ; info@sparkskada.io
10	4IR Solutions	Bethlehem, PA, USA	https://www.hivemq.com/news/hivemq-4ir-solutions-partner-enable-cloud-first-iiot-deployments/ : HiveMQ bundled into FactoryStack/PharmaStack for Ignition-to-cloud MQTT; IA technology provider page (https://inductiveautomation.com/partners/technology-providers/4ir-solutions/)	Pharma/life sciences, cloud-hosted multi-site manufacturing	https://www.4ir.cloud/contact-us ; James video https://inductiveautomation.com/resource/burnand-4ir-solutions)
11	BIJC Ltd	UK	https://www.bijc.co.uk/ : Premier Integrator; "Sparkplug B Specialists", UNS and broker design listed as services; 100+ custom modules, 15+ countries	Data centres, HV/LV power, oil & gas, manufacturing	Contact form on https://www.bijc.co.uk/
12	NeoMatrix	Massachusetts / New Hampshire, USA	https://neomatrixinc.com/blog/sparkplug-b-or-not-to-b-unified-namespace-architecture-considerations/ : UNS/Sparkplug architecture blog; Premier since 2016, top-20 seller 2023 (https://neomatrixinc.com/platforms/inductive-automation-integrator/)	Water/wastewater, biotech, manufacturing	https://neomatrixinc.com/platforms/inductive-automation-integrator-for-waste-water-facilities/ ; 603
13	Perceptive Controls (UpT Reliability)	Plainwell, MI, USA	https://inductiveautomation.com/resources/customerproject/cloudhosted-scada-with-mqtt-gives-water-agency-faster-access-to-data : AWS-hosted Ignition with MQTT Distributor/Engine and Transmission on groov EPIC RTUs for Waterford Twp (63 lift stations, 11 plants)	Municipal water/wastewater	https://perceptivecontrols.com/contact-us ; sales@perceptivecontrols.com
14	Hanley Energy	Ashburn, VA, USA and Co. Meath, Ireland	https://inductiveautomation.com/news/hanley-energy-ltd-achieves-premier-integrator-status : Premier; #2 top-selling 2025; Ignition-based EPMS for data centers, engineers in 11+ locations	Data centers (EPMS), energy	https://hanleyenergy.com/ (root seen)
15	Tamaki Control	Auckland, NZ (also UK, Idaho USA)	https://inductiveautomation.com/resources/icc/2024/integrator-panel-what-tech-and-trends-are-breaking-through : ICC 2024 integrator panel; top-10 seller; ANZCO 10-site enterprise SCADA	Multi-site food processing, manufacturing	https://www.tamakicontrol.com/detail ; Le Branch Manager)

#	Firm	HQ	Evidence (URL + reason)	Primary vertical(s)	Contact route
			(https://inductiveautomation.com/resources/customerproject/enterprise-scada-gives-new-zealand-meat-producer-standardized-control)		
16	NLS Engineering / NLS Energy (now Ulteig)	Stoney Creek (Hamilton), ON, Canada	https://inductiveautomation.com/resources/casestudy/solar-provider-puts-more-than-30-sites-on-a-single-scada-platform : 30+ solar sites on one Ignition platform; NextDAS product on Perspective; ICC Discover Gallery 2019 (https://icc.inductiveautomation.com/discover-gallery-detail/2019/nls-engineering)	Utility-scale solar, wind, BESS, water	https://www.nlsengineering.com/ (root seen acquisition note https://ulteig.com/insight/nls-engineering-expanding-into-canada/)
17	Brown Engineers	Little Rock, AR, USA	https://icc.inductiveautomation.com/discover-gallery-detail/2022/browngineersllc : early Premier Integrator, Golden Firebrand winner, system-wide SCADA migrations at Central Arkansas Water and Fort Smith (Ignition + Bedrock at four utilities)	Water/wastewater, electric, mission critical	http://www.browngineers.net/contact/ ; Brown
18	RoviSys	USA (HQ not confirmed)	https://inductiveautomation.com/resources/icc/2024/integrator-panel-what-tech-and-trends-are-breaking-through : ICC 2024 panel (Anh-Tuan Tran, CPG Group Manager); #5 top-selling 2025 with multi-site deployments across regions	CPG, multi-site manufacturing, process	https://www.rovisys.com/ (not seen in res integrator directory)
19	Thermo Systems	USA (HQ not confirmed)	https://inductiveautomation.com/blog/get-to-know-the-new-premier-integrators : Premier; HMI standard for global data-center site overview/monitoring; renewed Premier (https://thermosystems.com/thermo-systems-renews-premier-integrator-status-with-inductive-automation/)	Data centers, energy, life sciences	https://thermosystems.com/ ; IA directory https://inductiveautomation.com/integrator
20	ATS Global	International (Enterprise Integrator)	https://inductiveautomation.com/news/inductive-automation-designates-ats-enterprise-integrator : Enterprise Integrator (250+ engineers, global); #4 top-selling 2025; Ignition page https://www.ats-global.com/ats-partner-products/ignition-scada/	Automotive, F&B, life sciences, metals & mining	https://www.ats-global.com/ats-partner-pr-scada/
21	SL Controls	Ireland	https://slcontrols.com/en/sl-controls-achieves-premier-integrator-status-for-digital-transformation-platform-ignition/ : Premier; ICC 2024 panel (Brian McLaughlin, Systems Architect)	Pharma, medical device	https://slcontrols.com/ (root seen)
22	Agidens	Belgium	https://www.agidens.com/expertises/turnkey-solutions/process-automation/gold-level-certified-integrator-for-ignition-8-1/ : Premier Integrator with Ignition practice	Process, pharma, chemicals, turnkey	https://www.agidens.com/ (root seen)
23	Barry-Wehmiller Design Group	USA (HQ not confirmed)	https://inductiveautomation.com/blog/lessons-from-the-topselling-ignition-integrators-of-2025 : #6 top-selling 2025, hyperscale data center focus	Data centers, manufacturing	IA integrator directory (https://inductiveautomation.com/resource/integrator)
24	E Tech Group	USA (HQ not confirmed)	https://etechgroup.com/capabilities/ignition-expertise/ : Premier Integrator, multi-industry Ignition deployments (https://etechgroup.com/blog/general/e-tech-group-achieves-ignition-premier-integrator-status/)	Life sciences, F&B, water, discrete	https://etechgroup.com/capabilities/ignition
25	Industrial Networking Solutions (INS)	USA (HQ not confirmed)	https://inductiveautomation.com/resources/icc/2024/integrator-panel-what-tech-and-trends-are-breaking-through : ICC 2024 panel (Mo Moore, Director of Software Services); search summary described INS as early adopters of MQTT Sparkplug and a repeat Tamaki collaborator (no dedicated URL captured)	Industrial networking, distributed SCADA	Via IA integrator directory; Mo Moore nan page

Borderline candidates (5)

1. Automated Control Concepts (ACC), NJ, USA. Top-10 direct Ignition seller, Premier (<https://www.prweb.com/releases/premier-integrator-automated-control-concepts-named-in-top-10-list-for-inductive-automation-sales-of-ignition-882769406.html>). No MQTT content found.
2. Kanoa (kanoa.ai), USA. Ignition-based MES built on Cirrus Link MQTT modules, Cirrus Link technology partner (<https://cirrus-link.com/technology-partners/kanoa/>); a product vendor with an integrator partner network rather than an SI. Contact <https://kanoa.ai/contact>.
3. NZ Controls, New Zealand. New Premier Integrator (<https://inductiveautomation.com/news/nz-controls-becomes-inductive-automation-premier-integrator>). No MQTT content found.
4. InflexionPoint, Multi-Dimensional Integration (MDI), Merit Controls, USA. Ranked #7, #8, #10 top-selling 2025 (<https://inductiveautomation.com/blog/lessons-from-the-topselling-ignition-integrators-of-2025>); MDI cites large-scale enterprise SCADA/MES. No MQTT content found.
5. Fairfield Control Systems, UK. Certified Ignition integrator (<https://www.fairfields.co.uk/fairfields-are-now-an-ignition-systems-integrator/>); smaller, no MQTT content found. Also Martin CSI and APCO (Premier, <https://inductiveautomation.com/blog/get-to-know-the-new-premier-integrators>) and Coderise IoT (Suzhou, China, #9 top seller) if you want more volume.

Notes for outreach

- Strongest SparkBridge fit (published Sparkplug/UNS work on distributed sites): DMC, Streamline Control, Corso, CSE ICON, Grantek, NFM, SAGE, Vertech, Flexware, 4IR, BIJC, NeoMatrix, Perceptive.
- Vertical coverage: O&G (Streamline, CSE ICON, NFM), water (SAGE, Perceptive, Brown, NeoMatrix, NLS), renewables (NLS, Vertech, Streamline), data centers (Hanley, Thermo, BWDG, NFM, BIJC, Vertech), distributed manufacturing (DMC, Flexware, Corso, ATS, RoviSys, Tamaki). Logistics-specific Ignition MQTT integrators did not surface in search.
- Non-NA: SAGE (AU), Tamaki (NZ), BIJC (UK), Hanley (IE/US), SL Controls (IE), Agidens (BE), ATS (global).

- Useful directory hubs seen: <https://inductiveautomation.com/resources/video/find-an-integrator>, <https://cirrus-link.com/technology-partners/>, ICC session index <https://inductiveautomation.com/resources/icc>.

First outreach email

Audience: the technical lead at one of the 25 target integrators (see [integrator-targets.md](#)). Voice: company voice, terse, no hype, no em dashes. One ask. Sent from admin@greenguard-usa.com only after Dan names the recipient. Nothing in this program is sent cold without his say.

Subject

What changes when an Ignition deployment reaches 40 sites?

Body

Hello {first name},

GreenGuard USA has been building and benchmarking an alternative architecture for distributed Ignition deployments that have outgrown backend fan-in, remote tag provider scaling and per-site history pipelines.

SparkBridge separates the estate into three roles. Edge publishes once at each site and buffers to disk through an outage. Provider carries the data under per-site identities and deny-by-default permissions. Host turns many sites into one namespace and fans it out to consumers on MQTT 5 shared subscriptions, so adding a historian or a warehouse changes nothing at the sites.

We documented a 40-site reference architecture covering WAN outages, timestamp preservation, consumer scale-out and standby recovery, with every figure tied to the machine it was measured on:

<https://mqtt.greenguard-usa.com/sparkbridge/compare> <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>

We are not asking you to replace your stack. We would like to show the architecture to someone on your technical team and find out where it does, and does not, solve problems you have met in the field. A free architecture review of one of your estates is the usual way that starts: <https://mqtt.greenguard-usa.com/sparkbridge/architecture-review>

Would a 30-minute call with one of your Ignition architects make sense in the next two weeks?

{Dan's signature} GreenGuard USA

Follow-up, seven days later, if no reply

Subject: Re: What changes when an Ignition deployment reaches 40 sites?

Hello {first name}, one line in case the first note was buried. The 40-site comparison is here:

<https://mqtt.greenguard-usa.com/sparkbridge/compare>. If it is not relevant to your practice, a one-word reply saves us both a second follow-up.

Goal of the conversation

Technical validation, objections, and design partners. Not a sale on the first call.

Outreach sequence (four touches)

Rules: one named person per firm at a time; Dan has approved the row; plain text; the firm-specific hook from the CSV in sentence one; UTM only on the two links; log every send in HubSpot the same day. Company voice, no hype, no em dashes. Never attach files to touch 1 or 2.

Touch 1, day 0: the architecture question

Subject: What changes when an Ignition deployment reaches 40 sites?

Hello {first name},

{One sentence on their published work from the CSV hook. Example: "Your ICC 2025 session with Sila on a UDT-driven UNS over HiveMQ is the kind of estate we built SparkBridge for."}

GreenGuard USA has been building and benchmarking an alternative architecture for distributed Ignition deployments that have outgrown backend fan-in, remote tag provider scaling and per-site history pipelines. Sites publish once, so applications, historians and warehouses are added centrally without revisiting every plant.

We are looking for a small group of experienced Ignition integrators to pressure-test the architecture against real distributed estates. If you have a topology worth discussing, we would like to compare approaches.

Would a 30-minute call with one of your Ignition architects make sense in the next two weeks?

{signature} GreenGuard USA

LinkedIn version (300 characters): "{first name}, your {hook, short}. We built SparkBridge around one question: what changes when an Ignition estate reaches 40 sites? Looking for a few integrators to pressure-test it against real topologies. Worth a 30-minute call?"

Touch 2, day 3 to 5: the specific observation

Subject: Re: What changes when an Ignition deployment reaches 40 sites?

Hello {first name}, one specific thing from {project / post / case study}: {one observation that shows you read it, e.g. "you moved 63 lift stations onto MQTT Transmission into a cloud-hosted gateway"}. That is exactly where the second consumer usually forces a second engine gateway, and where we think the architecture should change instead. Happy to show you how the same 63 sites draw on our side. {first name}, is there someone on the team who owns that design?

Touch 3, day 8 to 10: one technical asset

Pick exactly one. Do not send more than one.

Asset	Send when
mqtt.greenguard-usa.com/sparkbridge/compare	They run Cirrus or Gateway Network today
40-site reference architecture PDF (03-reference-architecture/)	They mentioned a specific large estate
mqtt.greenguard-usa.com/sparkbridge/benchmarks	They asked "what has actually been measured"
One finished article from 14-articles/	Their pain matches the title (WAN outage, adding a consumer, fleet ops, timestamps)

Subject: Re: What changes when an Ignition deployment reaches 40 sites?

Hello {first name}, no pitch, one page: {asset link}. It compares the traditional, module-MQTT and SparkBridge shapes at 40 sites, problem by problem, and says where the standard alone gets you the same cell. If it is not relevant to your practice, a one-word reply saves us both a further note.

Touch 4, only after relevance is established: the partner program

Subject: SparkBridge Architecture Partners

Hello {first name}, thanks for the conversation. We are forming a small group of Architecture Partners, ten seats, invitation only, and thought your team may be a strong fit. It starts with a free architecture review of an estate you are designing; the terms are here: mqtt.greenguard-usa.com/sparkbridge/partners. Would you like me to set up the review?

Handling replies

- Any reply: log in HubSpot, create the deal, draft the response within one business day.
- "Not relevant": close lost, cancel remaining touches, thank them in one line.
- "Send me more": send exactly one asset (touch 3 table) and ask for the call again.
- Technical question you cannot answer: "I will document that and have engineering answer it"; ask Dan or Claude the same day.
- Silence after touch 4: 60-day hold, then one new architecture question, not a repeat.

Discovery call script and call report

Before the call

Read the CSV row and the HubSpot record. Know the firm's published multi-site project. Have the Architecture page's Choose Your Starting Point table and the cheat sheet open. Set the frame in the first minute: "This is a discovery call, not a demo. I want to understand how you build distributed estates today."

Integrator speaks 70%, you 30%.

The seven questions, every call (answers into HubSpot before the call ends)

- 1. How many Ignition gateways, and which are redundant pairs? (shape)
- 2. How do sites reach the center today: Gateway Network, MQTT, VPN, nothing? (shape)
- 3. What is the busiest central box doing right now, CPU and memory? (the pain number)
- 4. What happens to history and alarms when a site's WAN drops?
- 5. How many consumers of site data exist, and who is asking for the next one?
- 6. Who builds a new site gateway and how long does it take?
- 7. Is there a broker already, and who owns it?

One vertical question

Vertical	Ask
Multi-site manufacturing	Which plant is the hub, and what breaks when it goes down?
Water, wastewater, utilities	What links reach the lift stations, and what does a month of data cost?
Oil and gas	Who audits your EFM and batch records, and how do you prove a row never changed?
Data centers, facilities	Is the UPS and the chiller in the same screen as the process?
Pharma, food, validated	How many weeks does the validation package take, and who types it?
Integrator (always)	How many estates over ten sites are you scoping this year?

As time allows

Growth: at what site count does the architecture get hard; have customers added a historian or warehouse later; have you revisited sites to add a consumer; what becomes painful after 20 sites. Competitive: what MQTT stack do you use, why, what do you wish it did differently. Validation: does SparkBridge solve a problem you actually meet; where does it not fit; what would stop you recommending it.

Closing the call

Any estate over ten sites, any HA question, any broker they already run, any 8.3-only feature: offer the free architecture review and name the intake link. Otherwise agree one concrete next step with a date. Never end on

"interested".

Call report (within 24 hours, HubSpot)

Section	Field	HubSpot location
Account	Company, contact, role, vertical	Company + contact records; out reach_segment = vertical
Architecture	Estimated site count, current shape (traditional / module MQTT / other), MQTT stack, central applications, existing consumers	Deal description, first block
Pain	Primary, secondary, business consequence, current workaround	Deal description, second block
Fit	Strong / moderate / weak / unknown, with the one sentence that justifies it	Deal description, third block
Objections	Exact wording, one per line	Deal note + append to 09-objection-library.md
Competitive	What they run and why	Deal note + append to 10-competitive-cheat-sheet.md
Next step	Follow-up / technical review / demo / architecture review / partner discussion / not a fit, with date and owner	Deal stage + next-activity date

Exact words matter more than summaries. "We can't have another engine gateway per consumer" is data; "scaling concerns" is not.

Objection library

Seeded 2026-09-07 from SE training module 11. Add every new objection in the prospect's exact words. Review weekly with Dan. Pattern for every answer: answer, then the caveat, then stop.

Categories: Product (missing feature, integration, performance, security), Commercial (pricing, licensing, support, procurement), Market (no demand, existing solution sufficient, estate too small), Trust (new vendor, no references, longevity).

#	Category	Objection (exact words)	Frequency	Severity	Response	Caveat we say out loud	Product action
1	Commercial	"What happens to the price when we add tags?"			Nothing. Per gateway, per product. Tags, devices and clients are never counted; the 40-site examples do not move when the estate gets busier.	Edge has a 250,000-tag budget per gateway; past that you add a gateway, not a fee.	none
2	Market	"We already run Cirrus."			Keep the transport. What we add sits after the broker: shared-subscription consumers, history and warehouse on source timestamps, enrollment, fleet view, generated policy. Start with Host on your broker.	Mixed-vendor edge and host pairing is by specification; we run your exact pairing on the bench before you buy.	interop bench record per pairing
3	Commercial	"Why pay for Provider when we have HiveMQ / Mosquitto?"			You do not have to. Edge and Host connect through any conformant broker; Provider is for the customer with no broker or who wants generated per-site policy without ACL files and with no client count.	HiveMQ and Mosquitto 2.1 are the brokers that have run on our bench.	none
4	Commercial	"What happens if support lapses?"			The software keeps running exactly as before. You stop receiving new versions and support until you renew.	None.	none
5	Trust	"Early access and pilot: is this production software?"			Edge, Host, Provider, Vault, Inject, Passage, FleetOps, Sentinel, Flow and the drivers are released and live-proven on both Ignition lines. SparkRecord 0.1.0 is early access, SparkValidate and GitOps are pilots, labeled that way on every page.	Buy the labeled ones for the problem they solve today, not for the roadmap.	none
6	Trust	"Do you have a reference customer?"			Not a published one yet. Every number we print was measured on named hardware with reproduction steps. The partner program exists to produce the first case study; Founding Partners get a joint one.	Never name a customer or imply one. If a reference is a precondition, say so to Dan and route to the design partner conversation.	first measured deployment

#	Category	Objection (exact words)	Frequency	Severity	Response	Caveat we say out loud	Product action
7	Product	"8.1 or 8.3?"			Both. Every product ships signed on both lines and is proven on a live 8.3 gateway before release. Same price.	The 8.3 line is newer; ask engineering about any 8.3-only feature before promising it.	none
8	Trust	"Are you Sparkplug certified?"			We have run the Eclipse TCK harness and the Sparkplug Compatible listing is in progress. We do not use the word certified until it is granted.	That is the caveat.	listing request
9	Trust	"The gateway warned me about the module certificate."			Modules are signed with the SparkBridge code-signing certificate, not one issued by Inductive Automation, so the gateway asks to trust it once. Every release bundle includes the signature and the live proof record.	An IA-issued certificate is on the list; no date.	IA signing
10	Market	"Isn't this just MQTT with a markup?"			MQTT is the transport and the right one. What you pay for is what remains after it: one namespace built from births, consumers that scale on the shared stream, replay on source timestamps, enrollment, fleet, policy. The compare page lists it row by row and says where the standard alone gets you the same cell.	At five sites, use what Ignition gives you; we say that on the compare page.	none
11	Commercial	"We want a subscription / monthly price."			We sell one-time licenses with optional 20% support. There is no monthly option.	Volume pricing by site count is Dan's call, not a discount the MBA offers.	none
12	Trust	"Who supports this? You are a small company."			Direct engineering support is what the 20% buys: the people who wrote the modules. Every module has a manual, a live proof record and a review record in its release bundle. Development is free so you can prove it on your own estate before you spend.	Say the team size honestly; do not inflate it.	none
13	Product	"Can Passage / IEC-104 / DNP3 write to equipment?"			No. Passage is read only; the utility drivers read only. Edge write-back is off by default and allowed only for listed devices and points.	BACnet and Logix drivers do write; say which is which.	none
14	Product	"Your site says 8.3 dropped symbolic S7, but IA lists a Siemens			Both are true. Symbolic addressing left 8.3.0 over licensing and returned as a separately purchased module. SparkS7 gives symbolic S7-	State the timeline and the price line; do not characterize IA's driver. Real	none

#	Category	Objection (exact words)	Frequency	Severity	Response	Caveat we say out loud	Product action
		Enhanced driver."			1200/1500 on 8.1 and 8.3 from one \$695 per-gateway license, verified in PLCSIM Advanced with a real TIA project on both lines.	hardware pilot offered, not claimed.	
15	Product	"Is enrollment mTLS?"			2.9.0 issues a certificate per enrolled site from the Host CA and writes the broker credential.	The broker does not yet require the client certificate at connect; 3.0.0 enforces it. Say "certificate issuance".	3.0.0

Competitive cheat sheet (living document)

Seeded 2026-09-07 from SE training module 08. Honest by rule: name what each does well. Where a module-MQTT cell reads the same as ours, that is Sparkplug doing its job for both; say it.

Competitor / approach	Best at	Where SparkBridge differs	Do not claim	Heard in the field (date, firm)
Ignition-native (Gateway Network, remote providers, redundancy, store-and-forward)	Everything at five sites; included with Ignition; familiar to every integrator	At 40 sites: hub saturation, backend per application family, history reconciled by hand, provisioning by visit. Compare page 5-to-50 table is the honest ramp	That it is wrong; at five sites we recommend it	
Cirrus Link (Transmission, Engine, Distributor, Recorder)	The reference MQTT modules, sold through IA, mature, known everywhere; removes polling and most fan-in	What remains after the transport: consumer scale-out (one wildcard subscriber per engine vs shared subscriptions, four consumers about 20x one), enrollment code vs typed credentials, FleetOps vs status pages, SparkID generated rules vs ACL files, history on source timestamps through replay (Vault), exactly-once warehouse (Inject). Recorder maps to SparkVault. Host can be primary host for Transmission edges by specification	That any specific Transmission pairing has run on our bench unless it has	
HiveMQ (enterprise broker)	Broker; our reference Provider HA pair is two HiveMQ CE nodes	Not a competitor to Edge or Host. If they run HiveMQ Enterprise, Provider is optional and SparkID emits HiveMQ credentials. Say "keep it"	Anything negative	
Cedalo (Mosquitto Pro / Management Center)	Mosquitto 2.1: broker topic aliases, SQLite persistence, dynamic security; Pro adds HA, UI, identity, audit, priced per concurrent client	Provider has no client count; SparkID generates the policy. Bench: Mosquitto 2.1.2 with persistence plugin, four tests passing. If they love Mosquitto, keep it; we sell Edge and Host on top	Pricing specifics beyond "per concurrent client"	
EMQX	Broker, large scale, cloud-hosted options	Same as HiveMQ: keep the broker, add Edge and Host	That we have run EMQX on the bench (we have not, as of 2026-09-07)	
Kymera and other driver vendors	InfluxDB, SNMP and other Ignition modules	SparkInflux and SparkSNMP compete line for line, per gateway, both Ignition lines; compete on the proof in the release bundle and the price line	Anything characterizing their product	
Custom architectures (scripts, per-site DB pipelines, in-house UNS)	Fits the one customer exactly	The second consumer, the 41st site and the person who leaves. Fleet, drift, policy, provisioning are where custom stops	That it does not work today	

One-minute answer to "we already run Cirrus": "Good. Sparkplug is the right transport and your sites already publish. What we add sits after the broker: one Host namespace with consumers that scale on the shared

stream, history and warehouse delivery on source timestamps with exactly-once landing, an enrollment code for site 41, and a fleet view with drift and certificates. Your Transmission edges can stay; we start with Host on your broker. Where we have not run your exact broker on the bench, we say so and run it before you buy."

Weekly report, week of {date}

Send Friday to Dan. One page. Evidence, not activity.

Metric	This week	Total to date	Target at day 90
New targets researched			50
Tier A targets			
Rows approved by Dan			30
Personalized first touches			30
Replies			8 to 12
Discovery calls			5 to 8
Architecture reviews delivered			3 to 5
Partner candidates (touch 4 sent)			
Partners signed			2 to 3
Production opportunities (score 25+)			1 to 2

Top objection this week (exact words, firm):

Best-performing message (which touch, which hook):

Most promising vertical and why:

Competitive insight:

Product gap or feature request (theme: Operate / Trust / Move Data / Scale):

Website confusion observed:

Recommended action for next week:

Blockers needing Dan (approvals, review calls, pricing questions):

Leadership review, day {30 | 60 | 90}

Claude compiles the numbers from HubSpot the day before. The MBA presents.

1. What did we learn? (three sentences, each backed by a count or a quote)
2. What changed since the last review?
3. What should we stop doing?
4. What should we double down on?
5. Is the friction a product issue or a messaging issue? Evidence for each.

Funnel to date: targets approved / touched / replied / called / reviewed / partner / pilot. Vertical ranking with the reply and call counts behind it. Objection top five with frequency and the response that worked. Competitive: what prospects run today, by count. Product feedback, sorted by theme (Operate / Trust / Move Data / Scale) and by class (critical gap / strategic / nice-to-have / off-architecture). Decisions requested from Dan.

[illegible]

ARCHITECTURE COMPARISON

WHAT CHANGES WHEN IGNITION GROWS PAST A FEW SITES

Five sites can be run almost any way. Forty cannot. This page compares three ways to build a distributed Ignition estate on the problems that appear with scale: fan-in, backend count, outages, timestamps, adding a consumer, and running the fleet afterwards. It is written for the engineer who will own the result, and it names what each approach does well.

THE THREE SHAPES

TRADITIONAL, MODULE-BASED MQTT, SPARKBRIDGE

The same forty sites, drawn three ways. Every box is a server somebody has to license, patch and watch.

TRADITIONAL DISTRIBUTED IGNITION

```

40 site gateways
  |
Gateway Network, remote tag providers
  |
Backend 1 ---- Backend 2 ---- Backend 3
  |           |           |
History      Applications Integrations
(per backend) (per backend) (per system)
  
```

[● ASK AN ENGINEER](#)

Every consumer is another Gateway Network connection into every site, or another backend that holds its own copy of the estate. Site count multiplies backend load; consumer count multiplies site load.

SPARKBRIDGE

```
40 x Edge (publish once, buffer to disk)
|
Provider HA pair (identity, policy, routing)
|
Host pool (one namespace, shared consumers)
|
+-----+-----+-----+-----+
Vault Record Passage Inject Your tools
```

Sites publish once. Consumers subscribe at Host. Adding a consumer, a site, or a Host changes one layer and leaves the other two alone.

Module-based MQTT for Ignition sits between the two: sites publish over Sparkplug and buffer through outages, which removes polling and most of the fan-in. What it leaves to custom engineering is everything after the transport: consumer scale-out, provisioning, configuration drift, fleet health and security policy. MQTT is the transport. Operating the estate is the problem.

PROBLEM BY PROBLEM

WHERE EACH ARCHITECTURE SPENDS YOUR ENGINEERING TIME

Rows are the problems that arrive with site count. Cells say who solves them: the product, or your team.



PROBLEM AT SCALE	TRADITIONAL DISTRIBUTED IGNITION	MODULE-BASED MQTT	SPARKBRIDGE
Gateway Network fan-in	One hub gateway carries every site connection and is the first thing to saturate	Removed: sites publish to a broker	Removed: sites publish to Provider; Host consumers share the stream
Remote tag provider scaling	Each remote provider is a live subscription load on the site	Removed at the site; consumer side is the engine's tag count	Host holds 500,000 wire metrics in under 4 GiB; SparkFlow streams past the tag ceiling
Backend server proliferation	A backend per application family, each with its own copy of the estate	One engine gateway; a second consumer usually means a second engine	One Host pool; consumers are modules on Host or subscribers of the same stream
Database connection sprawl	Each backend writes its own history and reads its own DB	History still written per gateway that historizes	SparkVault writes once, centrally, on source timestamps; SparkInject lands the warehouse copy exactly once
Historian duplication	Site history plus central history, reconciled by hand	Depends on where history is enabled	History is one consumer of the backbone. Adding it changes nothing at the sites



PROBLEM AT SCALE	TRADITIONAL DISTRIBUTED IGNITION	MODULE-BASED MQTT	SPARKBRIDGE
WAN outages	Gateway Network reconnects; history during the gap depends on site-side store-and-forward configuration	Store-and-forward at the site, replayed on reconnect	Disk-backed buffer at 2,697 messages/s durable, replayed in order, 400,000 of 400,000 measured, with the historical flag carried so trends backfill
Timestamp integrity	Arrival time unless every hop is configured to preserve source time	Source timestamps carried by Sparkplug	Source timestamps carried through buffer, replay, history and warehouse delivery. A timestamp that survives the trip
Consumer scaling	Another backend, another set of site connections	Another subscriber on the broker, one wildcard subscriber per engine	MQTT 5 shared subscriptions: four Host consumers deliver about 20x one. Publish once, add consumers when you need them
Site provisioning	Manual gateway build, GAN approval, provider setup per site	Broker URL, credentials and certificates typed at each site	One-time enrollment code: the site bootstraps identity over a signed channel, credentials written owner-only. Certificate issuance and fleet registration are on the roadmap
Configuration drift	Per-gateway settings, reviewed by visiting each one	Per-gateway module settings	Signed configuration bundles distributed fleet-wide; FleetOps shows drift per gateway; GitOps / Enterprise Governance (pilot) adds review and rollback

PROBLEM AT SCALE	TRADITIONAL DISTRIBUTED IGNITION	MODULE-BASED MQTT	SPARKBRIDGE
Fleet operations	Gateway status pages, one at a time	Broker client list plus gateway status pages	FleetOps: versions, drift, certificates, backups per gateway in one view; Host /fleet and Prometheus metrics
Security policy	Gateway Network certificates and per-gateway users	Broker ACL files written by hand	SparkID generates deny-by-default rules from intent; Provider enforces per-site identities; Sentinel scores the result

Where a module-based MQTT cell reads the same as SparkBridge, that is the Sparkplug standard doing its job for both. The difference is what remains for your team to build afterwards.

5 TO 50

SCALING FROM FIVE SITES TO FIFTY

What each step costs in infrastructure, not licenses. License arithmetic is on [pricing](#).

SITE COUNT	TRADITIONAL DISTRIBUTED IGNITION	SPARKBRIDGE
5	One hub gateway, remote tag providers, fine	5 Edge, 1 Provider, 1 Host. Also fine
20	Hub CPU and memory climb with every remote provider; history moves to a second backend	Same three roles. Host memory grows about 300 bytes per wire metric per consumer

SITE COUNT	TRADITIONAL DISTRIBUTED IGNITION	SPARKBRIDGE
40	Hub is the overloaded box; a second hub splits the estate and every consumer now needs both	Provider HA pair, Host active plus warm standby (takeover measured at 244 ms across 25 nodes). One namespace
50 and beyond	Backend per consumer family; integration projects per new tool	Add Host consumers on the shared stream; add SparkFlow when tag count is the ceiling. Sites are not touched

ADD A CONSUMER

ADDING A CONSUMER WITHOUT TOUCHING FORTY PLANTS

DAY ONE

40 sites --> Provider --> Host --> Existing application

A YEAR LATER

40 sites --> Provider --> Host --> Existing application
--> SparkVault (history)
--> SparkInject (warehouse)
--> Passage (OPC-UA clients)
--> Analytics and AI

No changes at the forty sites. Each new consumer is a subscription at Host or a module on it. In the traditional shape each of those four lines is a new backend or a new set of site connections, and usually a project with a budget line.



OUTAGE

WAN OUTAGE RECOVERY, MEASURED

MOMENT	WHAT SPARKBRIDGE DOES	MEASURED
Link drops	Edge diverts to a disk buffer; Host marks the site's points bad, not stale-but-current	2,697 messages/s durable with batched fsync
Link returns	Edge reconnects within a bounded backoff, rebirths, replays in order with source timestamps and the historical flag	Reconnect capped at 10 s; 400,000 of 400,000 replayed, zero loss, 40,035 updates/s on the reference rig
Host fails	Warm standby promotes after a debounce; edges never rebirth	244 ms from offline STATE to online STATE across 25 nodes
Afterwards	SparkVault backfills the real curve; SparkInject lands each row once	Zero duplicate or missing rows in the record lane soak

Environment and method for every figure: [Scale and Resilience Report](#).

RUNNING IT

FLEET OPERATIONS AFTER GO-LIVE

The questions that arrive in month two, and where the answer lives.



Which sites are offline or buffering?

Host /fleet view with store-and-forward and capacity headroom per edge; Prometheus metrics

Which versions are deployed and which drifted?

FleetOps inventory and drift, per gateway

Which certificates expire soon?

FleetOps expiring list; Sentinel finding with the fix

Can site A publish as site B?

No by construction: Provider binds each credential to one group and node. Sentinel proves it

How much headroom is left?

Host /capacity reports its own ceiling from benchmark evidence and the shard count it runs

Who changed what?

Signed configuration bundles today; review, rollback and audit trail in GitOps / Enterprise Governance (pilot)

COST

ARCHITECTURE COST, COMPARED HONESTLY

Ignition gateway licenses are the same in every shape and are not counted here. The comparison is what sits on top of them.

ITEM

TRADITIONAL DISTRIBUTED IGNITION

SPARKBRIDGE, 40 SITES

Central servers
to run

Hub plus one backend per
application family, commonly
three or more

Provider pair and Host pair, four small
servers, or two with both roles



ITEM	TRADITIONAL DISTRIBUTED IGNITION	SPARKBRIDGE, 40 SITES
Software on top of Ignition	None, or per-site historian licenses	\$40,300 on your own broker, \$49,780 with Provider HA and SparkVault, list, once. Detail
Adding site 41	Gateway build, GAN approval, provider setup, history setup	\$995 and an enrollment code
Adding a warehouse feed	An integration project per backend	SparkInject, \$2,995 per connector family, no site changes
Support	Your integrator's rate	Optional, 20% of list per year, direct to the engineers who built it
Metering	None	None. No per-tag, per-device or per-client counts

Module-based MQTT list prices are not reproduced here because the vendors publish them through resellers and they change. Bring the quote you hold and we will put it beside these numbers for your site count.

DRAW YOUR ESTATE BEFORE YOU BUY ANYTHING

Send the topology and we return a diagram, the failure domains and the sizing. It costs nothing and it is how most conversations with us start.

REQUEST AN ARCHITECTURE REVIEW

READ THE BENCHMARK REPORT



DOWNLOAD THE 40-SITE REFERENCE ARCHITECTURE (PDF)

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

[Benchmarks](#) · [Roadmap](#) · [Changelog](#) · [Pricing](#) · [Partners](#) · [Contact](#)



HOW THIS RUNS, FROM ONE PLANT TO THE WHOLE COMPANY

Every layout below is built from the same two products: SparkBridge Edge at each site, SparkBridge Host in the middle. Add only the modules you need. Every server keeps the systems and databases you already run. Nothing gets torn out to get here.

— EIGHT LAYOUTS SAME TWO PRODUCTS, EVERY TIME NOTHING RIPPED OUT ON DAY ONE

HOW TO READ A DRAWING

● SIGNAL ORDER

Green-edged enclosure is an Ignition server. The chips inside are the modules installed on it: **green chips are SparkBridge**, grey chips are Ignition's own.

Double rule = an MQTT broker. Dashed outline = something you already own.

Rows are tiers: field, site servers, brokers, central servers, consumers. The text on the bus between rows is what travels on it.

— SHIPPED AND MEASURED — SHIPPED, AWAITING FIELD PROOF ... ON THE BENCH

Numbers are from the quiet-box gate of 2026-08-15 and the standby takeover run on a 25-node estate. Where a shape has a limit it is printed under the drawing.

CHOOSE YOUR STARTING POINT

CHOOSE YOUR STARTING POINT

Eight drawings follow, and most readers need one. Find your situation, open that drawing, and skip the rest.

YOUR SITUATION

ARCHITECTURE

One machine or one site, watched from somewhere else

[01 the wire](#)

One plant with central operations

[02 the plant](#)

10 to 50 distributed sites

[03 the data center](#)

Multiple enterprise consumers off one namespace (historian, OPC-UA, cloud)

[04 the whole line](#)

History and digital twin

[05 the twin](#)

100 to 200+ sites

[06 the ultra large estate](#)

Fleet operations: config, identity, audit for one team

[07 running it](#)

One Ignition server today, no second site yet

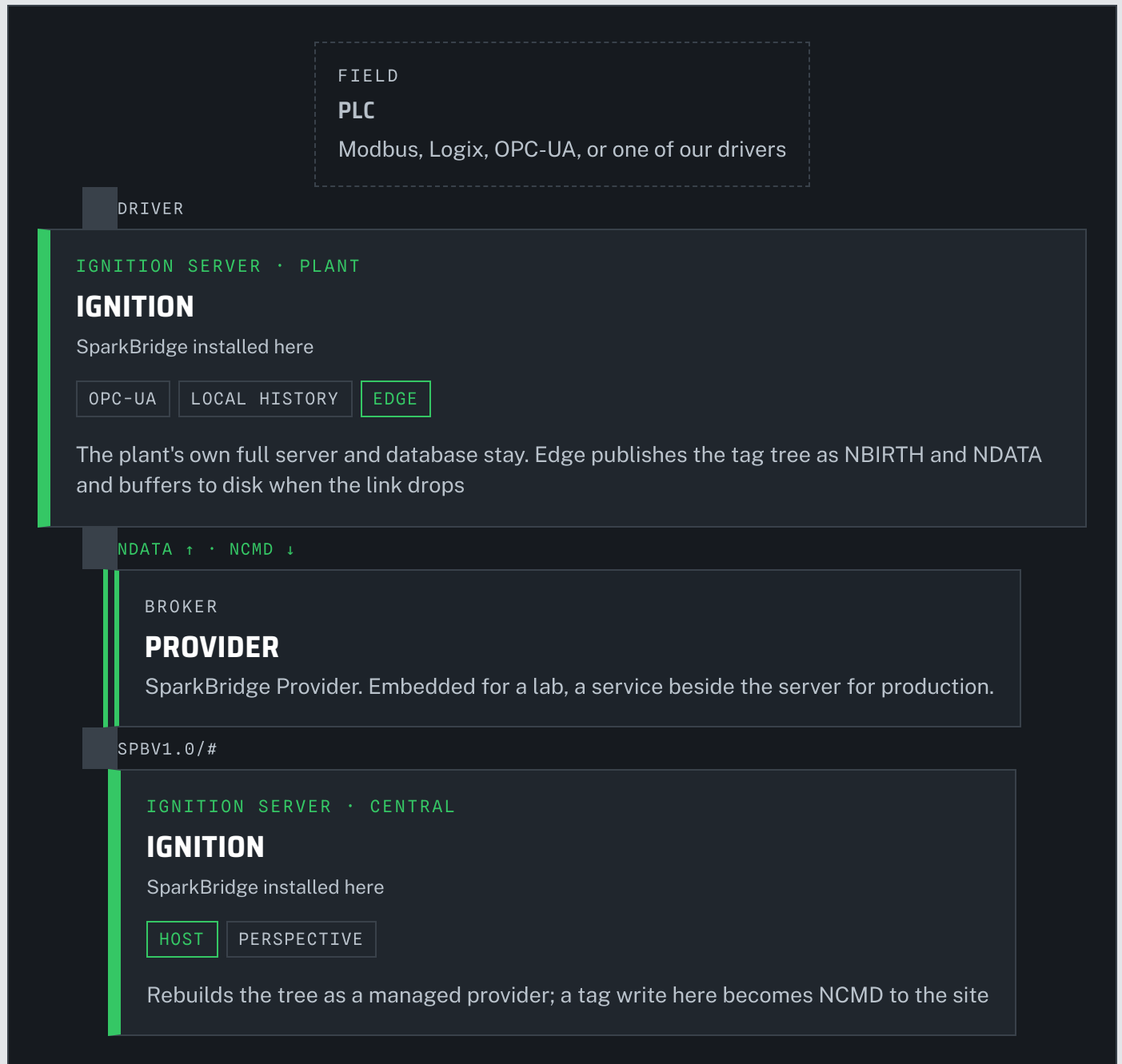
[08 one server](#)

The table at the end of the page, [Pick by What You Run Today](#), maps the same drawings from what is already installed.



ONE MACHINE, WATCHED FROM ANYWHERE

This is the smallest deployment, and every larger one on this page is built from it. A machine on your floor becomes a number your office can see. A change your office makes reaches the machine.



WHAT IT COSTS

SparkBridge Edge at the site, SparkBridge Host at the office. Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

WHAT YOU GET

A change on the floor shows up in your office in under a millisecond. Nothing is lost if the connection between them drops.

FOR YOUR ENGINEERS

SERVERS

Two Ignition servers: the plant's existing server (a full Ignition server with its own database, or Ignition Edge at a small onsite plant) and a central server

SPARKBRIDGE

Edge module on the plant server, Host module on the central server. Nothing else, and nothing removed.

ROUND TRIP

p50 0.44 ms, p99 0.97 ms end to end. A 4 h 11 min outage replayed with zero values lost, every one on the timestamp the PLC gave it.

LICENSE

SparkBridge Edge \$995 per site gateway, standalone. SparkBridge Host \$1,495 per Host gateway, standalone. Bought once.

[THE LIFE OF ONE CONNECTION](#) [DOWNLOAD](#)

02 • THE PLANT

KEEP THE SERVER YOU ALREADY PAID FOR

Your plant already runs its own server and its own database, with local screens and local alarms. SparkBridge does not replace any of that. It replaces the slow, fragile link that copies data to your office, and hands the plant the whole company's view back.

TODAY • HISTORY WRITTEN TWICE, TAGS FETCHED OVER THE GATEWAY NETWORK

PLANT
ONE SITE

PLANT IGNITION SERVER

PLANT DATABASE

IGNITION

- DRIVERS
- PERSPECTIVE
- ALARMS
- HISTORY SPLITTER

Full server. Splitter writes every value to the local database and to the central database.

SQL SERVER · POSTGRES

Local history, alarm journal, local reports

SQL ROWS OVER THE WAN · GAN TAG PROVIDER

HOST

CENTRAL DATABASE

SQL · SQLTH_PARTITIONS

One connection and one credential per plant. Every plant's rows land here on the central clock.

CENTRAL IGNITION SERVER

IGNITION

- GAN REMOTE PROVIDERS
- PERSPECTIVE

Reads live tags over the Gateway Network, history from the central database

WITH SPARKBRIDGE · ONE WIRE UP, THE PLANT KEEPS ITS DATABASE

PLANT ONE SITE

PLANT IGNITION SERVER

IGNITION

SparkBridge installed here

- DRIVERS
- PERSPECTIVE
- ALARMS
- LOCAL HISTORY
- EDGE

HOST · OPTIONAL

Same server, same screens. Edge publishes the tree; the local history provider keeps writing the plant database. Host, if installed, brings the other plants in as a local provider.

PLANT DATABASE

UNCHANGED

Local history, alarm journal, local reports. Still here when the WAN is not.

NBIRTH · NDATA ↑ · NCMD ↓

BROKER

BROKER

PROVIDER PAIR



SparkBridge Provider, redundant pair. One plant credential, scoped to its own group by SparkID. No database credential leaves the plant.

SPBV1.0/# · SQL ↓ ONCE

HOST

CENTRAL IGNITION SERVER

IGNITION

SparkBridge installed here

HOST

SPARKVAULT

PERSPECTIVE

Live tree from Host; SparkVault writes central history once, on the plant's clock, and serves Perspective trends as a tag-history provider

CENTRAL DATABASE

TIMESCALEDDB

One writer. Every plant in one open hypertable, original timestamps, `is_historical` marked on replayed rows.

WHAT STAYS

Your plant server, its screens, its alarms, its database and its local reports. Nothing gets ripped out on day one.

WHAT CHANGES

One reliable wire replaces the slow WAN link to your office, so live data and history arrive from the same source instead of two.

WHAT IT COSTS

Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

WHAT STAYS

The plant Ignition server, its drivers, Perspective, alarms, the plant database and the local history provider. The splitter can stay too, writing only its local leg, or be swapped for a plain local provider once the remote leg is off.

WHAT CHANGES

The remote leg of the history splitter and the Gateway Network remote provider are both replaced by one Sparkplug wire. Live tags and central history now come from the same stream. ●

WHY THE WIRE WINS

The splitter ships SQL rows: one database connection and credential per plant, rows stamped on arrival, and a central schema of sqlth_ partitions nobody wants to query directly. Edge ships values with the timestamp the PLC gave them, buffers to disk through an outage, and SparkVault writes them once into one open table. A 4 h 11 min outage replayed with zero values lost, every one on its original timestamp.

THE PLANT GETS MORE BACK

Install Host on the plant server too and the rest of the estate shows up there as a managed provider: sibling plants, SparkCalc results, corporate setpoints. A plant screen can show the neighbour's line with no Gateway Network link.

CUTOVER ORDER

Install Edge, leave the splitter running both legs, let SparkVault fill in parallel for one history partition, compare, then switch the remote leg off. Nothing is ripped out on day one; the plant's own history is never at risk because it never moves.

SMALL ONSITE PLANTS

A skid, a lift station, a packaging cell with no database of its own runs Ignition Edge instead. Same Edge module, same wire; its history lives at central only, with the Edge server's own disk buffer covering an outage.

LICENSE

Edge \$995 per site gateway, standalone, on the plant server. Host \$1,495 per Host gateway, standalone, if you also want the estate view there, SparkCalc included. On the Host server: Host \$1,495, plus SparkVault \$3,995 and Passage \$1,995.

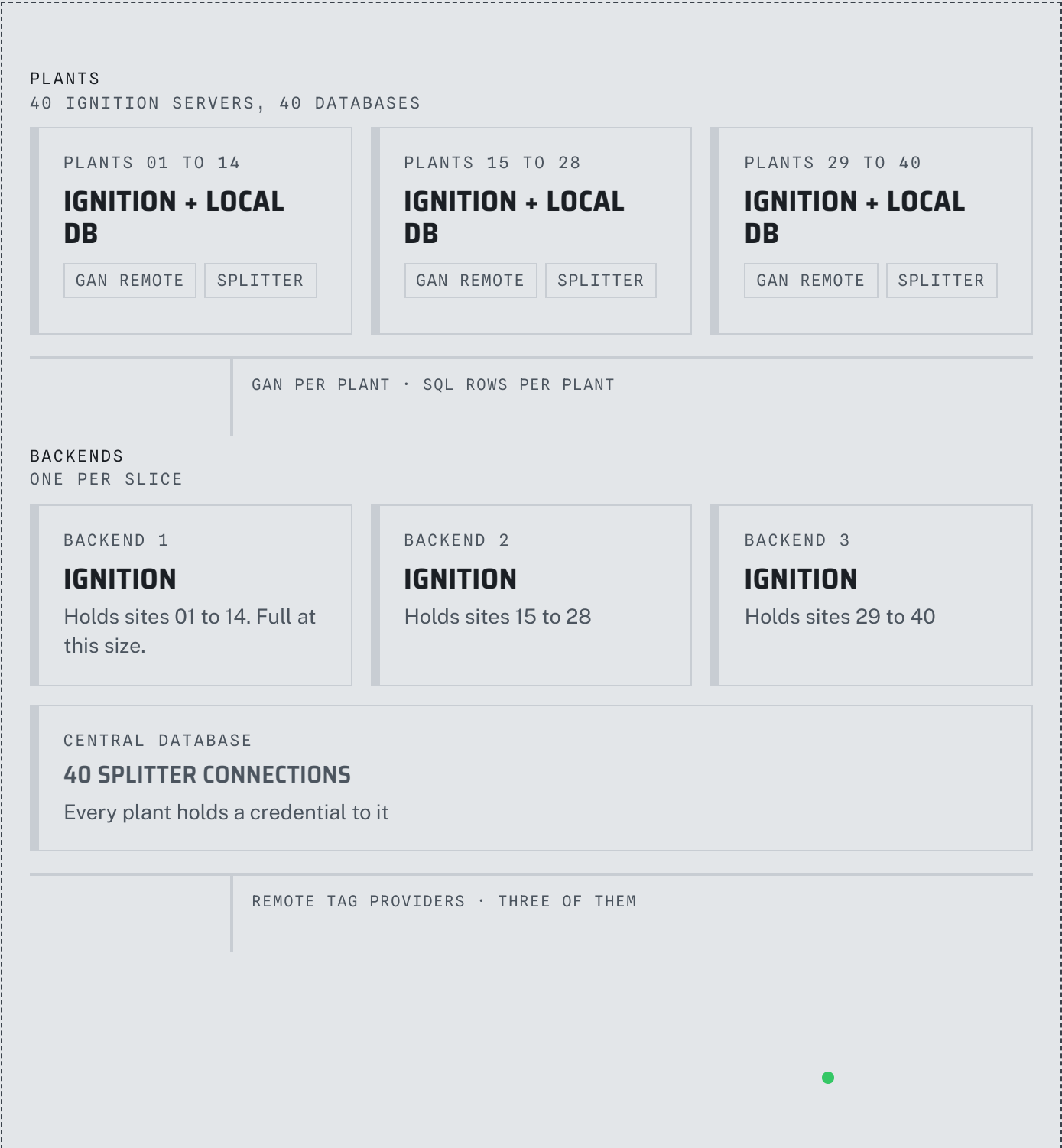
-
-
- Honest limits: reports written against the central sqlth_ tables have to be re-pointed at SparkVault's table or the Timescale hypertable. The plant's own reports do not change.
- The alarm journal stays in the plant database; alarm state travels as tags, the journal rows do not. Host on a plant server is a full Host license.
-
-

SPARKVAULT STORE AND FORWARD, MEASURED

ONE VIEW OF EVERY SITE, NOT A DIFFERENT SCREEN FOR EACH

Large operators run several separate servers because one cannot hold every site, and every team has to know which server owns which site. That slows down every decision and every new hire. SparkBridge turns those separate servers into one view, with one Host gateway that grows as you add sites.

TODAY · PER-BACKEND FAN-IN OVER THE GATEWAY NETWORK



APPS

APPLICATION SERVERS

IGNITION - PERSPECTIVE

Every screen has to know which backend owns which site. Add a site, touch every consumer.

WITH SPARKBRIDGE · ONE NAMESPACE, A POOL OF HOSTS

PLANTS

40 IGNITION SERVERS, 40 DATABASES

PLANTS 01 TO 40

IGNITION + LOCAL DB

SparkBridge installed here

DRIVERS

LOCAL HISTORY

EDGE

Each plant keeps its database and publishes its own group; the splitter's remote leg and the GAN link both retire

SMALL ONSITE PLANTS

IGNITION EDGE

SparkBridge installed here

EDGE

Skids and cells with no database; history lives at central

NBIRTH · NDATA · QOS 1

BROKERS

DATA CENTER

BROKER A

PROVIDER

SparkBridge Provider. Retains STATE and birth certificates

BROKER B

PROVIDER

SparkBridge Provider, paired with A; edges fail over between them

\$SHARE/HOSTS/... · SPBV1.0/#

HOST

IGNITION HOST POOL

CENTRAL 1 · PRIMARY

IGNITION

SparkBridge installed here

CENTRAL 2 · STANDBY

IGNITION

SparkBridge installed here

HOST · 4 CONSUMERS

SPARKVAULT

Owens the host id; publishes STATE, sends commands. SparkVault writes central history once.

HOST · STANDBY

Same state, same id, commands fenced until it claims

ONE MANAGED PROVIDER · EVERY SITE

APPS

APPLICATION SERVERS

IGNITION · PERSPECTIVE

One remote provider, one tree, every plant.
Add a plant and it appears.

CENTRAL DATABASE

TIMESCALEDDB

One writer, no plant credentials, original timestamps

WHAT YOU GET

Forty plants, one screen. Add a plant and it shows up automatically, no rework for the team watching it.

UPTIME

If the primary Host gateway fails, a standby takes over in about a quarter of a second. Nobody at any site notices.

WHAT IT COSTS

Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

SERVERS

40 plant Ignition servers with their 40 databases, untouched; 2 brokers; 2 central Ignition servers in a host pool; one central database; your existing application servers

SPARKBRIDGE

Edge on every plant server; Host and SparkVault on every server in the central pool



CAPACITY

At least 340,000 updates/s sustained through to visible tags on one central server; 480,000 offered without bending. Four shared consumers ingest about 20 times what one does. 500,000 metrics resident in under 4 GiB.

TAKEOVER

Standby claims the host id in 244 ms end to end, 5.7 ms mechanical, with zero rebirths. The sites never notice.

SIDEWAYS

Add a central server to the pool for more consumers, or partition the namespace by group across servers with ordering preserved. Both are live changes, no rebirth wave.

LICENSE

Edge \$995 per site gateway, standalone. Host \$1,495 per pool gateway, standalone.

-
-
- Honest limits: standby caps you at two writers for at most the claim-stale window, not zero. Shared consumers trade per-edge ordering for throughput; partition by group if
- ordering matters. Run the standby against one broker server, since convergence is per
- broker.
-
-

PERFORMANCE QUANTIFIED STANDBY AND SCALE-OUT

04 • THE WHOLE LINE

EVERY TOOL YOUR COMPANY NEEDS, OFF ONE NAMESPACE

This is the data center from section 03 with every optional module turned on: your history, your security audit, your reporting tools, your cloud warehouse, all reading the same feed instead of five.

FIELD

MODERN
PLCS

LEGACY
RTUS · 505 RACKS

REMOTE



Logix, Modbus,
OPC-UA

IEC-104, DNP3, TI
505

WELLHEADS · LIFT
STATIONS

Cellular and satellite

DRIVERS

SITES

PLANT AND EDGE SERVERS

PLANTS

IGNITION + LOCAL DB

SparkBridge installed here

OPC-UA

LOCAL HISTORY

EDGE

Full servers, as in 03

SMALL ONSITE PLANTS

IGNITION EDGE

SparkBridge installed here

EDGE

UTILITY SITES

IGNITION EDGE

SparkBridge installed here

IEC-104

DNP3

TI 505

EDGE

METERED SITES

IGNITION EDGE

SparkBridge installed here

MODBUS

EDGE

SPARKLINK

NBIRTH · NDATA ↑ · NCMD ↓ · SIGNED CONFIG ↓

BROKERS

DATA CENTER

BROKER A

PROVIDER

SparkBridge Provider.
Rules from SparkID: only
the host id may command,
each site scoped to its
own group

BROKER B

PROVIDER

SparkBridge Provider,
paired with A

NOT A GATEWAY

SPARKID

Generates credentials.xml,
extension config, NATS
ACLs, standby bindings;
rotates with overlap

SPBV1.0/# · \$SHARE · STATE · CLAIM

HOST

IGNITION HOST POOL

CENTRAL 1 · PRIMARY

IGNITION



SparkBridge installed here

HOST · 4 CONSUMERS

SPARKVAULT

SPARKCALC

PASSAGE

SPARKINJECT

SPARKFLOW

FLEETOPS CONSOLE

PERSPECTIVE

One managed provider with every site. Vault stores it, Calc derives from it under its own group, Passage serves it as OPC-UA, Inject and Flow send it out.

CENTRAL 2 · STANDBY

IGNITION

SparkBridge installed here

HOST · STANDBY

SPARKVAULT

Same state; takes the id in 244 ms

SQL · OPC-UA · NATS · STREAMING

CONSUMERS

VIA SPARKVAULT

TIMESCALEDB

Open history on the plant's clock; replay any day

VIA PASSAGE

SCADA · MES · DCS

Browse the namespace over OPC-UA

VIA SPARKFLOW

NATS · THE COMPANY

Subjects, last value, model bucket

VIA SPARKINJECT

SNOWFLAKE

Streaming, exactly once

CLAMPED ACROSS ALL OF IT

SENTINEL

Audits the brokers and the host from outside with read-only probes, reports what is exposed, and feeds the Compliance binder. Never in the data path.

WHAT YOU GET

One license for every site, one office server reporting on security, cost, history and cloud delivery. No custom integration project to reach any of it.

WHAT IT COSTS

Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts

the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

SERVERS

Plant Ignition servers with their local databases, Ignition Edge at small onsite plants, 2 brokers, 2 central Ignition servers, a SparkID workstation, your database, NATS and warehouse

ON EACH PLANT OR EDGE SERVER

SparkBridge Edge, plus our IEC-104 / DNP3 / TI 505 drivers where the rack needs them, plus SparkLink where the link is metered

ON EACH CENTRAL SERVER

SparkBridge Host, SparkVault, SparkCalc, Passage, SparkInject, SparkFlow, FleetOps. All are Ignition modules on the same server; none needs a box of its own.

OFF THE SERVERS

SparkID and Sentinel are tools you run from a workstation against the estate; nothing installs on a gateway for them

LICENSE

Per site gateway: Edge \$995 standalone, SparkLink included. Per Host gateway: Host \$1,495 standalone, SparkCalc and SparkID included. Other modules on Host: Passage \$1,995, Sentinel and FleetOps \$4,995 each, SparkInject \$2,995 per connector family, SparkVault \$3,995. Provider \$1,995 per gateway, managed broker and policy layer, works with Edge and Host or with any conformant Sparkplug host. SparkFlow \$995, SparkSNMP \$695 and drivers \$695 each carry their own license. Per gateway, bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr optional.

[THE PRODUCT LINE](#) [EDGE AND HOST PRICING](#)

Both your history and your live model live on the central server. Neither touches a site. History is kept on your own clock, so an outage never leaves a hole and any past day can be replayed. The live model is the same data, always current, for the tools that need it.

BROKERS

LIVE ESTATE

PROVIDER PAIR

SparkBridge Provider. Every site's NBIRTH, NDATA and templates

REPLAY NAMESPACE

REPLAY/<GROUP>

SparkVault re-publishes history here as its own edge, never under a site's id

SPBV1.0/# IN · REPLAY/ OUT

HOST

IGNITION HOST POOL

CENTRAL 1 · PRIMARY

IGNITION

SparkBridge installed here

HOST

SPARKVAULT

SPARKFLOW

PASSAGE

SPARKCALC

TAG HISTORY

PERSPECTIVE

SparkVault is also an Ignition tag-history provider, so Perspective trends read estate history with no new tooling. SparkCalc backfills derived values over that history as historical, not live.

SQL ↓ · KV PUTS · OPC-UA

THE TWIN

THREE FACES

HISTORY

TIMESCALEDDB

Hypertable, original timestamps, is_historical preserved through store and forward. Replay any day at 1x or faster.

LIVE MODEL

NATS · UNS-LAST · UNS-MODEL

Last value per metric, plus the structure: templates and birth properties per node and device, read at GET /model

STRUCTURED FACE

OPC-UA ADDRESS SPACE

group / edge / device / metric, live subscriptions, allow-listed writes down

SIMULATION, MES, ANALYTICS, AI

USES

REPLAY

WALK AN INCIDENT

Re-run the hour it happened through the real screens and alarms

SIMULATE

TEST BEFORE CUTOVER

Play history at speed against a candidate host or a new dashboard

MODEL

COMPANY SYSTEMS

Subscribe to the structure, not just the numbers

WHAT YOU GET

Every site's history, kept on your own clock, any past day replayable. A live model of the plant for the tools that want one, always current.

WHAT IT COSTS

Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

SERVERS

The central pool from 02 and 03. No new Ignition server; the database and NATS are yours.

HISTORIAN

SparkVault on the central server; each plant's own database keeps its local history exactly as before. History for every site in one open table, on the plant's clock. 40,035 updates/s replay on loopback; a 4 h 11 min outage backfilled with zero loss.

TWIN

SparkFlow's last-value and model buckets plus Passage's OPC-UA address space plus SparkVault history. Nothing is copied into a second tool; each face reads the same namespace.

DISCIPLINE

Replayed and computed values always publish under their own group with their own sequence, so a simulation or a derived number can never be mistaken for a live measurement.

LICENSE

SparkVault \$3,995 on the central server; Passage \$1,995; SparkCalc included with Host. SparkFlow \$995 carries its own license. SparkBridge Provider handles the broker layer for this drawing; any conformant Sparkplug broker also works.

TWO HUNDRED PLANTS, ONE NAMESPACE, NO CEILING

Past a certain size, no single server can hold everything. SparkFlow, its own license, is how the largest companies scale past that ceiling: each region keeps what its own operators watch, and the whole company still reads one namespace with nothing left out.

PLANTS

200 IGNITION SERVERS

REGION EAST • PLANTS 001 TO 070

IGNITION

SparkBridge installed here

DRIVERS

LOCAL HISTORY

EDGE

Full plant servers with their own databases, as in 03

REGION CENTRAL • PLANTS 071 TO 140

IGNITION

SparkBridge installed here

DRIVERS

LOCAL HISTORY

EDGE

REGION WEST • PLANTS 141 TO 200

IGNITION EDGE • IGNITION

SparkBridge installed here

DRIVERS

EDGE

Mixed: full plants and small Edge sites

NBIRTH • NDATA ↑ • NCMD ↓ • PER REGION

BROKERS

OT TIER, PER REGION

EAST

PROVIDER PAIR

SparkBridge Provider.
Sparkplug, MQTT 5,

HOST

PROVIDER PAIR

SparkBridge Provider.
Sparkplug, MQTT 5,

WEST

PROVIDER PAIR

SparkBridge Provider.
Sparkplug, MQTT 5,

\$share. Rules from
SparkID.

\$share. Rules from
SparkID.

\$share. Rules from
SparkID.

\$SHARE/HOSTS · SPBV1.0/#

REGIONAL IGNITION HOST POOLS

EAST POOL · PRIMARY + STANDBY

IGNITION

SparkBridge installed here

HOST · 4 CONSUMERS

SPARKFLOW · SEPARATE LICENSE

SPARKCALC

PERSPECTIVE

Tags: the ~150,000 points East operators watch. Stream: all 1.8 million points East decodes.

CENTRAL POOL

IGNITION

SparkBridge installed here

HOST

SPARKFLOW · SEPARATE LICENSE

Same shape. Namespace partitioned by group; ordering kept per edge.

WEST POOL

IGNITION

SparkBridge installed here

HOST

SPARKFLOW · SEPARATE LICENSE

UNS.<GROUP>.<EDGE>.<DEVICE>.<METRIC> · EXACTLY ONCE

BACKBONE NATS, APP TIER

PER REGION

NATS · JETSTREAM

Stream TELEMETRY on uns.>, KV uns-last (value, ts, quality per point), KV uns-model (structure from birth certificates). Each region owns its JetStream domain and keeps acking with the hub down.

COMPANY

NATS HUB

Sources TELEMETRY_EAST, TELEMETRY_CENTRAL, TELEMETRY_WEST from the regions over TLS leaf links. Regions dial out; the hub never dials in. Msg-Id survives the mirror, so a hub-side historian dedups exactly like a regional one.

DURABLE CONSUMERS · KV READS · UNS.CMD ↓

COMPANY

CONSUMERS

CORPORATE HISTORIAN

IGNITION

SparkBridge installed here

SPARKVAULT · JETSTREAM INGEST

SPARKINJECT

PERSPECTIVE

Vault drains the hub stream on a durable consumer into one Timescale history for all 200 plants; Inject streams the same to the warehouse

CORPORATE PERSPECTIVE

IGNITION · DISPATCHER

SparkBridge installed here

SPARKFLOW DISPATCHER

PERSPECTIVE

Screens bind to points on demand from uns-
last with no tag behind them. On the bench:
the SDK hook is stubbed, the dispatcher core
is written.

DIRECT

ANALYTICS · AI · MES

Subscribe to a subject filter, open with the last value of everything, replay any window in order

WHAT YOU GET

Every plant in the company in one view, with no ceiling on how large that view can grow.

MEASURED

7.25 million messages delivered across a real network, in order, exactly once, zero lost through a physical link cut.

WHAT IT COSTS

SparkFlow carries its own license, per gateway, bought once and kept.

FOR YOUR ENGINEERS

SERVERS

200 plant Ignition servers, 3 broker pairs, 3 regional host pools of 2, one corporate historian server, one corporate Perspective server, 4 NATS servers. The plants are the drawing in 03 repeated.

WHERE THE CEILING GOES

No Ignition server anywhere holds five million tags. Each regional pool holds the slice its operators watch as tags and streams everything it decodes. The company namespace lives in NATS: one subject per point, a last value and a model for each, no tag license behind any of it.

MEASURED

7.25 million messages across a real network at 48,000/s, 7.25 million received, in order, exactly once. A 60 s physical link cut at a leaf: 447

held, 447 reconciled at the hub, zero lost.

COMMANDS

Routed rather than mirrored. `uns.cmd.<group>.<edge>...` request and reply crosses the leaf link natively; only the region's bridge subscribes to its own command subtree, and only the issuers SparkID names may publish there. A region that cannot confirm a write times the requester out rather than reporting success.

TIERS, HONESTLY

SparkBridge Provider stays the OT tier: Sparkplug conformance, MQTT 5, shared subscriptions, TCK, all measured there; any conformant Sparkplug broker also works. NATS sits north of it as the backbone. SparkFlow is the certified translation between the two, not a replacement for either.

LICENSE

SparkFlow \$995 per Host gateway running it: three regional pools and the corporate historian, each its own license, on top of anything else on that server. Edge \$995 per plant, Host \$1,495 per pool server. NATS is open source and yours.

-
-
- Honest limits: a single region, one NATS domain, one host pool, SparkFlow, SparkVault
- ingest, the leaf partition, is shipped and gated. The multi-region hub with per-region
- accounts is a design built on those gates. Cross-domain sourcing across an account
- boundary is the one spike we still owe before a customer install, and that's why the hub
- is drawn dashed. The Perspective dispatcher is on the bench.
-
-

[SPARKFLOW](#) [SPARKVAULT](#) [SPARKID](#)

FORTY SITES, RUN AND PROVEN FROM ONE CHAIR

This is what running the estate from section 03 looks like day to day. One team pushes a change to every site at once. Access is generated instead of typed by hand. An outside audit checks the whole thing on demand.



OPS WORKSTATION

NOT A GATEWAY

FLEET PUBLISHER

P-256 signing key; site documents in git; publishes signed, sequenced, retained config

NOT A GATEWAY

SPARKID

One declaration of who may read and who may command; emits broker rules, NATS ACLs, standby bindings; dry-run diff against the live broker

NOT A GATEWAY

SENTINEL

Anonymous connect, command-topic reachability, passive capture. Reports and scores; never remediates.

[CONFIG](#) ↓ · [RULES](#) ↓ · [PROBES](#)

BROKERS

BROKER PAIR

PROVIDER

SparkBridge Provider. sparkbridge/config/<group>/<edge> retained; credentials and topic scoping from SparkID

[CONFIG](#) ↓ · [NODE INFO IN NBIRTH](#) ↑

SITES AND CENTRAL IGNITION SERVERS

SITES 01 TO 40

IGNITION EDGE

SparkBridge installed here

[EDGE](#) · [TRUSTED KEY](#)

Verifies, refuses replay below its sequence floor, applies without restart

CENTRAL POOL

IGNITION

SparkBridge installed here

[HOST](#) [FLEETOPS CONSOLE](#)

Module version, config version, buffered bytes and drops for every site on one grid; drift and cert expiry ledger

WHAT YOU GET

A change reaches every site at once, from one place, verified before it applies. Access is generated, not typed by hand, so nobody has to remember who can do what.

WHAT IT COSTS

Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts

the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

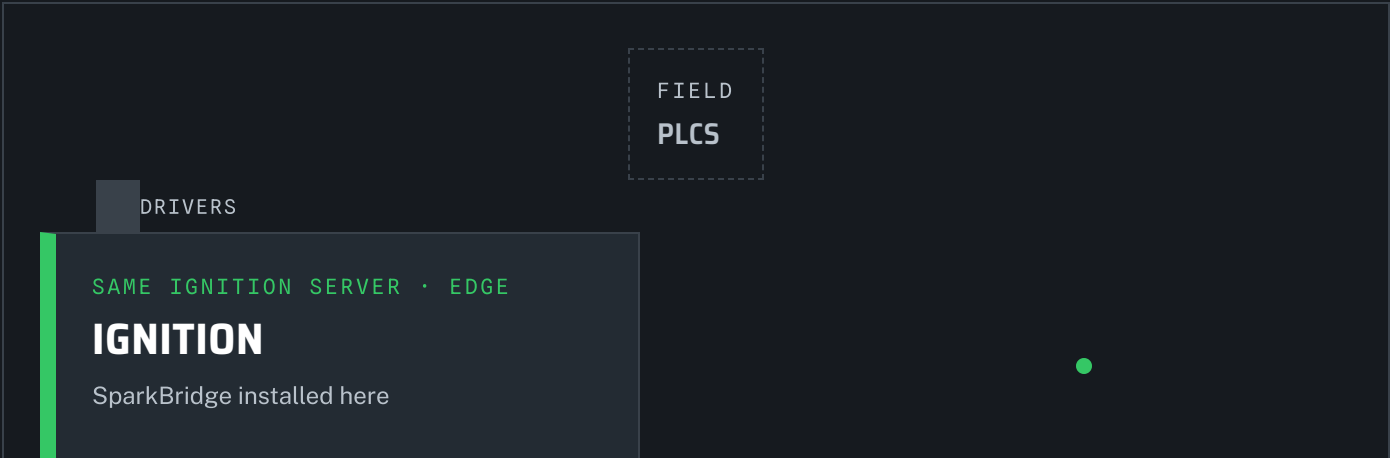
FOR YOUR ENGINEERS

SERVERS	No new Ignition servers. One ops workstation runs the three tools.
TRUST	A site with no trusted key subscribes to nothing. The signing key is a root credential for the estate; it lives on the workstation, never on a gateway.
PROOF	A Sentinel run becomes an auditor-ready binder mapped to NERC CIP and IEC 62443 through the Compliance add-on
LICENSE	FleetOps \$4,995, SparkID included with Host, Sentinel \$4,995; Sentinel compliance pack \$2,495 add-on.
FLEETOPS SPARKID SENTINEL	

08 • THE SMALL END

ONE PLANT, ONE SERVER, ROOM TO GROW

Everything above also fits on a single server for a single plant. Start here. The day you add a second site, nothing about the shape changes: it becomes site one of the data center.



DRIVERS

LOCAL HISTORY

EDGE

Publishes plant data to the required Provider broker. The physical server stays one enclosure.

SPARKPLUG

BROKER LAYER

PROVIDER

Embedded Moquette for a single node, or a HiveMQ CE companion service for MQTT 5 and shared subscriptions.

LOCAL SPARKPLUG

SAME IGNITION SERVER • HOST

IGNITION

SparkBridge installed here

HOST

SPARKVAULT

PASSAGE

SPARKCALC

PERSPECTIVE

Consumes through Provider, keeps history, serves OPC-UA and computes in one enclosure. Point it at a central broker later and it becomes site 01 of the data center.

SQL • OPC-UA

HISTORY

TIMESCALEDB

OTHER SYSTEMS

OPC-UA CLIENTS

WHAT YOU GET

One server does everything a plant needs: moves the data, keeps the history, serves it out. It grows into the data center in section 03 with no rework.



WHAT IT COSTS

Per gateway. Bought once and kept. Every paid module runs on Ignition's two-hour trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds. Support 20%/yr, optional.

FOR YOUR ENGINEERS

SERVERS

One Ignition server

BROKER

Embedded through the Provider module for a lab; SparkBridge Provider as a service beside the server for production, which is also what makes the later move to 03 a configuration change. Any conformant Sparkplug broker also works.

LICENSE

Edge \$995 per site gateway, standalone, SparkLink included. Add Host \$1,495 and SparkVault \$3,995 on this server only if it is also acting as the Host gateway with its own database.

[PRICING](#)

[DOWNLOAD](#)

WHICH DRAWING IS MINE

PICK BY WHAT YOU RUN TODAY

YOU RUN TODAY

YOUR DRAWING

WHAT CHANGES

One Ignition server, one plant

08 one server

Install the modules; nothing else moves

A site server and a central server on the Gateway Network

01 the wire

Edge on the site, Host at the center, a broker between; the GAN link retires

Full plant servers with local databases and a history splitter

02 the plant

Plant server and database stay; the splitter's remote leg becomes the

to a central database

Sparkplug wire and SparkVault writes central history

Several Ignition backends because one cannot hold every site

03 the data center

Backends become a host pool behind one broker pair; consumers read one tree

The above, plus a historian, an OPC-UA face or a cloud warehouse to feed

04 the whole line and 05 the twin

Modules on the central pool; no new servers

Hundreds of plants and millions of points, more than any Ignition server can hold as tags

06 the ultra large estate

Regional host pools hold what operators watch; SparkFlow streams everything onto NATS; the company reads one namespace

Forty sites and one team to run them

07 running it

An ops workstation, signed config, generated identity, outside audit

What is not on this page. No drawing mixes another vendor's Sparkplug modules with ours. That pairing has never run on our bench, so we are not going to imply compatibility we have not measured.

DRAW YOURS WITH US

Start from the written [40-site reference architecture \(PDF\)](#) or [request a review of your own estate](#).

Send us the server list, the site count, and what has to read the data. You get back one of these drawings with your server names on it. Try it on one site: every paid module runs on Ignition's two-hour trial until a key file names it, and a key lifts the pause within thirty seconds.

CONTACT

SEE THE PRODUCTS

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

[Benchmarks](#) · [Roadmap](#) · [Changelog](#) · [Pricing](#) · [Partners](#) · [Contact](#)



PRICING

PRICED PER GATEWAY, KEPT FOR GOOD

Free for development and testing, with no time limit. When a gateway goes into production you buy one license for it and keep it. Nothing is charged per tag, per device or per connected client. Support is optional at 20% per year.

THE CORE

EDGE, HOST, PROVIDER

Edge at each site. Host once. Provider, or any conformant broker you already run, in between.

SPARKBRIDGE EDGE

\$995

Per site gateway.
SparkLink included.

Moves plant data off each site and holds it safely when the link drops.

SPARKBRIDGE HOST

\$1,495

Per Host gateway.
SparkCalc and SparkID included.

Collects every site's data into one place. The modules below run on it.

PROVIDER

\$1,995

Per gateway. Managed broker and policy layer.

SparkBridge's Sparkplug-aware broker for routing, id c through compatible

● [ASK AN ENGINEER](#)

BUY, \$995

BUY, \$1,495

broker infrastructure
you already operate.

BUY, \$1,995

HOST MODULES

MODULES ON HOST

Each runs only on a SparkBridge Host gateway, on its own license. Add one when the need shows up. What each one does is on the [Host page](#).

MODULE	WHAT IT DOES	PRICE	
SparkCalc	Rates, totals and efficiencies calculated once on live data	Included	
SparkID	Who can read and who can send commands, generated from intent	Included	
Passage	Serves live plant data over OPC-UA to the tools you already own	\$1,995	BUY
SparkInject	Production data delivered into your warehouse, exactly once. Per connector family; Snowflake first, Databricks and BigQuery follow	\$2,995	BUY
SparkVault	History kept on the plant's own clock, in an open database	\$3,995	BUY
SparkRecord early access - 0.1.0	Tamper-evident record historian for structured industrial events. Sparkplug records, MQTT JSON events and SparkFlow	\$7,995	BUY

MODULE	WHAT IT DOES	PRICE	
	events landed exactly once with a SHA-256 audit chain. All destinations included		
<u>FleetOps</u>	Every gateway in one view: versions, drift, certificates, backups	\$4,995	BUY
<u>Sentinel</u>	Scores the plant data network, with the evidence and the fix	\$4,995	BUY
<u>Sentinel compliance pack</u>	The same findings as an auditor-ready binder. Add-on to Sentinel	\$2,495	CONTACT
<u>GitOps / Enterprise Governance pilot</u>	Configuration as code for the whole estate: signed bundles, review, rollback, audit trail	\$9,995	BUY
<u>SparkFlow</u>	Every point streamed to the whole company, past the tag-count ceiling. Its own license	\$995	BUY

DRIVERS

DEVICE DRIVERS

Licensed per gateway. Each works with or without SparkBridge. A site connecting one kind of equipment needs that driver and usually not the others.

PRODUCT	WHAT IT DOES	PRICE	
<u>SparkLogix</u>	Allen-Bradley Logix driver. Direct connect, live tag browse, no translator box	\$695	BUY
<u>SparkSNMP</u>	Network, power and server equipment brought into Ignition. One license covers both Ignition lines	\$695	BUY



PRODUCT	WHAT IT DOES	PRICE	
<u>IEC 60870-5-104</u>	Utility and substation equipment, connected directly	\$695	BUY
<u>DNP3</u>	Utility equipment, connected directly	\$695	BUY
<u>TI 505 · CTI 2500</u>	Older Siemens and CTI equipment, connected directly	\$695	BUY
<u>SparkADS</u>	Beckhoff ADS/AMS driver, browses TwinCAT symbols directly	\$695	BUY
<u>SparkS7</u>	Siemens S7-1200/1500 symbolic driver	\$695	BUY
<u>Spark61850</u>	IEC 61850 MMS client driver	\$695	BUY
<u>SparkBACnet</u>	BACnet/IP driver	\$695	BUY

ENHANCEMENTS

IGNITION ENHANCEMENTS

Licensed per gateway. Each works with or without SparkBridge.

PRODUCT	WHAT IT DOES	PRICE	
<u>SparkInflux</u>	InfluxDB history storage for teams already standardized on it	\$1,495	BUY
<u>SparkValidate pilot release</u>	Alarm review and a validation document set, available as a gateway module or Validate Desk CLI	\$2,495 gateway \$995 Validate Desk CLI	GATEWAY CLI

PRODUCT	WHAT IT DOES	PRICE	
<u>SparkNotify</u>	Alarm notification to Teams, Slack, WhatsApp, Telegram and Twilio	\$695	BUY
<u>SparkGantt</u>	Perspective Gantt chart component	\$695	BUY

WHAT AN ESTATE COSTS

FORTY SITES, THREE WAYS

Three real deployment shapes at list price. Required infrastructure first, optional consumers second, so the required number is visible on its own.

\$40,300 40 SITES ON A BROKER YOU ALREADY RUN

39 X EDGE	\$38,805
1 X HOST	\$1,495
BROKER	Your existing conformant MQTT broker, \$0 from us
REQUIRED TOTAL	\$40,300, once
SUPPORT, OPTIONAL	\$8,060 per year (20% of list)
SITE 41	\$995

\$49,780 40 SITES WITH PROVIDER HA AND HISTORY



39 X EDGE	\$38,805
-----------	----------

2 X PROVIDER (HA PAIR)	\$3,990
------------------------	---------

2 X HOST (ACTIVE AND WARM STANDBY)	\$2,990
---------------------------------------	---------

1 X SPARKVAULT	\$3,995
----------------	---------

REQUIRED TOTAL	\$49,780, once
----------------	----------------

SUPPORT, OPTIONAL	\$9,956 per year (20% of list)
-------------------	--------------------------------

SITE 41	\$995
---------	-------

\$69,765

A COMPLETE OPERATING
ARCHITECTURE FOR A
DISTRIBUTED IGNITION
ESTATE

39 X EDGE	\$38,805
-----------	----------

2 X PROVIDER (HA PAIR)	\$3,990
------------------------	---------

2 X HOST (ACTIVE AND WARM STANDBY)	\$2,990
---------------------------------------	---------

REQUIRED SUBTOTAL	\$45,785, once
-------------------	----------------

1 X SPARKVAULT	\$3,995
----------------	---------

1 X FLEETOPS	\$4,995
--------------	---------

1 X SENTINEL	\$4,995
--------------	---------

1 X GITOPS / ENTERPRISE GOVERNANCE	\$9,995, pilot availability
---------------------------------------	-----------------------------

TOTAL	\$69,765, once
-------	----------------

SUPPORT, OPTIONAL	\$13,953 per year (20% of list)
-------------------	---------------------------------



This is drawing [07 running it](#): central history, fleet operations, security posture and signed configuration on the Host pair. The 39 sites and the wire are identical to the second shape.

Optional consumers added to any shape, each on its own license and none of them touching the sites: FleetOps \$4,995, Sentinel \$4,995, SparkInject \$2,995 per connector family, SparkRecord \$7,995, Passage \$1,995.

Nothing is charged per tag, per device or per connected client, so none of these numbers move when the estate gets busier. Nothing stops running if support lapses. Development and testing gateways are free and are not counted. Every figure here is list, before volume pricing.

Not sure which shape fits? [Request an architecture review](#) and we will size it from your topology.

HOW LICENSING WORKS

THE TERMS, BRIEFLY

UNIT	One license per production gateway, per product
TERM	Perpetual. Bought once and kept
DEVELOPMENT	Free, no time limit, nothing held back
SUPPORT	Optional, 20% of list per year. Covers updates and direct engineering support
METERING	None. Tags, devices and connected clients are never counted
ACTIVATION	One key file per gateway, applied with no restart

MOVING A LICENSE

Follows the role, not the hardware. Move it when you rebuild a server

VOLUME

Every price here is list. Tell us your site count and we will run the numbers

INTEGRATORS

Architecture Partners take 20% off list with deal registration; the first five Founding Partners take 40% on their first production estate.
[Partner program](#)

WHAT HAPPENS IF SUPPORT LAPSES?

The software keeps running exactly as it did before. You stop receiving new versions and support until you renew.

CAN I ADD MODULES LATER?

Yes. Buy each module as your site needs it, on its own license, on a Host gateway you already own.

IS DEVELOPMENT REALLY FREE?

Yes. Every paid module runs on Ignition's two-hour module trial until a key file names it; Reset Trial restarts the clock, and a key lifts the pause within thirty seconds with no restart. SparkLink, SparkCalc and SparkID never pause. You buy a license once a gateway goes into production.

TRY IT ON ONE SITE

SEE THE PRODUCTS

Development use is free with no time limit.

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

[Download](#) · [Technology](#) · [Benchmarks](#) · [Roadmap](#) · [Changelog](#) · [Pricing](#) · [Partners](#) · [Contact](#)

SCALE AND RESILIENCE REPORT

EVERY NUMBER, WITH THE MACHINE IT CAME FROM

This is the canonical record behind every performance figure on this site. Each result names its rig, its method and the command that reproduces it. Where two captures disagree, the later quiet-box gate is authoritative and the earlier one is kept for history. Publish methodology, not just numbers.

ENVIRONMENT

THE THREE RIGS

Nothing here ran on cloud hardware. These are the actual machines. Bigger hardware moves the ceilings; the report states floors.

RIG	CPU	RAM	OS AND KERNEL	JVM	BROKER	USED FOR
Linux reference rig	Intel Core i7-9700T, 6 cores, 6 threads, 2.0 GHz	8 GB	Ubuntu 26.04 LTS on WSL2, kernel 6.18	OpenJDK 17.0.20	Embedded HiveMQ CE 2026.5, 4 shared consumers	2026-08-15 quiet-box gate: capacity floor, offered ramp, latency.

[ASK AN ENGINEER](#)

RIG	CPU	RAM	OS AND KERNEL	JVM	BROKER	USED FOR
Mac bench gateway	x86_64, 4 logical / 2 physical cores	not recorded	macOS 12	Temurin JDK 17	HiveMQ CE 2026.5 on loopback	Gap-closure capture (projected end-to-end, latency, store-and- forward), TCK session, standby takeover, live gateway projection on Ignition 8.1.38
Mac 8- core dev box	8 cores, model not recorded	not recorded	macOS	JDK 17	Embedded Moquette, all in one JVM	R3 gate 2026-08-11: codec, single-edge saturation, soak, fan-in, multi-broker

All-in-one-JVM runs put edge, broker and host in one process, so the numbers exclude network round trips and include GC contention between roles. The live gateway rows run the shipped modules on a real Ignition 8.1.38 gateway against a separate HiveMQ CE process.

METHODOLOGY

HOW THE LOAD IS SHAPED



PARAMETER	SETTING
Sites (edge nodes)	One edge per 50,000 wire metrics: 1, 5 and 10 edges for the 50k, 250k and 500k tiers; 20 edges on the opt-in 1M rig. One device BIRTH per edge
Metrics	50,000 / 250,000 / 500,000 wire metrics resident in Host state; 1,000,000 opt-in
Update rates	Capacity: a ramp that raises offered load step by step until delivered rate stops tracking offered, so the knee is found rather than asserted. Latency: paced at about 10,000 messages/s so the path is measured unsaturated. Fan-in: 600,000 updates/s offered throughout
Payload	1,000 metrics per NDATA in the capacity ramp; 50 metrics per message in the codec test; 1 metric per message in the latency test. Sparkplug B protobuf, alias-only after BIRTH
Projection cost	Real per-leaf tag-write cost injected at 50 microseconds per leaf so the end-to-end figure includes writing values into tags, not just decoding them
Duration	Warm-up phase discarded before every measurement (a cold JIT reads as latency on any code). Steady windows of 5 s per ramp step; 20 s soak for heap; node-churn soak configurable to 30 minutes or more
Quiet box	Nothing else running. A concurrent Gradle test JVM halves the 500k tier, which is why every published gate says quiet-box

RESULTS

RESULTS BY DIMENSION

The headline row per dimension is the one quoted elsewhere on the site. Earlier captures are listed so the trend is visible.



DIMENSION	RESULT
Sustained capacity, projected through to visible tags	at least 339,527 updates/s, lowest of the 50k, 250k and 500k tiers 44,769 updates/s at 250k, 4 projector shards; 19,507 at 50k, 1 shar 341k / 340k / 340k across tiers
Offered ramp absorbed without bending	480,000 updates/s, ramp exhausted, ceiling not found
End-to-end latency, edge to broker to host, warm and paced	p50 0.427 ms · p99 0.933 ms p50 0.216 · p95 0.366 · p99 0.517 · p99.9 1.489 · max 3.576 ms, 40,
Host resident state	500,000 wire metrics in under 4 GiB; about 300 bytes per metric per

DIMENSION

RESULT

Wire ingest,
no projection

59,980 / 59,910 / 59,902 updates/s at 50k / 250k / 500k, 0 loss

Codec
throughput,
single thread

3.9M metric-updates/s, encode plus decode round trip

Store-and-
forward
replay after a
broker
outage

40,035 updates/s, loopback

400,000 of 400,000 replayed in 20.33 s = 19,671 updates/s, zero los

Outage
buffering,
durable

2,697 messages/s with fsync every 32 records; about 90 messages/s w

Host scale-
out with
MQTT 5
shared
subscriptions,
ingest only

11,315 (1 broker x 1 consumer) to 222,000 (1 x 4) updates/s, about



DIMENSION

RESULT

Multi-broker scale-out, 16-edge fleet 211 to 215,695 updates/s delivered across 8 brokers against 1, about

Single edge, saturated about 100,000 updates/s, 0 loss

Soak, 20 s about 215,000 updates/s, heap flat, leak-free

Warm-standby Host takeover **244 ms** from offline STATE to promoted online STATE, 25 nodes, 200 m

Live gateway projection, scalar 20,000 leaves across 4 device BIRTHs ready in 1.03 s, about 20,000

Live gateway projection, templates 1,000 x 15 template members ready in 1.80 s, about 8,300 members/s,

The capacity row is a floor, not a peak. The 2026-08-15 ramp was exhausted at 480,000 updates/s without the run bending, so the ceiling on that rig is unknown and higher. Every capacity claim on this site is "at least", never "up to". The gap-closure figures are lower because that rig has 4 logical cores and the projector shards by core count.



WHAT WAS BROKEN ON PURPOSE

Each row is a test in the shipped suite. The first group runs in ordinary CI on every build; the second is the opt-in load suite.

INJECTED FAILURE	REQUIRED BEHAVIOUR	TEST
Broker killed mid-session, restored later	Edge buffers to disk, reconnects, rebirths, replays in order with source timestamps and the historical flag; zero loss	StoreForwardTest
Disk append and fsync failures while buffering	Counted, surfaced as degraded, recovered when the disk returns; a flush after an open failure does not clear the degraded state	DiskFullStoreForwardTest, DiskBufferPartialWriteRecoveryTest
Garbage payloads, malformed topics, oversized messages, skewed timestamps	Host survives and keeps bridging a legitimate edge	ChaosTest, MalformedStructuredPayloadTest
50,000 mutated protobuf payloads	No crash, no hang	DecoderFuzzTest
Untrusted TLS certificate	Connection rejected	TlsNegativeTest
Publish failures on the live path	Counted and retried without blocking the tag subscription	LivePublishFailureTest, PublishPipelineFlushFailureTest

INJECTED FAILURE	REQUIRED BEHAVIOUR	TEST
Primary Host crash	Standby promotes after debounce; edges do not rebirth	standbyDrill harness, documented in HA.md
Node churn for 30 minutes or more	Heap, thread count and node inventory stay bounded; dead nodes evicted	LongSoakTest (load suite)
5,000 distinct edge nodes on one Host	Bounded heap	LargeFleetTest (load suite)
Rebirth storm from sequence gaps under fan-in	Rate-limited per node; loss at 4 edges fell from 76% to 5% after the fix this test found	LoadTest fan-in scenario

CONFORMANCE

CONFORMANCE IS THE BASELINE

Eclipse Sparkplug TCK 3.0.0, ratified binary, 2026-08-15: 182 Edge and 52 Host assertions passed, zero FAIL lines across ten runs. Host MultipleBroker and the BROKER profile were not run, stated on the [conformance record](#) with the unedited logs.

REPRODUCE



RUN IT YOURSELF

The suite ships in the product repository. Run it on a quiet box and compare against the tables above.

LOAD SUITE

● 4 TO 6 MINUTES

```
./gradlew :common:loadTest --console=plain
./gradlew :common:loadTest --tests io.sparkbridge.perf.CapacityLoadTest
./gradlew :common:loadTest -PloadHeap=9g -Pcapacity1m --tests
io.sparkbridge.perf.CapacityLoadTest
./gradlew :common:gatewayBench -PgatewayBench      # against a live gateway
./gradlew :common:standbyDrill                    # two-JVM takeover
# knobs: -Pshards=N -Dsparkbridge.host.rebirthCooldownMs -
Dsparkbridge.mqtt.maxReconnectMs
```

Results and per-scenario notes are written to the console and to build/reports. Send us yours and we will add the rig to this page.

SIZE YOUR ESTATE FROM THESE NUMBERS

Send site count, tags per site and update rates and we return the sizing with the ceiling it is based on.

[REQUEST AN ARCHITECTURE REVIEW](#)

[BACK TO THE SUMMARY TABLE](#)

[40-SITE REFERENCE ARCHITECTURE \(PDF\)](#)

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

[Benchmarks](#) · [Roadmap](#) · [Changelog](#) · [Pricing](#) · [Partners](#) · [Contact](#)

PUBLIC ROADMAP

WHAT IS PLANNED, AND WHAT IS ALREADY PARTLY THERE

No dates. Each item states what ships today, what is being built, and what would make it done. Product pages describe only what is released; this page is the only place planned work appears.

CAPABILITY	SHIPS TODAY	PLANNED	STATUS
Capacity intelligence	Host /capacity reports its own ceiling from benchmark evidence and the shard count it runs; /fleet shows capacity headroom per edge; Prometheus metrics	Shipped in 2.8.0: per-site buffer health (outage coverage from each site's own store-and-forward figures, kept current between BIRTHs), a seven-day ingest trend at /capacity/history, and one expansion recommendation naming the layer to grow, the number behind it, and the benchmark row it rests on. Every figure is a measured lower bound. Proven live on the reference gateway 2026-09-04. Shipped in 2.9.0: per-site buffer message depth and degraded flag on the wire, a degraded-buffer recommendation that names the site, sparklines	Shipped 2.9.0

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CAPABILITY	SHIPS TODAY	PLANNED	STATUS
		/capacity/view, and a clock-skew clamp on the refresh watermark; a 2.8.0 Edge keeps reporting to a 2.9.0 Host. Planned: per-shard queue depth at /capacity/view, the pool-wide figure stands in for now.	
Zero-touch enrollment	One-time enrollment code: a new site bootstraps identity over a signed, TLS-required channel; credentials written owner-only	Shipped in 2.8.0: the Host answers the code itself, keeps a site registry, issues the site's broker identity inside the signed document, and writes the authorization scopes both brokers read. Enrollment sessions are confined to the two enrollment topics on HiveMQ and the embedded broker. Proven live 2026-09-04. Shipped in 2.9.0: the Host issues a certificate for each enrolled site from its own CA, writes the broker credential and reloads the broker without a restart, registers the site with FleetOps, and the Edge settings page (8.1 only) carries the enrollment field and status. Honest limit: the broker does not yet require the client certificate at connect. Planned: client-cert enforcement and cert-to-identity binding, certificate renewal and revocation, a two-round name exchange for true zero-touch enrollment, and Edge settings on the 8.3 line.	Shipped 2.9.0
Data lineage	Single-metric trace inside Host: five hops from receipt to tag write, with per-	Shipped in 2.8.0: a sixth hop at the MQTT client boundary, source timestamp kept apart from arrival time with the store-and-forward  replay flag, consumer hops	Shipped 2.8.0

CAPABILITY	SHIPS TODAY	PLANNED	STATUS
	stage timing, over an admin route	appended by historian and cloud adapters over a write route, a durable ring of the last 1,000 traces that survives a restart, and a lineage page that shows every stage with "not measured" where a boundary is not instrumented. Proven live 2026-09-04. Still to come: edge and broker hops, SparkVault and SparkInject instrumentation.	
Data contracts	SparkRecord validates records against a versioned schema and refuses narrowing changes	Shipped in 2.8.0: a namespace registry on Host (prefix, owner, required metadata, strict or additive or open), a shape adopted per node and device from its first BIRTH, and every later BIRTH held to it. Removals, type changes, suspected renames and missing metadata raise an alarm-able status tag, a Prometheus counter and a registry row; additive mode accepts additions and bumps the version. Proven live 2026-09-04. Still to come: bad-quality projection in strict mode, a settings tab, and a CI check command.	Shipped 2.8.0
GitOps / Enterprise Governance	Signed configuration bundles distributed fleet-wide with anti-rollback sequencing	Shipped in 2.8.0 in a first form: a bundle directory as the source of truth (one signed document per site, every document checked before signing), drift on the Host between the bundle and the version each site declares on its own BIRTH, apply to one site as a canary or to all with an approvals gate, rollback as an older bundle at a newer sequence, and an	Shipped 2.8.0

CAPABILITY	SHIPS TODAY	PLANNED	STATUS
		append-only audit trail. Proven live 2026-09-04. Still to come: the Git adapter with commit-signature verification, signature-based two-key approvals, and the policy documents.	
SparkRecord	Shipped 0.1.0 early access 2026-09-05: tamper-evident record historian with a per-stream SHA-256 chain, Sparkplug B, MQTT JSON and NATS sources, PostgreSQL, TimescaleDB, SQL Server and object-store sinks, at-least-once delivery with a disk spill buffer	0.2.0: writer lifecycle and ownership ledger extracted into separately tested classes, schema migration for existing tables, Snowflake sink validated live, manual MQTT acknowledgement through the shared transport	Shipped 0.1.0
SparkInject connector families	Kafka, BigQuery, Databricks, Azure Data Explorer, S3 and Snowflake modules ship in the 2.7.0 bundle; Kafka, BigQuery, Databricks and ADX proven against live services	Snowflake live validation; further families as partners ask for them	Shipped 2.7.0
SparkNotify channels	Teams, Slack, WhatsApp,	Email channel deprecated; Ignition  ships its own email notification	Shipped 0.2.0

CAPABILITY	SHIPS TODAY	PLANNED	STATUS
	Telegram, Twilio SMS and voice, Discord, PagerDuty (0.2.0, 2026-09-05)	module	
Host MultipleBroker TCK profile	Single-server- with-failover Host plus warm standby; profile not run	Decision on simultaneous multi- server Host sessions, then the profile run	Under review
Sparkplug Compatible listing	TCK record published with the ratified binary	Listing request to the Eclipse Sparkplug Working Group	Administrative

Roadmap items are not commitments to a date and are not sold. If one of them decides a purchase for you, say so and we will tell you plainly where it stands.

ASK ABOUT ANY ROW

- ASK AN ENGINEER
- WHAT HAS ALREADY SHIPPED

USE CASES

FIND YOUR SITUATION

The same architecture serves a wellhead, a bottling line, a pump station. Almost nobody buys everything on day one. Most plants buy one product for one problem, then add the next one when it shows up.

ROUTING

START HERE

Find the row that sounds like your plant. The last column is what fixes it and what it costs: list price, per gateway, bought once and kept.

IF THIS IS YOUR SITUATION	WHAT CHANGES FOR THE BUSINESS	WHAT YOU BUY
A remote site on cellular or satellite. The link drops, and the data bill keeps climbing.	The site reports changes instead of restating the same reading, keeps everything on disk while the link is down, and gives you a plain list of what is actually spending the budget.	SparkBridge Edge \$995, with SparkLink included free. Standalone; nothing else installs at the site.

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IF THIS IS YOUR SITUATION	WHAT CHANGES FOR THE BUSINESS	WHAT YOU BUY
Forty sites landing on one central server that is already stretched thin and cannot ride out a network outage.	Each site holds its own data during an outage and reports only what changed. The central server stops working harder every time someone adds a new dashboard.	SparkBridge Host \$1,495 once, plus SparkBridge Edge \$995 per site. See the products
A trend that runs flat through an outage and then jumps. Nobody can defend the report built on it.	Every reading keeps the time it actually happened. A backlog lands where it belongs, and the trend shows what the plant really did.	SparkVault \$3,995. Host gateway only.
Analysts reconciling duplicate rows in the warehouse by hand, and never quite trusting the totals.	Every reading lands exactly once in your own warehouse. A restart or a dropped connection can no longer leave a gap or a duplicate.	SparkInject \$2,995. Host gateway, per connector family.
A historian or SCADA package that will never be replaced, and still needs the same live data.	Your existing tools connect the way they already do. No custom connectors, no integration project.	Passage \$1,995. Read only. It cannot write or command.
The same efficiency number, hand built in a PLC, a dashboard and a spreadsheet, giving three different answers.	The number is calculated once, in one place, then used everywhere. Everyone in the building looks at the same figure.	SparkCalc Included with Host. Runs at a site or centrally.
Nobody can say what the plant network is exposing, and an auditor is asking for evidence.	A scored report of what is wrong, ranked by risk, with the evidence behind each finding and the fix for it. It	Sentinel \$4,995. Compliance pack \$2,495.

IF THIS IS YOUR SITUATION	WHAT CHANGES FOR THE BUSINESS	WHAT YOU BUY
	only observes and changes nothing.	
One compromised site could write another site's data, or send a command to equipment it has no business touching.	A read permission never quietly includes the ability to control equipment. Access is scoped to one site on purpose.	<u>SparkID</u> Included with Host. Host gateway.
Forty gateways in the field and nobody knows which version, which drifted, or which certificate expires next month.	One view of every gateway in the estate. A fix is made once and applied everywhere instead of forty separate visits.	<u>FleetOps</u> \$4,995. Worth it from roughly ten sites up.
An older Siemens or CTI 505 rack sitting behind a translator box with its own license and its own patching.	The translator box comes out, along with its license and its patching.	<u>TI 505 / CTI 2500 driver</u> \$695. No SparkBridge required.
Substation equipment speaking IEC 104, reached today through a protocol converter.	A direct connection from the gateway to the equipment, with the converter and its failure points removed.	<u>IEC 60870-5-104 driver</u> \$695. Reads only, no control path.
Utility equipment on DNP3, with another vendor's converter in the path.	The same fix as the other drivers: a direct connection, reads only, one less box to maintain.	<u>DNP3 driver</u> \$695. No SparkBridge required.



IF THIS IS YOUR SITUATION	WHAT CHANGES FOR THE BUSINESS	WHAT YOU BUY
A new dashboard or an analytics project is coming, and it must not slow down the plant.	A new consumer of the data adds no load at the source. Cost tracks what the plant is doing, not how many people are watching.	<u>SparkBridge Host</u> \$1,495, or <u>Passage</u> \$1,995 for tools that already connect over industry standards.

All prices are US dollars, list, per gateway, bought once and kept. Not a subscription. Support is 20% per year, optional. Architecture Partners take 20% off list with deal registration, Founding Partners 40% on their first production estate (partner program), and volume discounts apply by site count. Development and testing is free, no time limit.

WIDE AREA

REMOTE SITES, EXPENSIVE LINKS

Cellular, satellite or radio links, where you pay for the data and the connection drops.

The problem

POLLING WASTES THE CONNECTION

Polling spends your data allowance asking questions whose answer is usually "nothing changed." An outage leaves a gap in the history that no later poll can fill. Nobody can name the tag that is actually spending the money. In a real three day capture, one vibration tag with no change threshold set was taking 50.7% of a site's monthly budget, at 47 times the next largest consumer.

What changes

SEND CHANGES, STORE DURING OUTAGES

Normal traffic tracks what the plant is doing. If the link drops, the site  saves readings to disk with their original timestamps and sends them when it reconnects. The history shows

the real curve, not a flat line and a jump. The budget becomes a measured list you can act on. Readings you mark critical are protected in the sending path itself, so no budget rule can quietly degrade them.

Products in play: SparkBridge Edge, [SparkLink](#) for the data budget, and [SparkVault](#) if the history has to survive the outage as well as the link.

CONSOLIDATION

MANY SITES, ONE SERVER

A real example: one central server carrying 40 sites, sitting at 50 to 75% CPU, using 14 GB of memory, unable to ride out a network outage, getting worse every time someone added a dashboard.

280k to 300k metrics. That estate's consolidated published metrics, sized from real gateway backups.

500k in under 4 GiB. The measured single-host capacity, with per-metric cost flat to a million. The fleet fits with margin on every dimension.

Cost does not grow with consumers. Adding a dashboard, a historian or an ML pipeline adds a subscriber, not load at the source.

The estate stays visible. Forty gateways is forty version numbers, forty certificate expiries and forty backup schedules. On a 40-site rig, one authorized certificate rotation reached every site in 937 ms, and the same instruction captured and replayed was refused by all forty.

Products in play: SparkBridge Host and SparkBridge Edge, with [FleetOps](#) once the site count is too large to track by hand.



UNS

ONE SOURCE OF PLANT DATA

This just means every system reads the same up-to-date picture of the plant from one place. You get it by publishing properly, not by buying something else.

The namespace populates itself. Each site describes itself at birth, so a new consumer subscribes and discovers the estate rather than being configured for it.

Readable, not just fast. The same values are also published on a plain, readable path, for people and tools that were never going to read a machine format.

One number, calculated once. Efficiency, throughput and cost per unit are computed in one place and published back as plant data, each value carrying the time of its oldest input and going bad when any input does. That is what makes a reported figure defensible.

Everything you already own can read it. The same live values are served to the historian, SCADA package or analytics tool you are not going to replace, with source timestamps intact.

Products in play: [Passage](#) for the systems you already own, [SparkCalc](#) for the derived numbers, [SparkVault](#) for the history behind them, and [SparkInject](#) when the warehouse is the destination.

OT IT

PLANT TO IT, SAFELY

The security question that matters here is what a compromised site can do to the rest of the estate.



Inbound firewall rules	The site connects outbound only. No inbound port into the plant network is required for data to flow.
A compromised site publishing as another	Each site has its own identity, bound to its own name. An unlisted identity is denied outright.
A compromised site issuing commands	Site identities cannot send commands at all. Command injection is structurally unavailable, not just discouraged.
A compromised site forging enterprise data	Every identity is authorized per data path and denied by default, whether the deployment is on our system or a third party's.
Credentials on the box	Secrets live in a permission-restricted file, not in a place any local user on the machine can read.
Rules typed by hand	Permissions are generated from stated intent, one line per action, and the change is shown as a difference before it is applied. A change that would lock you out of your own system is refused.
Proving any of this to an auditor	A scored assessment with ranked findings, the evidence behind each one and the fix. The first run against our own estate found four genuine defects, one of them critical.

Products in play: [SparkID](#) for the permissions, [Sentinel](#) for the assessment and the evidence.

WHAT IT COSTS

WHAT A WHOLE SOLUTION LOOKS LIKE

Three worked examples at list price. Every figure below comes from the price list. Nothing here is per tag, per device or per connected client.



One remote site on a metered link. A single wellhead or pump station on cellular, where the link drops, the allowance costs money, and the history has to survive both.

That is SparkBridge Edge, \$995, standalone; SparkLink is included. History is kept where the data lands: SparkVault runs on the Host gateway, so if this site is also your only server, it is that server that takes SparkVault at \$3,995.

Nothing central is required. One site is a complete deployment. Development use is free while you decide.

ONE SITE, AT LIST

SPARKBRIDGE EDGE	\$995
SPARKLINK	Free, included
AT THE SITE	\$995, standalone
SPARKVAULT, ON THE CENTRAL SERVER	\$3,995

A 40-site estate. Thirty-nine sites publishing to one Host gateway that hosts the estate and serves everything downstream of it.

At list that is $39 \times \$995 + 1 \times \$1,495$, which is \$40,300, once, before any volume discount. There is no per-tag, per-device or per-connected-client charge. The number does not move when the tag count does or when the third dashboard arrives.

Support is 20% per year, optional. Volume discounts apply by site count and are quoted. Architecture Partners take 20% with deal registration; the first five Founding Partners take 40% on their first production estate. [Partner program](#).

FORTY SITES, AT LIST

39 × SPARKBRIDGE EDGE	\$38,805
1 × SPARKBRIDGE HOST	\$1,495
TOTAL	\$40,300, once

BEFORE	Volume discount by site count
--------	-------------------------------

PER TAG OR PER CLIENT	Nothing
-----------------------	---------

A utility adding two substations on IEC 104. The equipment is already there and already speaks the protocol. What is in the way is a converter per site, with its own license, its own patching and its own failure mode.

The driver has its own license and its own price. It does not require SparkBridge, so this is the whole cost.

The same figure applies to a DNP3 site, or to an older Siemens or CTI 505 rack on the TI 505 driver.

TWO SUBSTATIONS, AT LIST

IEC 60870-5-104 DRIVER, SITE ONE	\$695
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IEC 60870-5-104 DRIVER, SITE TWO	\$695
-------------------------------------	-------

SPARKBRIDGE REQUIRED	No
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TOTAL	\$1,390, once
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CONTROL PATH	None. The drivers read only
--------------	-----------------------------

START WITH ONE SITE

Each of these shapes is drawn out on the [architecture page](#), including the 40-site and 200-plant layouts.

Try it on one site. Development use is free, no time limit.



CONTACT

SEE PRICING

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

[Benchmarks](#) · [Roadmap](#) · [Changelog](#) · [Pricing](#) · [Partners](#) · [Contact](#)



DISTRIBUTED IGNITION ARCHITECTURE REVIEW

SEND THE TOPOLOGY. GET THE DIAGRAM, THE RISKS AND THE SIZING BACK.

Before anyone installs a module, we review the estate as it is and as it is planned. You get an architecture diagram, the scaling risks, the failure domains, the bottlenecks and a recommended shape with its infrastructure requirements. It costs nothing, it is not a sales call, and you keep the document whether or not SparkBridge is in the answer.

[REQUEST A REVIEW](#)[SEE THE COMPARISON FIRST](#)

YOU PROVIDE

SITES	How many today, how many planned
TOPOLOGY	Hub and backend gateways, Gateway Network links, remote tag providers
TAGS AND METRICS	Per site, roughly. A gateway backup answers this exactly if you prefer to send one
UPDATE RATES	Typical and worst case
WAN	Link type per site: fiber, cellular, satellite, radi

[● ASK AN ENGINEER](#)

HISTORY	Where it is written today and who reads it
APPLICATIONS	Everything that consumes the data now, and what is coming
PAIN POINTS	What already hurts. The overloaded box usually has a name
WE RETURN	
ARCHITECTURE DIAGRAM	Your estate drawn as it is and as recommended
SCALING RISKS	Which component saturates first and at what site count
FAILURE DOMAINS	What stops when each server or link fails, and what keeps going
BOTTLENECK ANALYSIS	Fan-in, projection, history writes, consumer load, with the measured ceilings they are judged against
RECOMMENDED ARCHITECTURE	Edge, Provider and Host placement, HA shape, which consumers belong where
INFRASTRUCTURE ESTIMATE	Server count, cores, memory and buffer sizing from the <u>published benchmarks</u>
LIST PRICE	For the recommended shape, per gateway, once

HOW IT RUNS

WHAT HAPPENS AFTER YOU SEND IT

STEP WHAT WE DO

- 1 We reply to confirm what you sent and ask for anything missing. Most reviews need one short call with the engineer who owns the estate.



- 2 GreenGuard USA engineers draw the current and recommended architectures and size them from the measured figures on the [benchmark report](#), not from a spreadsheet of guesses.
- 3 You receive the document. We walk through it once if you want. Where SparkBridge does not solve a problem you have, the document says so.
- 4 If you proceed, the review becomes the deployment plan. Development use of every module is free with no time limit, so the pilot costs nothing to start.

We built the modules and run the benchmarks ourselves. The review is done by the same people, which is why it is free: it is the fastest way for both sides to find out whether the architecture fits.

SEND THE TOPOLOGY

The button opens an email with the intake list already in it. Fill in what you know and leave the rest.

[REQUEST A REVIEW](#)

[SEE WHAT AN ESTATE COSTS](#)

[40-SITE REFERENCE ARCHITECTURE \(PDF\)](#)

SparkBridge · Sparkplug B / 3.0 · MQTT 3.1.1 and MQTT 5 · Ignition 8.1.19+

[Benchmarks](#) · [Roadmap](#) · [Changelog](#) · [Pricing](#) · [Partners](#) · [Contact](#)



SparkBridge product themes (internal)

Status: internal product-strategy document, 2026-09-05. Not published. The public roadmap (/sparkbridge/roadmap) stays detailed and item-by-item; this file is the filter every new item passes through before it reaches that page.

Why this exists

The public roadmap lists ten capabilities. Each is justified on its own, but a list of ten justified items is not a strategy. Without a frame, the next customer request becomes the next roadmap row and the architectural thesis erodes one reasonable-sounding feature at a time.

The thesis, in one line: **publish once at the site, add consumers at the Host, and run the whole estate from one chair**. Sites and consumers scale on separate lines. Nothing at a site changes when a consumer is added.

The four themes

Every shipped module, every roadmap row, and every incoming request lands in exactly one theme.

1. Operate

Run a 40 to 200 site estate from one team without visiting sites.

Item	Status 2026-09-05
Capacity intelligence	Shipped 2.8.0; 2.9.0 wires sf.depthMessages / sf.degraded, sparkline, credential-cut scenario
Zero-touch enrollment	Shipped 2.8.0 slice 1; 2.9.0 slice 2 (CSR/CA + mTLS, HiveMQ credential emit, Edge Settings tab, FleetOps registration)
FleetOps	Shipped; true fleet management (config templates, drift detection) is 3.1 program, NOT NOW
GitOps / Enterprise Governance	Shipped 2.8.0 first form; Git adapter + two-key approvals after 3.0, partner-driven
Lifecycle management (upgrade in place, retire, replace)	Partner proof track: upgrade-in-place QA 2.7 -> 2.8 -> 2.9; no module yet

2. Trust

Prove where a number came from and that nobody changed it on the way.

Item	Status
Data lineage (trace)	Shipped 2.8.0 six hops; slice 2 wire hops (sb_trace, MQTT5 user property) in 3.0.0
Data contracts	Shipped 2.8.0; strict-mode bad-quality projection, settings tab, CI check after 3.0
Sentinel	Shipped; compliance pack \$2,495
SparkID	Shipped, included with Host

Item	Status
SparkValidate	0.1.0 early access candidate; ISA-18.2 / GAMP5 / Part 11 / 62443 documents
Policy-as-code	Not started; the natural home is GitOps policy documents (after 3.0) rather than a new module

3. Move Data

Land the same published data in every system the company runs, exactly or at-least once, with source timestamps intact.

Item	Status
SparkRecord	0.1.0 early access 2026-09-05; 0.2.0 lifecycle extraction + Snowflake live (merged 5493f69, unreleased)
SparkInject families	Kafka, BigQuery, Databricks, ADX, S3, Snowflake all in 2.7.0 bundle; ingest-reconnect fix on branch, awaiting live proof
SparkFlow	Shipped 2.5.0, \$995; NATS streaming for the ultra large estate
SparkVault	Shipped; central history
Passage	Shipped; OPC-UA face
SparkNotify	0.2.0 Discord + PagerDuty shipped 2026-09-05; Email channel removed in 3.0.0
Interoperability (Sparkplug Compatible listing, other vendors' hosts)	Listing is administrative; mixed-vendor pairing has never run on the bench and the site says so

4. Scale

Grow past what one Ignition server or one broker can hold, without a new integration project.

Item	Status
Host MultipleBroker TCK profile	Under review: decision on simultaneous multi-server Host sessions
Device sharding / host pools	Shipped (drawing 03); DEVICE_SHARDING_INVESTIGATION.md
Federated estates / regional architectures	Drawing 06 shape; partner-driven
Sparkplug Compatible listing	Administrative; Eclipse WG request in Dan queue

Feature-accumulation check (external review 2026-09-06)

The reviewer's five questions, asked before any new product, driver, connector or convenience module:

1. Does it reinforce Edge -> Provider -> Host?
2. Does it create a reusable platform capability?
3. Does it benefit multiple customers?
4. Does it strengthen one of the four pillars?
5. Does it reduce operational complexity at scale?

A new driver passes 3 and rarely 1, 2 or 5: drivers stay a separate per-gateway line and never enter the SparkBridge roadmap table. Same for isolated Ignition conveniences (Gantt).

Intake rule for new requests

A request goes on the roadmap only if all four answers are yes:

1. **Which theme?** Name one. If it does not fit any, it is a parking-lot item, not a roadmap row.
2. **Does it advance the thesis?** It must make publish-once, add-consumers-later, or run-from-one-chair more true. A feature that adds a per-site touch, a per-consumer change at the site, or a new licensing dimension (per tag, per client) is rejected on that ground alone.
3. **Is it measurable?** State the number that will be printed on the site when it ships and the bench that will produce it. No number, no row.
4. **Who asked, and how many?** One customer request is evidence, not a mandate. Two design or founding partners asking for the same thing inside the same theme is a mandate.

Requests that pass go to the roadmap table with the theme recorded in the commit message. Requests that fail go to docs/internal/PARKING-LOT.md with the theme they almost fit and the reason they failed, so the answer to the customer is written once.

Ownership

Someone has to hold this frame against the queue. Until a product owner exists, Dan is the product owner and this file is the checklist; Claude applies it when drafting roadmap rows and flags any row that fails a question rather than adding it. The Solution Architect role (the one that draws estates with partners) feeds requests in; it does not decide what ships.

Sequencing, unchanged

The release ordering in ~/ .claude/plans/what-remains-for-sparkbridge-tranquil-blossom.md stands: 2.9.0 additive (Operate: enrollment + capacity), SparkRecord 0.2.0 (Move Data), 3.0.0 breaking batch (Trust: trace wire hops; Move Data: manual-ack transport; hardening), with the partner proof track in parallel feeding site should-adds 4 through 9.

CRM conventions and product calendar

HubSpot

- Contact properties: outreach_campaign = sparkbridge-fap-2026-09; outreach_segment = vertical (manufacturing, water-utilities, data-centers, oil-gas, pharma, other); lifecycle stage lead.
- One contact per named person, one company record per firm. Company name exactly as in the CSV firm column.
- Deals: pipeline "SparkBridge Partners", stages contacted, call booked, review delivered, partner signed, pilot live, production measured. The pipeline must be created once in the HubSpot UI (the MCP integration cannot create pipelines). Until then, deals go in the default Sales Pipeline with the tag "SparkBridge FAP" in the deal name.
- Deal created at first reply, not at first send.
- Every deal carries: owner, next action, next meeting date, architecture status, technical blocker, commercial blocker, target deployment date.
- Call reports go in the deal description in the three blocks defined in 08-discovery-call-and-report.md. Objections as deal notes, exact words.
- Email sends: log manually the same day if the mailbox is not connected to HubSpot.

Gmail and Drive

- Outreach from the MBA's greenguard-usa.com mailbox. Drafts prefixed [FAP] (first touch) and [FAP-2] (follow-up) if Claude drafts them.
- Target sheet: "SparkBridge Partner Targets" in Drive > Claude uploads > Industrial & Drivers. Dan's approve column is the approval record.
- This bundle lives in the SparkBridge repo at docs/gtm/mba-handoff and as a zip in the same Drive folder.

Product calendar during the sprint

When	What	What it means for the MBA
Now	SparkBridge 2.9.0 released 2026-09-06 on Ignition 8.1 and 8.3; SparkRecord 0.1.0 early access; SparkS7 0.2.1; 27 product repos	Quote the pricing page; labels as the site shows them
Target week of 2026-10-12	SparkBridge 3.0.0, breaking: manual-ack transport, client certificate required at the broker with identity binding, two-round enrollment, renewal and revocation	Partners starting now install 3.0.0. Do not sell 2.9.0 enrollment as mTLS.
After 3.0.0	SparkRecord 0.2.0, SparkInject 0.2.0, SparkNotify 0.3.0	Early access and pilot labels may change; check the site
Early December (internal)	3.1 fleet management program	Not sold, not dated externally

Externally there are no dates. The public roadmap has none on purpose. Any "when does X ship" question is answered with the roadmap page and "I will document that requirement".

Who does what

Task	Owner
Row approval	Dan
Research, scoring, outreach, CRM, follow-up, scheduling	MBA
Architecture review pre-read, written review draft, weekly numbers	Claude
Architecture review call, partner terms, pricing exceptions, anything sent to a customer that commits the company	Dan

Why Distributed Ignition Architectures Get Harder at 20 Sites

Status: outline. Engineering document, not an SEO article. Company voice, no em dashes, every number tied to PERFORMANCE.md or the benchmark page.

- Thesis: the hub gateway is a fan-in point; every remote tag provider is a live subscription load on both ends.
- Evidence: the 40-site estate baseline (hub at 50 to 75% CPU with a 14 GB sawtooth) from the migration plan; Irving 8,700 leaves, Mesa 22,400, fleet average ~13k leaves per site, 400 to 500k unscoped.
- Mechanism: GAN subscription aggregation, why front-end user count does not multiply load but site count does.
- Diagram: traditional vs SparkBridge shape from the compare page.
- Numbers to cite: Host holds 500k wire metrics in under 4 GiB; ~300 bytes per metric per consumer.

Sources: PERFORMANCE.md, GATEWAY_BENCH.md, docs/HA.md, docs/TCK.md, docs/products/*.md, <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>

The 40-Site Ignition Reference Architecture

Status: outline. Engineering document, not an SEO article. Company voice, no em dashes, every number tied to PERFORMANCE.md or the benchmark page.

- Thesis: three roles, two HA pairs, one namespace. The document behind the downloadable PDF.
- Content: topology, sizing per PERFORMANCE.md tiers, Provider HA on HiveMQ CE, Host active plus warm standby, consumer placement, buffer sizing per WAN type.
- Diagram: the second pricing example (\$49,780) drawn as servers.
- Numbers: 244 ms standby takeover; 2,697 msg/s durable buffering; 40,035 upd/s replay; capacity floor 339,527 upd/s on the six-core rig.

Sources: PERFORMANCE.md, GATEWAY_BENCH.md, docs/HA.md, docs/TCK.md, docs/products/*.md, <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>

What Actually Happens to Historical Data During a WAN Outage?

Status: outline. Engineering document, not an SEO article. Company voice, no em dashes, every number tied to PERFORMANCE.md or the benchmark page.

- Thesis: arrival-time history is wrong history. Walk one reading through a four-hour outage.
- Mechanism: disk buffer, is_historical flag, source timestamp propagation into the managed provider, SparkVault backfill.
- Evidence: 400,000 of 400,000 replayed; the SparkVault four-hour-eleven-minute disconnect with zero loss.
- Diagram: the products-page journey (one timestamp that never moved).

Sources: PERFORMANCE.md, GATEWAY_BENCH.md, docs/HA.md, docs/TCK.md, docs/products/*.md, <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>

Polling vs Publish/Subscribe at Enterprise Scale

Status: outline. Engineering document, not an SEO article. Company voice, no em dashes, every number tied to PERFORMANCE.md or the benchmark page.

- Thesis: polling cost grows with tags times consumers; report-by-exception grows with process activity.
- Mechanism: NBIRTH pays for the name once, NDATA is alias-only; metadata folding removes about a third of steady-state points.
- Numbers: 3.9M metric-updates/s codec; wire ingest 59.9k upd/s per consumer; 20x with four shared consumers.

Sources: PERFORMANCE.md, GATEWAY_BENCH.md, docs/HA.md, docs/TCK.md, docs/products/*.md, <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>

Adding a New Data Consumer Without Touching 40 Plants

Status: outline. Engineering document, not an SEO article. Company voice, no em dashes, every number tied to PERFORMANCE.md or the benchmark page.

- Thesis: publish once, add consumers when you need them. Shared subscriptions are the mechanism.
- Gotchas worth publishing: naive \$share breaks Sparkplug two ways (alias tables live in BIRTH; seq is per edge), and how SparkBridge subscribes BIRTH/DEATH/STATE non-shared.
- Diagram: add-a-consumer from the compare page.
- Numbers: 11,315 to 222,000 upd/s ingest, 1x1 to 1x4.

Sources: PERFORMANCE.md, GATEWAY_BENCH.md, docs/HA.md, docs/TCK.md, docs/products/*.md, <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>

Why Fleet Operations Become the Real Problem After Deployment

Status: outline. Engineering document, not an SEO article. Company voice, no em dashes, every number tied to PERFORMANCE.md or the benchmark page.

- Thesis: month two questions: which sites are offline, buffering, drifted, expiring.
- Content: FleetOps inventory and drift, Host /fleet headroom columns, Sentinel findings with fixes, signed configuration bundles, the enrollment code.
- Honest caveat: certificate issuance and reconciliation are on the roadmap, say so.

Sources: PERFORMANCE.md, GATEWAY_BENCH.md, docs/HA.md, docs/TCK.md, docs/products/*.md, <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>

Timestamp Integrity: When Did the Machine Actually See the Value?

Status: outline. Engineering document, not an SEO article. Company voice, no em dashes, every number tied to PERFORMANCE.md or the benchmark page.

- Thesis: a timestamp that survives the trip. Source time vs receipt time at every hop.
- Mechanism: per-metric timestamps in Sparkplug, batching skew bound (250 ms), replay preserving capture time, SparkRecord keeping source and receipt separately.
- Future: the lineage roadmap item, framed as what we do not have yet.

Sources: PERFORMANCE.md, GATEWAY_BENCH.md, docs/HA.md, docs/TCK.md, docs/products/*.md, <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>

How to Scale Ignition Consumers Horizontally

Status: outline. Engineering document, not an SEO article. Company voice, no em dashes, every number tied to PERFORMANCE.md or the benchmark page.

- Thesis: consumers scale on the shared stream; producers never change.
- Content: shard-by-edge dispatch preserving per-node order, projector shards by core count, control-message divert, the 2.3x sharding gain and why the 50k tier is the control.
- Numbers: 19,507 vs 44,769 upd/s; ramp bends at 55,464 on four logical cores; 480,000 absorbed on six.

Sources: PERFORMANCE.md, GATEWAY_BENCH.md, docs/HA.md, docs/TCK.md, docs/products/*.md, <https://mqtt.greenguard-usa.com/sparkbridge/benchmarks>